Discrete Mathematics: Exam Companion

Diskret støddfrøði (005019.23)
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Notation bridge: slides vs. Kolman vs. past MC papers

The slides write $\neg p$, $p \to q$, $p \leftrightarrow q$. Kolman and the past MC papers write $\sim p$, $p \Rightarrow q$, $p \Leftrightarrow q$ — same meaning. On the slides $\mathbb{N} = \{1, 2, 3, \dots\}$; the MC papers and some books include 0. If a question hinges on it, check which convention it uses.

Yellow shading marks the rules, laws and theorems.

Exam method subsections are step-by-step recipes for the typical MC question types (own notes). **Not on the slides** = from Kolman; **Slide erratum** = a mistake on a slide. The multiplication table and a list of primes are on the last page before the index.

Contents

1 Fundamentals 5

1.1 Sets 5

1.2 Operations on sets 5
1.2.1 Set laws (Theorem 1.2.1) . . . 6
1.2.2 Inclusion–exclusion . . . 6

1.3 Sequences and characteristic functions 6

1.4 Properties of integers 7
1.4.1 Divisibility rules (no calculator) . . . 7
1.4.2 Prime factorization by trial division (exam method) . . . 8
1.4.3 Is n prime? (exam method) . . . 8
1.4.4 Number of divisors . . . 8
1.4.5 GCD and LCM . . . 8
1.4.6 Mod-n and number bases . . . 10

1.5 Matrices 11
1.5.1 Boolean (0/1) matrices . . . 11

2 Logic 13

2.1 Propositions and logical operations 13

2.2 Conditional statements and equivalence 13
2.2.1 Theorem 2.2.1 (fundamental equivalences) . . . 14
2.2.2 Theorem 2.2.2 (conditionals) . . . 14
2.2.3 Theorem 2.2.3 (quantifiers) . . . 14
2.2.4 Theorem 2.2.4 (tautologies behind the inference rules) . . . 15

2.3 Methods of proof 15

2.4 Mathematical induction 16

2.5 From pattern to conjecture to proof **16**

3 Counting **17**

3.1 Permutations and combinations **17**
3.1.1 Multiplication principle . . . 17
3.1.2 Factorial . . . 17
3.1.3 Working with factorials without a calculator (exam method) . . . 17
3.1.4 Order matters vs. order does not matter . . . 18
3.1.5 Pascal's triangle (look-up table) . . . 18
3.1.6 Arranging letters with repeats . . . 19
3.1.7 Repetition allowed: stars and bars . . . 19
3.1.8 Exactly, at least, at most . . . 19
3.1.9 When the MC options are products of primes . . . 20
3.1.10 Which formula? The four cases . . . 20
3.1.11 Worked slide examples . . . 20

3.3 Pigeonhole principle **21**

3.4 Probability **21**
3.4.1 Adding fractions (exam method) . . . 22

3.5 Recurrence relations **22**
3.5.1 Exponent rule . . . 23
3.5.2 The four-step recipe for distinct roots (exam method) . . . 23
3.5.3 Linear and homogeneous — recognising the type . . . 23
3.5.4 Working backwards: from characteristic equation to recurrence . . . 24
3.5.5 Sigma notation and the geometric series . . . 24

4 Relations and Digraphs **26**

4.1 Relations and digraphs **26**
4.1.1 Computing R^2 from a list of pairs (exam method) . . . 26
4.1.2 R^∞: the connectivity relation (exam method) . . . 27
4.1.3 Slide results for 4.1–4.3 . . . 28

4.4 Properties of relations **28**

4.5 Equivalence relations **29**

4.7 Operations on relations **29**

5 Functions **31**

5.1 Functions **31**
5.1.1 Checking a function from a table (exam method) . . . 31
5.1.2 Injective or surjective from a formula (exam method) . . . 32
5.1.3 Inverses . . . 32

5.2 Functions for computer science **33**

5.3 Growth of functions **33**
5.3.1 Limits and L'Hôpital . . . 34

6 Order Relations and Structures **35**

6.1 Partial orders and posets **35**
6.1.1 Comparability, chains, duals . . . 35
6.1.2 Product and lexicographic order . . . 35
6.1.3 Exam method: is a relation a partial order? . . . 36
6.1.4 Hasse diagrams . . . 37
6.1.5 Exam method: from a matrix to a Hasse diagram . . . 37
6.1.6 Topological sorting . . . 38
6.1.7 Isomorphism of posets . . . 38

6.2 Extremal elements of posets **39**
6.2.1 Upper and lower bounds, LUB and GLB . . . 39
6.2.2 Exam methods: greatest in a lexicographic order, bounds from a diagram, product order . . . 40
6.2.3 Topological sorting: algorithm SORT . . . 41
6.2.4 Exam method: which list is *not* a topological sorting? . . . 41
6.2.5 Theorems . . . 42
6.2.6 Exercises covered . . . 42

6.3 Lattices **42**
6.3.1 Properties of lattices . . . 43
6.3.2 Exam method: is this Hasse diagram a lattice? . . . 44
6.3.3 Sublattice . . . 44
6.3.4 Bounded lattice . . . 45
6.3.5 Distributive lattice . . . 45
6.3.6 Modular lattice . . . 45
6.3.7 Complement . . . 46
6.3.8 Exercises covered . . . 47

6.4 Finite Boolean algebras **48**
6.4.1 Exam method: is it a Boolean algebra? . . . 48
6.4.2 Which D_n are Boolean algebras? . . . 49
6.4.3 Shape of the Hasse diagram of D_n (exam method) . . . 49
6.4.4 Fast square-free check (no factoring needed) . . . 50
6.4.5 Lookup: every D_n seen in questions and examples . . . 51
6.4.6 Calculating in a Boolean algebra . . . 51
6.4.7 Exercises covered . . . 53

6.5 Functions on Boolean algebras **53**
6.5.1 Logic diagrams . . . 54
6.5.2 Exercises covered . . . 55

6.6 Circuit design **55**
6.6.1 Karnaugh maps . . . 56
6.6.2 Grouping rules and slide examples . . . 57

7 Trees **58**

7.1 Rooted trees **58**
7.1.1 Exam method: which relation (or matrix) is a (directed) tree? . . . 59

7.2 Labeled trees **60**

7.3 Tree searching **61**
7.3.1 Exam method: traversal and Polish notation 62

7.4 Undirected trees **63**
7.4.1 Spanning trees 63
7.4.2 Spanning trees by vertex merging 64

7.5 Minimal spanning trees **64**
7.5.1 Prim in the matrix 65
7.5.2 Exam method: spanning-tree and MST questions 66

Exam review and 2023/2024 practice **67**

Multiplication Table and Primes **68**

Index **69**

Question looks like	**First thing to ask**	**Go to**
Size of a union of sets	Do the sets overlap?	§1.2.2, p. 6
gcd, lcm, $sa + tb$	Can I divide repeatedly?	§1.4.5, p. 8
Are two statements equivalent?	Same truth value on every row?	§2.2, p. 13
Prove $p \Rightarrow q$	Is $\neg q \Rightarrow \neg p$ easier?	§2.3, p. 15
Show something is impossible	Does the opposite lead to a clash?	§2.3, p. 15
Statement for all n	Does $P(k)$ give $P(k+1)$?	§2.4, p. 16
How many selections?	Does order matter? Repetition?	§3.1.10, p. 20
"How many to be sure…"	Objects vs. categories, worst case plus one	§3.3, p. 21
Chance of an event	Equally likely? Overlap? Complement easier?	§3.4, p. 21
Closed form of a recurrence	Linear, homogeneous, degree 2?	§3.5.2, p. 23
Compare running times	Which term dominates as $n \to \infty$?	§5.3, p. 33
Equivalence relation or partial order	Reflexive, symmetric/antisymmetric, transitive?	§4.5, p. 29
Injective, surjective, invertible	Table, formula, or matrix?	§5.1, p. 31
Lattice, Boolean algebra, Hasse diagram, trees	See the one-page recap	p. 67

Chapter 1
Fundamentals

1.1 Sets

Source: Kolman 1.1

Definition. A **set** is a well-defined collection of objects: for any object you can decide whether it is in the set. Order and repetition don't matter: $\{1,2,3\} = \{3,2,1\}$. We write $x \in A$ or $x \notin A$.

$\mathbb{Z}^+$	positive integers $\{1,2,3,\dots\}$
$\mathbb{N}$	natural numbers $\{1,2,3,\dots\}$ (slides' convention)
$\mathbb{Z}$	integers $\{\dots,-2,-1,0,1,2,\dots\}$
$\mathbb{Q}$	rationals $\{a/b \mid a,b \in \mathbb{Z},\ b \neq 0\}$
$\mathbb{R}$	real numbers
$\emptyset$, U	empty set, universal set

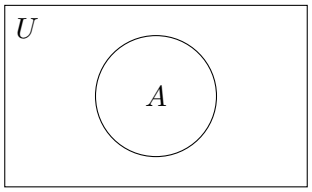
$$\mathbb{Z}^+ \subseteq \mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}$$

Terms. $A \subseteq B$: every element of A is in B. $A \nsubseteq B$: at least one element of A is not in B. $A \cap B = \emptyset$: disjoint. $|A|$ is the **cardinality** (number of elements). The **power set** $P(A)$ is the set of all subsets, including $\emptyset$ and A, and $|P(A)| = 2^{|A|}$.

Example. $A = \{1,2,3\}$: $|A| = 3$, $P(A) = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1,2\}, \{1,3\}, \{2,3\}, \{1,2,3\}\}$, $|P(A)| = 2^3 = 8$.

Set-builder notation. $P(x)$ is a sentence about x (a **predicate**); $\{x \mid P(x)\}$ is read "the set of all x such that $P(x)$". E.g. $\{x \mid x \in \mathbb{Z} \text{ and } x^2 < 12\} = \{-3,-2,\dots,3\}$.

Venn diagram: the universal set U is a rectangle, a set A a circle inside it.



1.2 Operations on sets

Source: Kolman 1.2

$A \cup B = \{x : x \in A \vee x \in B\}$	union (OR)
$A \cap B = \{x : x \in A \wedge x \in B\}$	intersection (AND)
$A - B = \{x : x \in A \wedge x \notin B\}$	difference
$A \oplus B = (A - B) \cup (B - A)$	symmetric difference (XOR)
$\overline{A} = U - A$	complement

Example. $A = \{a,b,c,e,f\}$, $B = \{b,d,r,s\}$: $A \cup B = \{a,b,c,d,e,f,r,s\}$. $A = \{a,b,c,e,f\}$, $B = \{b,e,f,r,s\}$: $A \cap B = \{b,e,f\}$.

1.2.1 Set laws (Theorem 1.2.1)

Commutative	$A \cup B = B \cup A$	$A \cap B = B \cap A$
Associative	$A \cup (B \cup C) = (A \cup B) \cup C$	$A \cap (B \cap C) = (A \cap B) \cap C$
Distributive	$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$	$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
Identity	$A \cup \emptyset = A$	$A \cap U = A$
Domination	$A \cup U = U$	$A \cap \emptyset = \emptyset$
Idempotent	$A \cup A = A$	$A \cap A = A$
Complement	$A \cup \overline{A} = U$	$A \cap \overline{A} = \emptyset$
	$\overline{\emptyset} = U$	$\overline{U} = \emptyset$
De Morgan	$\overline{A \cup B} = \overline{A} \cap \overline{B}$	$\overline{A \cap B} = \overline{A} \cup \overline{B}$

De Morgan: when the complement moves inside, $\cup \leftrightarrow \cap$ swap.

1.2.2 Inclusion–exclusion

$$|A \cup B| = |A| + |B| - |A \cap B|$$
$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$$

Add singles, subtract pairwise overlaps, add back the triple overlap.

Example (1.2 Ex. 7). $A = \{a, b, c, d, e\}$, $B = \{c, e, f, h, k, m\}$: $|A \cup B| = 5 + 6 - 2 = 9$.

Example (1.2 Ex. 8). $A = \{a, b, c, d, e\}$, $B = \{a, b, e, g, h\}$, $C = \{b, d, e, g, h, k, m, n\}$: $|A \cup B \cup C| = 5 + 5 + 8 - 3 - 2 - 4 + 2 = 11$.

Example (1.2 Ex. 10, commuter survey). $|B| = 30$, $|T| = 35$, $|A| = 100$, $|B \cap T| = 15$, $|B \cap A| = 15$, $|T \cap A| = 20$, $|B \cap T \cap A| = 5$. Respondents $= 30 + 35 + 100 - 15 - 15 - 20 + 5 = 120$ (answer not on slides).

1.3 Sequences and characteristic functions

Source: Kolman 1.3

A **sequence** is a list in a definite order; unlike a set, order and repeats matter. $(1, 2, 3, 2, 2, 3, 1)$ is a sequence; swapping two elements gives a different sequence. It can be finite or infinite.

- **Explicit formula**: a_n computed directly from n, e.g. $a_n = 2n$ gives $2, 4, 6, 8, \ldots$
- **Recursive formula**: a_n from earlier terms, e.g. $a_n = a_{n-1} + a_{n-2}$ gives $1, 1, 2, 3, 5, 8, \ldots$

Definition. The **characteristic function** of $A \subseteq U$ is $f_A(x) = 1$ if $x \in A$, 0 if $x \notin A$.

Theorem 1.3.1

$f_{A \cap B} = f_A f_B \qquad f_{A \cup B} = f_A + f_B - f_A f_B \qquad f_{A \oplus B} = f_A + f_B - 2 f_A f_B$

Example (1.3 Ex. 12). $U = \{1, \ldots, 6\}$, $A = \{1, 2\}$, $B = \{2, 4, 6\}$, $C = \{4, 5, 6\}$ as bit strings: $f_A = (1, 1, 0, 0, 0, 0)$, $f_B = (0, 1, 0, 1, 0, 1)$, $f_C = (0, 0, 0, 1, 1, 1)$.

1.4 Properties of integers

Source: Kolman 1.4

Theorem 1.4.1 (division with remainder)

For integers m, n with $n > 0$ there are unique q, r with $m = qn + r$, $0 \le r < n$. (m dividend, n divisor, q quotient, r remainder.) E.g. $17 = 5 \cdot 3 + 2$.

The remainder is *never* negative — careful with negative n:

Example. $47 = 6 \cdot 7 + 5$, so $q = 6$, $r = 5$.
$-47 = (-7) \cdot 7 + 2$, so $q = -7$, $r = 2$ (not $q = -6$, $r = -5$).
$5 = 0 \cdot 7 + 5$: when $n < m$ the quotient is 0 and $r = n$.

$m \mid n$ exactly when the remainder is 0. In code this is `n % m` — Python's `%` follows the same rule (`-47 % 7 == 2`), C# and C do not (`-47 % 7 == -5`).

Divisibility. $n \mid m$ if $m = qn$ for some integer q. $4 \mid 24$, $5 \mid 135$, $4 \nmid 25$.

Theorem 1.4.2. For integers a, b, c: $a \mid b \wedge a \mid c \Rightarrow a \mid (b + c)$ and $a \mid (b - c)$; $a \mid b \Rightarrow a \mid bc$; $a \mid b \wedge b \mid c \Rightarrow a \mid c$.

Primes. $p > 1$ whose only positive divisors are 1 and p: $2, 3, 5, 7, 11, 13, 17, 19, 23, 29, \ldots$ (note that 1 is *not* prime).

Theorem 1.4.3 (unique factorisation). Every $n > 1$ is uniquely $p_1^{k_1} p_2^{k_2} \cdots p_s^{k_s}$ with primes $p_1 < \cdots < p_s$, $k_i \ge 1$. $30 = 2 \cdot 3 \cdot 5$, $1890 = 2 \cdot 3^3 \cdot 5 \cdot 7$.

1.4.1 Divisibility rules (no calculator)

$a \mid b$ means "a divides b": $b = a \cdot k$ for some integer k.

d	$d \mid n$ when ...	Why it works
2	last digit is even	10 is a multiple of 2, so only the last digit matters
3	digit sum is a multiple of 3	$10, 100, 1000, \ldots$ are each 1 more than a multiple of 3 ($9, 99, 999, \ldots$)
4	last two digits form a multiple of 4	100 is a multiple of 4
5	last digit is 0 or 5	10 is a multiple of 5
6	divisible by both 2 and 3	$6 = 2 \cdot 3$ and $\gcd(2, 3) = 1$
7	chop off the last digit b, subtract $2b$ from what is left; the result is a multiple of 7	$n = 10a + b$ and $n - 21b = 10(a - 2b)$, with $7 \mid 21$ and $7 \nmid 10$
8	last three digits form a multiple of 8	1000 is a multiple of 8
9	digit sum is a multiple of 9	same reason as 3
10	last digit is 0	
11	alternating digit sum (from the right: $+ - + \cdots$) is 0 or a multiple of 11	$10, 1000, \ldots$ are 1 *less* than a multiple of 11, $100, 10000, \ldots$ are 1 more, so the signs alternate

Combining rules: if $\gcd(m, k) = 1$ then $mk \mid n \iff m \mid n$ and $k \mid n$. So $12 \mid n$ iff $3 \mid n$ and $4 \mid n$ — but 2 and 6 do *not* give 12, since $\gcd(2, 6) \neq 1$.

1.4.2 Prime factorization by trial division (exam method)

1. Try the primes in increasing order: $2, 3, 5, 7, 11, \ldots$
2. When a prime p divides the current number, write p down, divide, and try the *same* p again on the quotient.
3. When p no longer divides, move on to the next prime.
4. Stop when the quotient is 1, or when p^2 exceeds the quotient — the quotient is then itself prime.

Why stop at $\sqrt{n}$: if $n = ab$ with $1 < a \le b$, then $a \cdot a \le a \cdot b = n$, so $a \le \sqrt{n}$. A composite number always has a prime factor $\le \sqrt{n}$; if no prime up to $\sqrt{n}$ divides n, then n is prime. Trying primes in increasing order also means every factor found is automatically prime — any smaller factor would already have been divided out.
Layout: write it as a "division ladder", the number on the left and the prime that divides it on the right.

Example (495).

495	3
165	3
55	5
11	11
1	

- 2: last digit 5 is odd — no.
- 3: $4 + 9 + 5 = 18$ — yes, $495 = 3 \cdot 165$.
- 3 again: $1 + 6 + 5 = 12$ — yes, $165 = 3 \cdot 55$.
- 3 again: $5 + 5 = 10$ — no, move on.
- 5: last digit 5 — yes, $55 = 5 \cdot 11$.
- 11: next prime 7 has $7^2 = 49 > 11$, so 11 is prime. Stop.

$$495 = 3^2 \cdot 5 \cdot 11$$

Check by multiplying back: $9 \cdot 5 = 45$, $45 \cdot 11 = 495$ ✓

Example (1001). 2: odd. 3: digit sum 2. 5: last digit 1. 7: $100 - 2 \cdot 1 = 98 = 7 \cdot 14$ — yes, $1001 = 7 \cdot 143$. On 143: 7 again gives $14 - 2 \cdot 3 = 8$, no. 11: $3 - 4 + 1 = 0$ — yes, $143 = 11 \cdot 13$, and 13 is prime.

$$1001 = 7 \cdot 11 \cdot 13$$

1.4.3 Is n prime? (exam method)

Test only the primes $p \le \sqrt{n}$. If none divides n, then n is prime.

Example. 211: $\sqrt{211} \approx 14.5$ (since $14^2 = 196$, $15^2 = 225$), so test $2, 3, 5, 7, 11, 13$. Odd; digit sum 4; last digit 1; $21 - 2 = 19$; $1 - 1 + 2 = 2$; $13 \cdot 16 = 208$, remainder 3. None divide, so 211 is **prime**.
221: same primes to test, and $13 \cdot 17 = 221$, so it is **not** prime — and $13 \le \sqrt{221} \approx 14.9$ as promised.

1.4.4 Number of divisors

If $n = p_1^{e_1} \cdots p_k^{e_k}$, every divisor of n is $p_1^{f_1} \cdots p_k^{f_k}$ with $0 \le f_i \le e_i$: $e_i + 1$ choices for each exponent, so by the multiplication principle

$$\text{number of positive divisors of } n = (e_1 + 1)(e_2 + 1) \cdots (e_k + 1).$$

Example. $495 = 3^2 \cdot 5^1 \cdot 11^1$ has $(2 + 1)(1 + 1)(1 + 1) = 12$ divisors: 1, 3, 5, 9, 11, 15, 33, 45, 55, 99, 165, 495.

1.4.5 GCD and LCM

$\gcd(a, b)$ is the largest k with $k \mid a$ and $k \mid b$.

Theorem 1.4.4. If $d = \gcd(a, b)$ then (1) $d = sa + tb$ for some integers s, t; (2) $d \mid a$ and $d \mid b$; (3) any common divisor c of a and b satisfies $c \mid d$.

Euclidean algorithm

a	b	q	r
190	34	5	20
34	20	1	14
20	14	1	6
14	6	2	2
6	2	3	0

$\gcd(190, 34) = 2$

```
gcd(a, b):
  while b != 0:
    r = a % b
    a = b
    b = r
  return a
```

Also $\gcd(12, 30) = 6$ (answer not on slides). If $\gcd(a, b) = 1$, then a and b are **relatively prime** (no common prime factor).

Method 1 — prime factorizations. Line up the factorizations over the same primes (exponent 0 where a prime is missing) and take the **smallest** exponent of each prime:

$$\gcd(a, b) = \prod p_i^{\min(e_i, f_i)} \qquad \text{where } a = \prod p_i^{e_i},\ b = \prod p_i^{f_i}.$$

Example. $495 = 3^2 \cdot 5^1 \cdot 7^0 \cdot 11^1 \cdot 13^0$ and $1001 = 3^0 \cdot 5^0 \cdot 7^1 \cdot 11^1 \cdot 13^1$. Only 11 appears in both, so $\gcd(495, 1001) = 11$.

Method 2 — Euclidean algorithm. No factoring needed, so it is the fast method for large numbers. It rests on

$$\gcd(a, b) = \gcd(b, r) \quad \text{where } a = qb + r, \qquad \text{and} \qquad \gcd(a, 0) = a.$$

Divide the larger by the smaller, replace the pair by (smaller, remainder), and repeat. The last *nonzero* remainder is the gcd. It is a loop: `while b != 0:  a, b = b, a % b`, then the answer is `a`.

Example ($\gcd(120, 84)$).

$$\begin{aligned} 120 &= 1 \cdot 84 + 36 \\ 84 &= 2 \cdot 36 + 12 \\ 36 &= 3 \cdot 12 + 0 \end{aligned}$$

The last nonzero remainder is 12, so $\gcd(120, 84) = 12$.

Writing the gcd as $sa + tb$. There are always integers s, t with $\gcd(a, b) = sa + tb$. To find them, run Euclid, then solve each line for its remainder and substitute backwards, starting from the line that produced the gcd:

Example (continued). From the lines above: $12 = 84 - 2 \cdot 36$ and $36 = 120 - 1 \cdot 84$.

$$\begin{aligned} 12 &= 84 - 2 \cdot 36 \\ &= 84 - 2\,(120 - 84) \\ &= 3 \cdot 84 - 2 \cdot 120. \end{aligned}$$

So $s = -2$, $t = 3$ with $a = 120$, $b = 84$. Check: $252 - 240 = 12$ ✓. (Keep 84 and 120 as symbols while substituting — only collect terms at the end.)

Useful gcd rules.

- $\gcd(a, b) = \gcd(b, a)$ and $\gcd(a, 0) = a$.
- $\gcd(a, b) = \gcd(a, b - ka)$ for any integer k — the Euclidean step.
- If p is prime: $\gcd(p, n) = p$ if $p \mid n$, otherwise 1.
- Consecutive integers are always relatively prime: $\gcd(n, n + 1) = 1$.

$\operatorname{lcm}(a, b)$ is the smallest positive k with $a \mid k$ and $b \mid k$.

Theorem 1.4.6

$\gcd(a,b)\cdot\operatorname{lcm}(a,b)=ab$. Proof idea: $a=dx$, $b=dy$ with $\gcd(x,y)=1$, so $\operatorname{lcm}(a,b)=dxy$ and $d\cdot dxy=(dx)(dy)=ab$.

Method 1 — prime factorizations. Take the **largest** exponent of each prime:

$$\operatorname{lcm}(a,b)=\prod p_i^{\max(e_i,f_i)}.$$

Example. $120=2^3\cdot3\cdot5$ and $84=2^2\cdot3\cdot7$: lcm $=2^3\cdot3\cdot5\cdot7=840$ and gcd $=2^2\cdot3=12$, matching Euclid.

Method 2 — from the gcd.

$$\gcd(a,b)\cdot\operatorname{lcm}(a,b)=a\cdot b\qquad\Longrightarrow\qquad\operatorname{lcm}(a,b)=\frac{a}{\gcd(a,b)}\cdot b.$$

This holds because $\min(e,f)+\max(e,f)=e+f$ for every prime's exponents. Divide *before* multiplying to keep the numbers small.

Example. $\operatorname{lcm}(495,1001)=\dfrac{495}{11}\cdot1001=45\cdot1001=45045$.

Special cases: if $\gcd(a,b)=1$ then $\operatorname{lcm}(a,b)=ab$; if $a\mid b$ then $\gcd(a,b)=a$ and $\operatorname{lcm}(a,b)=b$.

1.4.6 Mod-n and number bases

$f_n(z)=r$ where $z=qn+r$, $0\le r<n$ (i.e. $z \bmod n$). $f_3(14)=2$ since $14=3\cdot4+2$; $f_7(153)=6$ since $153=7\cdot21+6$.

Theorem 1.4.7. For any base $b>1$ every positive n is uniquely $n=d_kb^k+d_{k-1}b^{k-1}+\cdots+d_1b+d_0$ with $0\le d_i<b$. Convert by repeated division by b; the remainders, read bottom-up, are the digits.

Example (1.4 Ex. 11, answers not on slides). $158=2\cdot64+1\cdot16+3\cdot4+2=2132_4$. $215=1\cdot125+3\cdot25+3\cdot5+0=1330_5$.

Base 10 → base b. Divide by b repeatedly, writing down the remainders, until the quotient is 0. The remainders read **bottom-up** are the digits (the first remainder is the last digit).

Example (495 to hex).

$$\begin{aligned}495&=30\cdot16+15\quad(\text{F})\\30&=1\cdot16+14\quad(\text{E})\\1&=0\cdot16+1\end{aligned}$$

Bottom-up: $495=(1\text{EF})_{16}$.

Base b → base 10. Either sum d_ib^i, or (fewer multiplications) go left to right: start at 0, and for each digit multiply by b and add the digit — Horner's rule, a single loop.

Example. $(1\text{EF})_{16}$: $0\cdot16+1=1$, $1\cdot16+14=30$, $30\cdot16+15=495$ ✓
$(2101)_3=2\cdot27+1\cdot9+0\cdot3+1=64$.

Binary ↔ hex shortcut. $16=2^4$, so each hex digit is exactly four bits. Group the bits from the *right* in fours:

$$(1\text{EF})_{16}=\underbrace{0001}_{1}\ \underbrace{1110}_{\text{E}}\ \underbrace{1111}_{\text{F}}=(111101111)_2.$$

The same trick works for octal with groups of three ($8=2^3$).

Example (MC practice: $(123)_{16}$ in base 8). **Via binary** (no arithmetic): $1 \to 0001$, $2 \to 0010$, $3 \to 0011$, so $(123)_{16} = (000100100011)_2 = (100100011)_2$. Regroup from the right in threes: $100\,|\,100\,|\,011 = 4, 4, 3$, i.e. $(443)_8$.
Check via decimal: $(123)_{16} = 256 + 32 + 3 = 291$, and $291 = 4 \cdot 64 + 4 \cdot 8 + 3$ ✓. The distractors are the same digits shuffled, so a reversed grouping (starting from the left) lands on one.

Binary addition. Add column by column from the right, carrying when a column reaches 2 (as $1 + 1 = 10_2$):

$$\begin{array}{r} 1011 \\ +\,1100 \\ \hline 10111 \end{array} \qquad (11 + 12 = 23)$$

1.5 Matrices

Source: Kolman 1.5

An $m \times n$ matrix has m rows and n columns. **Square**: $m = n$. **Diagonal**: all off-diagonal entries 0. **Identity** I_n: diagonal with 1s. **Transpose**: $A^T = [a_{ji}]$ (swap rows and columns).

Addition (Thm 1)	$A + B = B + A$, $(A + B) + C = A + (B + C)$, $A + 0 = A$
Multiplication (Thm 2)	$A(BC) = (AB)C$, $A(B + C) = AB + AC$, $(A + B)C = AC + BC$ **but generally** $AB \neq BA$
Transpose (Thm 3)	$(A^T)^T = A$, $(A + B)^T = A^T + B^T$, $(AB)^T = B^T A^T$

Example (1.5 Ex. 7, answer not on slides).

$$\begin{bmatrix} 2 & 3 & -4 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 3 & 1 \\ -2 & 2 \\ 5 & -3 \end{bmatrix} = \begin{bmatrix} -20 & 20 \\ 14 & -4 \end{bmatrix}$$

Counterexample to commutativity: $A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ give $AB = B$ but $BA = 0$.

Inverse (Theorem 1.5.4)

A^{-1} satisfies $AA^{-1} = A^{-1}A = I$. For $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ with $ad - bc \neq 0$: $A^{-1} = \dfrac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$.

1.5.1 Boolean (0/1) matrices

- **Join** $A \vee B$: element-wise OR.
- **Meet** $A \wedge B$: element-wise AND.
- **Boolean product** $A \odot B$: like ordinary multiplication, with AND for $\times$ and OR for $+$: $c_{ij} = (a_{i1} \wedge b_{1j}) \vee \cdots \vee (a_{ik} \wedge b_{kj})$. Needs columns of A = rows of B.

Theorem 1.5.5: $\vee$ and $\wedge$ are commutative and associative; $\odot$ is associative (not commutative).

Example.

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix},\ B = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}: \qquad A \vee B = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix},\ A \wedge B = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix},\ A \odot B = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

Example (1.5 Ex. 12, answers not on slides).

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix},\quad B = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}: \qquad A \vee B = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix},\quad A \wedge B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Example.

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix} \odot \begin{bmatrix} 0 & 1 \\ 1 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, \qquad c_{11} = (1 \wedge 0) \vee (0 \wedge 1) \vee (1 \wedge 1) = 1, \text{ likewise the rest.}$$

Chapter 2
Logic

2.1 Propositions and logical operations

Source: Kolman 2.1

A **proposition** is a declarative sentence that is either true or false. "$2+3=5$" (T) and "7 is even" (F) are propositions; "Close the door" (command), "What time is it?" (question) and "$x+2=5$" (open sentence, x unknown) are not.

p	q	$\neg p$	$p \wedge q$	$p \vee q$	$p \to q$	$p \leftrightarrow q$
T	T	F	T	T	T	T
T	F	F	F	T	F	F
F	T	T	F	T	T	F
F	F	T	F	F	T	T

$p \to q$ is false *only* when a true p gives a false q.

Example (2.1 Ex. 5).

p	q	$p \wedge q$	$\neg p$	$(p \wedge q) \vee \neg p$
T	T	T	F	T
T	F	F	F	F
F	T	F	T	T
F	F	F	T	T

Similarly $\neg(p \vee q) \wedge r$ is true only in the row $p = F$, $q = F$, $r = T$.

2.2 Conditional statements and equivalence

Source: Kolman 2.2

The conditional family (p: it is raining, q: I get wet)

Conditional	$p \to q$	If it rains, I get wet.
Converse	$q \to p$	If I get wet, it is raining.
Inverse	$\neg p \to \neg q$	If it is not raining, I don't get wet.
Contrapositive	$\neg q \to \neg p$	If I don't get wet, it is not raining.
Biconditional	$p \leftrightarrow q$	It rains if and only if I get wet.

Only the **contrapositive** is equivalent to the original: $p \to q \equiv \neg q \to \neg p$. (Converse $\equiv$ inverse, but neither $\equiv$ the original.)

Example (MC practice: contrapositive of $A \neq B \Rightarrow A \cap B = \emptyset$). Recipe: **swap the two sides, negate both**. Here p is "$A \neq B$" and q is "$A \cap B = \emptyset$", so $\neg q$ is "$A \cap B \neq \emptyset$" and $\neg p$ is "$A = B$":

$$A \cap B \neq \emptyset \;\Rightarrow\; A = B.$$

The distractors are the converse ($A \cap B = \emptyset \Rightarrow A \neq B$: swapped, not negated), the inverse ($A = B \Rightarrow A \cap B \neq \emptyset$: negated, not swapped), and $\wedge/\vee$ forms, which are not conditionals at all. (Negating $=$ gives $\neq$ and vice versa; it is the sets' relation that flips, never the $\cap$ or $\cup$.)

Tautology: true in every row ($p \vee \neg p$). **Contradiction**: false in every row ($p \wedge \neg p$). **Contingency**: sometimes true, sometimes false ($p \to q$).

Definition. P and Q are **logically equivalent**, $P \equiv Q$, if $P \leftrightarrow Q$ is a tautology — i.e. their truth-table columns are identical in every row.

2.2.1 Theorem 2.2.1 (fundamental equivalences)

Commutative	$p \vee q \equiv q \vee p$	$p \wedge q \equiv q \wedge p$
Associative	$p \vee (q \vee r) \equiv (p \vee q) \vee r$	$p \wedge (q \wedge r) \equiv (p \wedge q) \wedge r$
Distributive	$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$	$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$
Idempotent	$p \vee p \equiv p$	$p \wedge p \equiv p$
Double negation	$\neg(\neg p) \equiv p$	
De Morgan	$\neg(p \vee q) \equiv \neg p \wedge \neg q$	$\neg(p \wedge q) \equiv \neg p \vee \neg q$

2.2.2 Theorem 2.2.2 (conditionals)

(a) $p \to q \equiv \neg p \vee q$ (b) $p \to q \equiv \neg q \to \neg p$
(c) $p \leftrightarrow q \equiv (p \to q) \wedge (q \to p)$ (d) $\neg(p \to q) \equiv p \wedge \neg q$
(e) $\neg(p \leftrightarrow q) \equiv (p \wedge \neg q) \vee (q \wedge \neg p)$ also $p \leftrightarrow q \equiv (p \wedge q) \vee (\neg p \wedge \neg q)$

Slide erratum. The slide lists (b) as $p \to q \equiv \neg(q \to \neg p)$, which is false ($\neg(q \to \neg p) \equiv p \wedge q$), and omits (a) and (d). The versions above are Kolman's. (d) — the negation of a conditional is $p \wedge \neg q$, *not* $\neg p \to \neg q$ — is a recurring MC trap.

2.2.3 Theorem 2.2.3 (quantifiers)

(a) $\neg(\forall x\, P(x)) \equiv \exists x\, \neg P(x)$ (b) $\neg(\exists x\, P(x)) \equiv \forall x\, \neg P(x)$
(c) $\exists x\, (P(x) \to Q(x)) \equiv \forall x\, P(x) \to \exists x\, Q(x)$ (d) $\exists x\, (P(x) \vee Q(x)) \equiv \exists x\, P(x) \vee \exists x\, Q(x)$
(e) $\forall x\, (P(x) \wedge Q(x)) \equiv \forall x\, P(x) \wedge \forall x\, Q(x)$
(f) $(\forall x\, P(x)) \vee (\forall x\, Q(x)) \to \forall x\, (P(x) \vee Q(x))$ is a tautology (one-way only)
(g) $\exists x\, (P(x) \wedge Q(x)) \to \exists x\, P(x) \wedge \exists x\, Q(x)$ is a tautology (one-way only)
Negating a quantified statement: flip $\forall \leftrightarrow \exists$ and negate the inside.

Slide erratum. The slide omits (a), prints (d) twice, and gives "$(\exists x\, P(x)) \to (\forall x\, Q(x)) \equiv \forall x\, (P(x) \to Q(x))$", which is false: take domain $\{1, 2\}$ with P, Q true only at 1 — the right side is true, the left is $T \to F$. The list above is Kolman's.

2.2.4 Theorem 2.2.4 (tautologies behind the inference rules)

(a) $(p \wedge q) \to p$ (b) $(p \wedge q) \to q$
(c) $p \to (p \vee q)$ (d) $q \to (p \vee q)$
(e) $\neg p \to (p \to q)$ (f) $\neg(p \to q) \to p$
(g) $(p \wedge (p \to q)) \to q$ *modus ponens* (h) $(\neg p \wedge (p \vee q)) \to q$ *disjunctive syllogism*
(i) $(\neg q \wedge (p \to q)) \to \neg p$ *modus tollens* (j) $((p \to q) \wedge (q \to r)) \to (p \to r)$ *hyp. syllogism*

(i) and (j) are from Kolman; the slides list (a)–(h), with $(p \to q) \to (\neg q \to \neg p)$ as their (f) (also a tautology).

Example (verifying (g)).

p	q	$p \to q$	$p \wedge (p \to q)$	$(p \wedge (p \to q)) \to q$
T	T	T	T	T
T	F	F	F	T
F	T	T	F	T
F	F	T	F	T

Last column all T $\Rightarrow$ tautology.

Example (MC practice: which tautology is used in "Mum Kathy is a stone; a stone cannot fly; $\therefore$ Mum Kathy cannot fly"?). **Method:** turn the sentences into symbols, then match the *shape* (premises above the line, conclusion below) to (a)–(j). Let p = "Mum Kathy is a stone" and q = "Mum Kathy cannot fly". The second premise, "a stone cannot fly", is a general rule applied to her: $p \to q$. So the argument is

$$\frac{\begin{array}{c} p \\ p \to q \end{array}}{\therefore q}$$

which is **modus ponens**, tautology (g): $(p \wedge (p \to q)) \to q$. (Look for the option where the premises are joined by $\wedge$ and the conclusion after $\to$ is a bare letter that appeared as the *consequent* of the conditional.) Wrong shapes to rule out: (h) has a negated premise and a $\vee$; (i) concludes a *negation*; (j) chains two conditionals.

2.3 Methods of proof

Source: Kolman 2.3

A proof shows that from premises $p_1, \ldots, p_n$ the conclusion q must follow, i.e. $(p_1 \wedge \cdots \wedge p_n) \to q$ is a tautology. Valid argument forms:

$$\frac{\begin{array}{c} p \\ p \to q \end{array}}{\therefore q}\ \text{(modus ponens)} \qquad \frac{\begin{array}{c} \neg q \\ p \to q \end{array}}{\therefore \neg p}\ \text{(modus tollens)} \qquad \frac{\begin{array}{c} p \to q \\ q \to r \end{array}}{\therefore p \to r}\ \text{(hypothetical syllogism)}$$

Three proof methods

Direct	To show $p \to q$: assume p, derive q.
Contrapositive	Prove $\neg q \to \neg p$ instead (equivalent).
Contradiction	To show p: assume $\neg p$, derive something of the form $r \wedge \neg r$; so p must hold.

Example (n^2 even $\iff$ n even). ($\Rightarrow$ direction by contrapositive) n odd $\Rightarrow n = 2k+1 \Rightarrow n^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1$ odd. ($\Leftarrow$ direct) $n = 2k \Rightarrow n^2 = 4k^2 = 2(2k^2)$ even.

Example (direct: odd + odd = even). $a = 2m + 1$, $b = 2n + 1 \Rightarrow a + b = 2(m + n + 1)$, even.

Example (contradiction: $\sqrt{2}$ is irrational). Suppose $\sqrt{2} = p/q$ with $\gcd(p, q) = 1$. Then $2q^2 = p^2$, so p^2 is even, so p is even: $p = 2k$. Then $2q^2 = 4k^2$, $q^2 = 2k^2$, so q is even too — contradicting $\gcd(p, q) = 1$.

2.4 Mathematical induction

Source: Kolman 2.4

Template (same shape as a recursive function: base case + recursive step)

1. **Base case**: show $P(n_0)$ (usually $n_0 = 1$ or 0).
2. **Induction hypothesis**: assume $P(k)$ for some $k \geq n_0$.
3. **Induction step**: using the hypothesis, prove $P(k+1)$.
4. **Conclude**: by induction, $P(n)$ holds for all $n \geq n_0$.

Strong induction: in step 2 assume *all* of $P(n_0), \ldots, P(k)$. Needed when $P(k+1)$ depends on several earlier cases — and then you need as many base cases as the recurrence looks back.

Example (2.4 Ex. 1: $1+2+\cdots+n = \frac{n(n+1)}{2}$). Base: $1 = \frac{1\cdot 2}{2}$. Step: $1+\cdots+k+(k+1) = \frac{k(k+1)}{2}+(k+1) = \frac{k(k+1)+2(k+1)}{2} = \frac{(k+1)(k+2)}{2}$.

Example (2.4 Ex. 2: De Morgan for n sets, $\overline{\bigcup_{i=1}^{n} A_i} = \bigcap_{i=1}^{n} \overline{A_i}$). Base $n = 1$ trivial. Step: $\overline{\bigcup_{i=1}^{k+1} A_i} = \overline{\left(\bigcup_{i=1}^{k} A_i\right) \cup A_{k+1}} = \overline{\bigcup_{i=1}^{k} A_i} \cap \overline{A_{k+1}} = \left(\bigcap_{i=1}^{k} \overline{A_i}\right) \cap \overline{A_{k+1}} = \bigcap_{i=1}^{k+1} \overline{A_i}$ (two-set De Morgan, then the hypothesis).

Example (2.4 Ex. 4: $n! \geq 2^{n-1}$). Base: $1! = 1 = 2^0$. Step: $(k+1)! = (k+1)k! \geq (k+1)2^{k-1} \geq 2 \cdot 2^{k-1} = 2^k$ (since $k+1 \geq 2$).

Example (strong induction). $a_1 = 2$, $a_2 = 4$, $a_3 = 8$, $a_{n+3} = a_{n+2} + a_{n+1} + a_n$. Claim $a_n \leq 2^n$. Base: $P(1), P(2), P(3)$ (three, because the recurrence looks back three). Step ($k \geq 3$): $a_{k+1} = a_k + a_{k-1} + a_{k-2} \leq 2^k + 2^{k-1} + 2^{k-2} = 7 \cdot 2^{k-2} < 8 \cdot 2^{k-2} = 2^{k+1}$.

Slide erratum. The conclusion slide says $a_n < 2^n$; the claim is $a_n \leq 2^n$ (equality holds at $n = 1, 2, 3$).

2.5 From pattern to conjecture to proof

Source: Kolman 2.5

A **conjecture** is a statement believed true from a pattern but not yet proved: examples → pattern → conjecture → proof.

Example (2.5 Ex. 2: $\sum_{m=0}^{n} 2^m = 2^{n+1} - 1$). Pattern: $1, 3, 7, 15, \ldots$ Base $n = 0$: $1 = 2^1 - 1$. Step: $S_{k+1} = S_k + 2^{k+1} = (2^{k+1} - 1) + 2^{k+1} = 2^{k+2} - 1$.

Example (divisibility: $16 \mid (19 \cdot 9^n - 8n - 3)$, $n \geq 0$). Base: $19 - 3 = 16$. Step: $19 \cdot 9^{k+1} - 8(k+1) - 3 = 9(19 \cdot 9^k - 8k - 3) + 64k + 16 = 9 \cdot (\text{multiple of } 16) + 16(4k+1)$.

Example (practice: $1 + 3 + \cdots + (2n-1) = n^2$). Base: $1 = 1^2$. Step: $k^2 + (2k+1) = (k+1)^2$.

Chapter 3
Counting

3.1 Permutations and combinations

Source: Kolman 3.1–3.2; own notes

3.1.1 Multiplication principle

If a task is done in steps, with n_1 ways to do the first step, n_2 ways for the second, and so on, independently of each other, the number of ways to do the whole task is $n_1 \cdot n_2 \cdots n_k$. It is nested loops: the inner body runs $n_1 \cdot n_2 \cdots n_k$ times. "This *and* that" → multiply; "this *or* that" (one or the other, not both) → add.

Example (MC practice: license plates, 2 letters from A–J (repeats allowed) then 3 distinct digits). Fill the five slots left to right and count the choices per slot: letters $10 \cdot 10$ (repeats allowed, so the pool never shrinks); digits $10 \cdot 9 \cdot 8$ (no repeats, so each pick removes one digit and order matters, i.e. $P(10,3)$). Total $10^2 \cdot 720 = \mathbf{72\,000}$. The letter part and the digit part are independent, hence multiplied.

3.1.2 Factorial

$n! = n \cdot (n-1) \cdots 2 \cdot 1$, with $0! = 1$. E.g. $4! = 4 \cdot 3 \cdot 2 \cdot 1 = 24$: the number of ways to put n things in a row.

3.1.3 Working with factorials without a calculator (exam method)

Key idea: $n! = n \cdot (n-1)!$ — the recursive definition, `fact(n) = n * fact(n-1)`. You may stop unrolling anywhere and leave the rest as a factorial; all of these are the same number:

$$11! = 11 \cdot 10! = 11 \cdot 10 \cdot 9! = 11 \cdot 10 \cdot 9 \cdot 8 \cdot 7!$$

1. **Unroll to the right place:** down to the factorial the MC options end with (e.g. 7!), or to the biggest factorial in the denominator.
2. **Cancel** the denominator against the unrolled numbers.
3. **No direct match?** Split the denominator into small factors (primes) and cancel them piece by piece.
4. **Compare options** by rewriting them all with the same tail factorial — then only the numbers in front matter; you never need the value of 7!.

Example (Direct match: $\frac{11!}{8}$). $\dfrac{11 \cdot 10 \cdot 9 \cdot 8 \cdot 7!}{8} = 11 \cdot 10 \cdot 9 \cdot 7!$

Example (Split into factors: $\frac{10!}{3!\cdot 2!} = \frac{10!}{12}$, options ending in 7!). 12 is not among $10, 9, 8$, but $12 = 3 \cdot 4$, with $9/3 = 3$ and $8/4 = 2$:

$$\frac{10 \cdot 9 \cdot 8 \cdot 7!}{3 \cdot 4} = 10 \cdot 3 \cdot 2 \cdot 7! = 60 \cdot 7!.$$

Example (Comparing options for $\frac{11!}{8}$ — all in terms of 7!). Target: $\frac{11\cdot 10\cdot 9\cdot 8}{8} \cdot 7! = 990 \cdot 7!$. $8! = 8 \cdot 7!$; $3 \cdot 8! = 24 \cdot 7!$; $3! \cdot 8! = 48 \cdot 7!$; $11! = 7\,920 \cdot 7!$; $11 \cdot 10 \cdot 9 \cdot 7! = 990 \cdot 7!$ ✓

The same trick inside $C(n,k)$: unroll $n!$ down to the bigger factorial in the denominator and it cancels completely — what remains is the "count down k numbers, divide by $k!$" shortcut:

$$C(12,4) = \frac{12!}{4! \cdot 8!} = \frac{12 \cdot 11 \cdot 10 \cdot 9 \cdot 8!}{4! \cdot 8!} = \frac{12 \cdot 11 \cdot 10 \cdot 9}{24} = 495.$$

3.1.4 Order matters vs. order does not matter

Choosing k things out of n (each thing at most once):

	Formula	Quick by hand
Order matters (permutation)	$P(n,k) = \dfrac{n!}{(n-k)!}$	multiply k numbers counting down from n
Order does not matter (combination)	$C(n,k) = \binom{n}{k} = \dfrac{n!}{k!\,(n-k)!}$	the same, then divide by $k!$

$C(n,k)$ is read "n choose k". **Why divide by $k!$:** $P(n,k)$ counts every ordering of the same k things separately, and k things can be ordered in $k!$ ways — so each unordered group was counted $k!$ times.

Shortcut: $C(n,k) = C(n,n-k)$ — choosing which k to take is the same as choosing which $n-k$ to leave. So $C(6,5) = C(6,1) = 6$. Also $C(n,0) = C(n,n) = 1$ and $C(n,1) = n$.

Example (Bag with 7 red, 6 blue, 8 black; draw 3 red, 5 blue and 2 black, order does not matter). Three independent choices, one per colour — "and", so multiply. Order does not matter, so each choice is a combination:

$$\begin{aligned}
\text{red: } C(7,3) &= \frac{7\cdot 6\cdot 5}{3\cdot 2\cdot 1} = \frac{210}{6} = 35\\
\text{blue: } C(6,5) &= C(6,1) = 6\\
\text{black: } C(8,2) &= \frac{8\cdot 7}{2\cdot 1} = \frac{56}{2} = 28
\end{aligned}$$

$$35\cdot 6\cdot 28 = 210\cdot 28 = 5\,880 \text{ ways.}$$

3.1.5 Pascal's triangle (look-up table)

Row n, column k is $C(n,k)$. Only $k \le 6$ is shown: for larger k use the symmetry $C(n,k) = C(n,n-k)$, so $C(12,8) = C(12,4) = 495$.

$n\backslash k$	0	1	2	3	4	5	6
0	1						
1	1	1					
2	1	2	1				
3	1	3	3	1			
4	1	4	6	4	1		
5	1	5	10	10	5	1	
6	1	6	15	20	15	6	1
7	1	7	21	35	35	21	7
8	1	8	28	56	70	56	28
9	1	9	36	84	126	126	84
10	1	10	45	120	210	252	210
11	1	11	55	165	330	462	462
12	1	12	66	220	495	792	924

Pascal's rule: $C(n,k) = C(n-1,k-1) + C(n-1,k)$ — each entry is the sum of the two above it (e.g. $C(7,3) = 15+20 = 35$); this is how a row is built from the one before. **Row sums:** $\sum_k C(n,k) = 2^n$ (every subset of an n-set has some size k, so this is the 2^n subsets count). **Symmetry:** each row reads the same backwards. **Beyond the table:** cancel factorials, $C(n,k) = \frac{n(n-1)\cdots(n-k+1)}{k!}$.

3.1.6 Arranging letters with repeats

A word of n letters, where the repeated letters occur $n_1, n_2, \ldots, n_k$ times, has

$$\frac{n!}{n_1!\, n_2! \cdots n_k!}$$

distinguishable arrangements (Kolman 3.1). **Why:** $n!$ treats the two M's as different (M_1, M_2); swapping them gives the same visible word, so each word was counted 2! times — divide it away, for each repeated letter. Letters that occur once give $1! = 1$ and can be ignored.

Example (MATHEMATICS). 11 letters; M, A and T occur twice each (H, E, I, C, S once):

$$\frac{11!}{2! \cdot 2! \cdot 2!} = \frac{11!}{8} = \frac{11 \cdot 10 \cdot 9 \cdot 8 \cdot 7!}{8} = 11 \cdot 10 \cdot 9 \cdot 7! = 4\,989\,600.$$

Matching expanded-factorial options: see "Working with factorials without a calculator" (3.1.3).

3.1.7 Repetition allowed: stars and bars

Choosing k things from n *kinds*, order does not matter, and the same kind may be chosen again:

$$C(n + k - 1,\ k).$$

Why: draw the k chosen things as stars $\star$ and separate the n kinds with $n-1$ bars $|$. Every choice is one row of k stars and $n-1$ bars — $n+k-1$ symbols — and a row is fixed by which k positions hold the stars.

Typical wording: *identical* objects distributed over kinds/colours/boxes, or "how many of each". "Identical" is the key word: if the objects were different (chair 1, chair 2, ...), each would pick a kind independently, giving n^k instead.

Example (8 identical chairs, 5 colours; how many ways to choose the colours?). Only how many chairs get each colour matters: $k = 8$ chairs, $n = 5$ colours, repetition allowed, order irrelevant. One choice, e.g. 3 red, 0 green, 2 blue, 3 black, 0 orange:

★★★ | | ★★ | ★★★ |

8 stars and 4 bars, 12 positions:

$$C(5 + 8 - 1,\ 8) = C(12, 8) = C(12, 4) = \frac{12 \cdot 11 \cdot 10 \cdot 9}{4 \cdot 3 \cdot 2 \cdot 1} = \frac{11\,880}{24} = 495.$$

Wrong options: 5^8 (chairs treated as different), ${}_8C_5$ (no repetition, numbers swapped).

Notation: Moodle and some books write ${}_nC_k$ and ${}_nP_k$ for $C(n,k)$ and $P(n,k)$ — the same thing.

3.1.8 Exactly, at least, at most

When the question puts a condition on a *special* group (aces, red balls, women on a committee), split everything into the special group and the rest, then count each part:

Wording	Method
exactly j special	$C(\text{special}, j) \cdot C(\text{rest}, k-j)$
at least one special	$\text{all} - \text{none} = C(\text{all}, k) - C(\text{rest}, k)$
at most j special	exactly 0 + exactly 1 + $\cdots$ + exactly j

Taking the other cards from "the rest" is what makes it *exactly*: no further special one can slip in.

Example (5-card hand from 52 cards with exactly one ace). Special: the 4 aces; rest: $52 - 4 = 48$ non-aces. Choose 1 ace *and* 4 non-aces — multiply:

$$C(4,1) \cdot C(48,4) = 4 \cdot {}_{48}C_4 = 4 \cdot 194\,580 = 778\,320.$$

Traps: $48 \cdot {}_{48}C_4$ (only 4 aces to choose from); $52 \cdot {}_{52}C_4$ (lets more aces in, and counts hands several times). For comparison, *at least* one ace would be $C(52,5) - C(48,5)$.

3.1.9 When the MC options are products of primes

Options like $7^2 \cdot 5 \cdot 3 \cdot 2^3$ are the answer in prime factorized form. Do *not* multiply out — factor each piece into primes and collect equal primes. Multiplying powers of the same base **adds the exponents**: $p^a \cdot p^b = p^{a+b}$.

Example (the bag example above).

$$C(7,3) = 35 = 5 \cdot 7$$
$$C(6,5) = 6 = 2 \cdot 3$$
$$C(8,2) = 28 = 2^2 \cdot 7$$

$$(5 \cdot 7)(2 \cdot 3)(2^2 \cdot 7) = 7^{1+1} \cdot 5 \cdot 3 \cdot 2^{1+2} = 7^2 \cdot 5 \cdot 3 \cdot 2^3.$$

Check: $49 \cdot 5 \cdot 3 \cdot 8 = 5\,880$ ✓. Compare the options prime by prime (exponent of 7, of 5, of 3, of 2); exactly one matches.

This form is easier without a calculator: the numbers stay small. Tip: cancel in prime form from the start, e.g. $C(8,2) = \frac{8 \cdot 7}{2} = \frac{2^3 \cdot 7}{2} = 2^2 \cdot 7$.

3.1.10 Which formula? The four cases

Choosing k from n always lands in one of four boxes. Ask: *does order matter?* and *can the same thing be picked again?*

	no repetition	repetition **allowed**
order matters	$P(n,k)$	n^k
order does not matter	$C(n,k)$	$C(n+k-1,\ k)$

Recognising the problem type:

- "order does not matter", "choose", "a group/committee/hand of" → $C(n,k)$
- "order matters", "arrange", "first/second/third place", "line up" → $P(n,k)$
- *different* objects each getting one of n options (each chair picks a colour, each digit of a PIN) → n^k
- *identical* objects split over n kinds, "how many of each" → $C(n+k-1,k)$
- several independent choices made together ("and") → multiply the counts

3.1.11 Worked slide examples

3.1 Theorem 2. Tasks $T_1, \ldots, T_k$ in sequence, T_i done in n_i ways: $n_1 n_2 \cdots n_k$ ways in all. **Theorem 3.** Sequences of length r from n objects, repeats allowed: n^r. Special case: each element of an n-set is in or out, so an n-set has $2 \cdot 2 \cdots 2 = \mathbf{2^n}$ subsets.

- **Password**, 8 characters, first a letter, the other seven letter or digit, repeats allowed: $26 \cdot 36^7 = 2\,037\,468\,266\,496$.
- **Three-digit codes**, repeats allowed: $10^3 = 1000$.
- **GILBREATH**, 4-letter arrangements, no repeats (9 distinct letters): ${}_9P_4 = 9 \cdot 8 \cdot 7 \cdot 6 = 3024$.

- **Prizes**: 10 students, 1st/2nd/3rd prize: ${}_{10}P_3 = 10 \cdot 9 \cdot 8 = 720$. **Arranging all** n: ${}_nP_n = n!$ (5 books: 120).
- **Distinguishable permutations** (Theorem 5, $n!/(k_1! \cdots k_t!)$): BANANA $= 6!/(3!\,2!) = 60$; MISSISSIPPI (M1, I4, S4, P2) $= 11!/(1!\,4!\,4!\,2!) = 34\,650$.
- **Hands**: $\binom{52}{5} = 2\,598\,960$. **Groups**: $\binom{12}{4} = 495$. Symmetry $\binom{n}{r} = \binom{n}{n-r}$, so $\binom{10}{3} = \binom{10}{7} = 120$ (choosing who is in = choosing who is left out).
- **Committee**, 3 women from 20 *and* 4 men from 30 (multiply): $\binom{20}{3}\binom{30}{4} = 1140 \cdot 27\,405 = 31\,241\,700$.
- **Ice cream**, 3 scoops from 5 flavours, repeats allowed, order irrelevant: $\binom{5+3-1}{3} = \binom{7}{3} = 35$.

Slide erratum. The committee slide prints 31,242,600; the product is $1140 \cdot 27\,405 = 31\,241\,700$.

3.3 Pigeonhole principle

Source: Kolman 3.3

3.3 Theorem 1: the pigeonhole principle

If $n+1$ or more objects (pigeons) are put into n boxes (pigeonholes), some box holds at least 2 objects.
Generalized: N objects in k boxes $\Rightarrow$ some box holds at least $\lceil N/k \rceil$. To force m objects into one box you need $(m-1)k+1$ objects.

Recognising it: (1) what are the objects, (2) what are the categories (the boxes), (3) are there more objects than boxes? If so, a crowded box is guaranteed. It is a *worst-case* argument: the answer to "how many must I pick to be sure" is "everything that can go wrong, plus one".

- **Socks**, black and white: 2 boxes, so 3 socks guarantee a matching pair.
- **Birthdays**: 366 students, 365 days $\Rightarrow$ two share a birthday.
- **Tutorial groups**: 30 students in 7 groups $\Rightarrow$ some group has $\geq \lceil 30/7 \rceil = 5$.
- **Bicycles**: 50 bicycles in 7 colours $\Rightarrow$ some colour has $\geq \lceil 50/7 \rceil = 8$.
- **Three in one month**: worst case 2 per month $= 2 \cdot 12 = 24$ people without three, so **25** people guarantee it.

3.4 Probability

Source: Kolman 3.4

Terms. Random experiment (e.g. rolling a die); outcome (a 4); **sample space** S = all outcomes ($\{1, \ldots, 6\}$); **event** $E \subseteq S$ (even: $\{2,4,6\}$); **disjoint** events share no outcomes.

Each outcome i has a probability p_i with $0 \leq p_i \leq 1$ and $\sum p_i = 1$ (something always happens), and $P(E)$ is the sum of the p_i over the outcomes in E. With equally likely outcomes this becomes $|E|/|S|$ ("favourable over possible"); with an *unfair* die add the given probabilities instead.

Probability rules

$0 \leq P(E) \leq 1$, $\quad P(S) = 1$, $\quad P(\emptyset) = 0$; $\quad$ mutually exclusive E_i: $P(E_1 \cup \cdots \cup E_k) = \sum P(E_i)$.
Equally likely outcomes: $P(E) = \dfrac{|E|}{|S|}$. $\quad$ Complement: $P(E) = 1 - P(\overline{E})$.
Overlap: $P(E \cup F) = P(E) + P(F) - P(E \cap F)$. $\quad$ Expected value: $E(X) = \sum_i x_i\, P(X = x_i)$.

Example (3.4 Ex. 8, unequal weights). $P(\{1\}), \ldots, P(\{6\}) = \frac{1}{12}, \frac{1}{12}, \frac{1}{3}, \frac{1}{6}, \frac{1}{4}, \frac{1}{12}$. $P(\text{even}) = \frac{1}{12} + \frac{1}{6} + \frac{1}{12} = \frac{1}{3}$.

Example (two dice, sum ≤ 3). $|S| = 36$, $E = \{(1,1),(1,2),(2,1)\}$, $P = \frac{3}{36} = \frac{1}{12}$.

Example (3.4 Ex. 9, four kings in a 4-card hand). $\binom{52}{4} = 270725$ hands, one of them all kings: $P = \frac{1}{270725}$.

Example (3.4 Ex. 10, 6 red + 4 green, draw 4). Exactly 2 red: $\frac{\binom{6}{2}\binom{4}{2}}{\binom{10}{4}} = \frac{90}{210} = \frac{3}{7}$. At most 2 red: $\frac{\binom{6}{0}\binom{4}{4}+\binom{6}{1}\binom{4}{3}+\binom{6}{2}\binom{4}{2}}{210} = \frac{1+24+90}{210} = \frac{23}{42}$. At most 3 red: $1 - \frac{\binom{6}{4}\binom{4}{0}}{210} = 1 - \frac{15}{210} = \frac{13}{14}$.
Example (3.4 Ex. 11, die thrown 3 times: all equal OR no 4). $|F| = 6$ (all equal), $|G| = 5^3 = 125$ (no 4), $|F \cap G| = 5$. $|F \cup G| = 6 + 125 - 5 = 126$, $P = \frac{126}{216} = \frac{7}{12}$.
Example (sum is 8 OR doubles, two dice). $|E| = 5$, $|F| = 6$, $E \cap F = \{(4,4)\}$: $P = \frac{5+6-1}{36} = \frac{5}{18}$.
Example (fair die). $E(X) = \frac{1+2+3+4+5+6}{6} = 3.5$.

3.4.1 Adding fractions (exam method)

Only fractions with the same denominator can be added: rewrite each over a **common denominator** (the lcm of the denominators), multiplying top and bottom by the same number, then add the numerators. Finally simplify by dividing top and bottom by their gcd.
Example (Unfair die: $p_1 = \frac{1}{6}$, $p_2 = \frac{3}{8}$, $p_3 = \frac{1}{12}$, $p_4 = \frac{1}{24}$, $p_5 = \frac{1}{12}$, $p_6 = \frac{1}{4}$; probability of an even number). Event $E = \{2,4,6\}$, so $P(E) = p_2 + p_4 + p_6 = \frac{3}{8} + \frac{1}{24} + \frac{1}{4}$. Common denominator $\operatorname{lcm}(8,24,4) = 24$:

$$\frac{3\cdot 3}{8\cdot 3} + \frac{1}{24} + \frac{1\cdot 6}{4\cdot 6} = \frac{9}{24} + \frac{1}{24} + \frac{6}{24} = \frac{16}{24} = \frac{16\div 8}{24\div 8} = \frac{2}{3}.$$

Sanity check — all six must sum to 1: $\frac{4+9+2+1+2+6}{24} = \frac{24}{24} = 1$ ✓. So also $P(\text{odd}) = 1 - \frac{2}{3} = \frac{1}{3}$.

3.5 Recurrence relations

Source: Kolman 3.5

A recurrence defines a_n from earlier terms; with **initial conditions** it fixes the sequence.
Backtracking (repeated substitution — unrolling the recursion):

- $a_n = a_{n-1} + 3$, $a_1 = 4$: $a_n = a_{n-2} + 2\cdot 3 = \cdots = a_1 + 3(n-1) = 4 + 3(n-1)$.
- $b_n = 2b_{n-1} + 1$, $b_1 = 7$: $b_n = 2^{n-1}b_1 + 2^{n-2} + \cdots + 2 + 1 = 7\cdot 2^{n-1} + 2^{n-1} - 1 = 4\cdot 2^n - 1$.

Linear homogeneous of degree k: $a_n = r_1 a_{n-1} + r_2 a_{n-2} + \cdots + r_k a_{n-k}$, constants r_i. The degree is how far back it reaches: $c_n = -2c_{n-1}$ (1), $a_n = 3a_{n-1} - 2a_{n-2}$ (2), $b_n = 2b_{n-1} + b_{n-3}$ (3).

Theorem 3.5.1 (degree 2)

$a_n = r_1 a_{n-1} + r_2 a_{n-2}$ has characteristic equation $x^2 - r_1 x - r_2 = 0$.
(a) Two distinct roots $s_1 \neq s_2$: $a_n = u\, s_1^n + v\, s_2^n$.
(b) One (repeated) root s: $a_n = u\, s^n + v\, n\, s^n$. **Not on the slides** (Kolman 3.5 Thm 1b).
Then solve for u, v from the initial conditions, e.g. $a_1 = A$, $a_2 = B \Rightarrow us_1 + vs_2 = A$, $us_1^2 + vs_2^2 = B$.
Roots give the form; initial conditions pick the specific sequence.

Why it works. If $s^2 = r_1 s + r_2$, multiply by s^{n-2}: $s^n = r_1 s^{n-1} + r_2 s^{n-2}$, so $a_n = s^n$ satisfies the recurrence. Because the recurrence is linear, any combination $us_1^n + vs_2^n$ does too.
Not on the slides. The roots can be complex (negative discriminant); the same formula (a) still applies with complex s_1, s_2 and u, v. Quadratic formula: $x = \dfrac{r_1 \pm \sqrt{r_1^2 + 4r_2}}{2}$.
Example (3.5 Ex. 7: $c_n = 3c_{n-1} - 2c_{n-2}$, $c_1 = 5$, $c_2 = 3$). $x^2 - 3x + 2 = (x-1)(x-2) = 0$, so $c_n = u + v2^n$. Then $u + 2v = 5$, $u + 4v = 3 \Rightarrow v = -1$, $u = 7$: $c_n = 7 - 2^n$.
Example (3.5 Ex. 9: Fibonacci $f_n = f_{n-1} + f_{n-2}$, $f_1 = f_2 = 1$). $x^2 - x - 1 = 0$, $s_{1,2} = \frac{1\pm\sqrt{5}}{2}$, $u = \frac{1}{\sqrt{5}}$, $v = -\frac{1}{\sqrt{5}}$: $f_n = \frac{1}{\sqrt{5}}\left(\frac{1+\sqrt{5}}{2}\right)^n - \frac{1}{\sqrt{5}}\left(\frac{1-\sqrt{5}}{2}\right)^n$.
Example ($a_n = 5a_{n-1} - 6a_{n-2}$, $a_1 = 5$, $a_2 = 13$). $(x-2)(x-3) = 0$; $2u + 3v = 5$, $4u + 9v = 13$ $\Rightarrow u = v = 1$: $a_n = 2^n + 3^n$.

MC shortcut: plug $n = 1, 2$ into each candidate formula and compare with the given initial terms — quicker than re-deriving, and catches n vs. $n-1$ offset distractors.

3.5.1 Exponent rule

Dividing powers of the same base: **subtract the exponents**.

$$\frac{r^n}{r^{n-2}} = r^{n-(n-2)} = r^2, \qquad \frac{r^{n-1}}{r^{n-2}} = r^{(n-1)-(n-2)} = r^1 = r, \qquad \frac{r^{n-2}}{r^{n-2}} = r^0 = 1.$$

Why this matters: guess a solution of the form $a_n = r^n$ and substitute it: $r^n = c_1 r^{n-1} + c_2 r^{n-2}$. Dividing every term by r^{n-2} (the rule above) gives $r^2 = c_1 r + c_2$ — the *characteristic equation* of step 1.

3.5.2 The four-step recipe for distinct roots (exam method)

1. **Characteristic equation.** Replace a_n by r^2, a_{n-1} by r and a_{n-2} by 1: $r^2 = c_1 r + c_2$, i.e. $r^2 - c_1 r - c_2 = 0$.
2. **Solve the quadratic** to get the roots r_1 and r_2.
3. **General solution:** $a_n = A\, r_1^n + B\, r_2^n$ (when $r_1 \neq r_2$).
4. **Fit the starting values.** Put $n = 1$ and $n = 2$ into step 3 to get two equations in A and B, and solve them.

Example ($a_n = 7a_{n-1} - 12a_{n-2}$, $a_1 = 1$, $a_2 = 7$). **Step 1.** $r^2 = 7r - 12$, so $r^2 - 7r + 12 = 0$.

Step 2. Two numbers that multiply to 12 and add to -7: -3 and -4. So $(r-3)(r-4) = 0$, giving $r_1 = 3$, $r_2 = 4$.

Step 3. $a_n = A \cdot 3^n + B \cdot 4^n$.

Step 4.

$$\begin{aligned} n = 1: &\quad 3A + 4B = 1 \\ n = 2: &\quad 9A + 16B = 7 \end{aligned}$$

Multiply the first equation by 3: $9A + 12B = 3$. Subtract it from the second: $4B = 4$, so $B = 1$. Then $3A + 4 = 1$, so $3A = -3$ and $A = -1$.

$$a_n = 4^n - 3^n$$

Check: $a_1 = 4 - 3 = 1$ ✓, $a_2 = 16 - 9 = 7$ ✓, $a_3 = 7 \cdot 7 - 12 \cdot 1 = 37 = 64 - 27$ ✓

Example (Repeated root: $a_n = 6a_{n-1} - 9a_{n-2}$, $a_1 = -3$, $a_2 = 9$). **Steps 1–2.** $r^2 - 6r + 9 = (r-3)^2 = 0$: one root, $r = 3$ (a perfect square — spot it when the discriminant $c_1^2 + 4c_2$ is 0).

Step 3 changes. With one root the form is $a_n = (A + Bn)\, 3^n$ — the extra n replaces the missing second root (Theorem 3.5.1b).

Step 4.

$$\begin{aligned} n = 1: &\quad 3(A + B) = -3 \Rightarrow A + B = -1 \\ n = 2: &\quad 9(A + 2B) = 9 \Rightarrow A + 2B = 1 \end{aligned}$$

Subtract: $B = 2$, then $A = -3$. So $a_n = (-3 + 2n)\, 3^n = -3 \cdot 3^n + 2n \cdot 3^n$.

Check: $a_3 = 6 \cdot 9 - 9 \cdot (-3) = 81$ and $(-3 + 6) \cdot 27 = 81$ ✓. MC trap: the options with $(-3)^n$ mix up the root 3 with the coefficient -3.

3.5.3 Linear and homogeneous — recognising the type

- **Linear**: each a term appears on its own, only multiplied by a constant — no a_{n-1}^2, no $a_{n-1} \cdot a_{n-2}$.
- **Homogeneous**: every term on the right contains an a — no lone constant or n, as in $a_n = 3a_{n-1} - a_{n-2} + 1$.
- **Constant coefficients**: the numbers in front of the a's are fixed numbers — *no n in them.* $a_n = (3 + n)\, a_{n-1} - a_{n-2}$, $a_n = 3a_{n-1} - n\, a_{n-2}$ and $a_n = 3n!\, a_{n-1}$ all fail this: the swap to x would leave an n in the equation, so there is no characteristic equation.
- **Degree**: how far back it reaches; a_{n-2} as the furthest back means degree 2 (a quadratic characteristic equation), a_{n-3} means degree 3.

3.5.4 Working backwards: from characteristic equation to recurrence

Undo step 1: get x^2 (or r^2) alone on the left — moving terms across $=$ **flips their sign** — then swap $x^2 \to a_n$, $x \to a_{n-1}$ and the plain number $\to a_{n-2}$. The letter (x or r) does not matter.

Example (Which recurrence has characteristic equation $x^2 - 3x + 1 = 0$?).

$$x^2 - 3x + 1 = 0 \quad \Longrightarrow \quad x^2 = 3x - 1 \quad \Longrightarrow \quad a_n = 3a_{n-1} - a_{n-2}.$$

Check forwards: $a_n \to x^2$, $a_{n-1} \to x$, $a_{n-2} \to 1$ gives $x^2 = 3x - 1$, i.e. $x^2 - 3x + 1 = 0$ ✓

Method for "which recurrence has this characteristic equation?": (1) rule out the wrong types first — n in a coefficient, a lone constant, a square or product; (2) swap the remaining options to x and compare.

Traps in the MC options:

- **Starting values are irrelevant** — the characteristic equation comes only from the $a_n = \dots$ part.
- **Shifted indices are the same relation**: $a_n = 3a_{n-1} - a_{n-2}$, $a_{n+1} = 3a_n - a_{n-1}$ and $a_{n+2} = 3a_{n+1} - a_n$ all give $x^2 - 3x + 1 = 0$. Read them as highest index $\to x^2$, next $\to x$, lowest $\to 1$.
- **Sign not flipped**: $a_n = 3a_{n-1} + a_{n-2}$ gives $x^2 - 3x - 1 = 0$ — wrong.
- **Coefficients swapped**: $a_n = a_{n-1} - 3a_{n-2}$ gives $x^2 = x - 3$, i.e. $x^2 - x + 3 = 0$ — wrong.
- n **in a coefficient**, e.g. $(3+n)\,a_{n-1}$ — looks like the right numbers, but the coefficients are not constant.
- A lone constant ($+1$) makes it not homogeneous; a square or product of a's makes it not linear; a_{n-3} makes it degree 3.

Checking a formula by computer: `discrete-math/code/recurrence.py` computes the terms straight from the recurrence and compares them with a formula, e.g.

```
python3 recurrence.py 7 -12 1 7 -formula "4**n - 3**n"
```

3.5.5 Sigma notation and the geometric series

Σ (capital sigma) means "add up". The numbers below and above it give where the counter starts and stops; the expression to the right is what gets added for each value:

$$\sum_{i=1}^{4} 3^{i-1} = 3^0 + 3^1 + 3^2 + 3^3 = 1 + 3 + 9 + 27 = 40.$$

It is a for-loop: `for i in range(1, 5):  total += 3**(i-1)`. Remember $r^0 = 1$, so the first term is 1.

Geometric series.

A sum where each term is the previous one times a fixed ratio r is a **geometric series**. For $r \neq 1$:

$$1 + r + r^2 + \cdots + r^{n-1} = \sum_{i=1}^{n} r^{i-1} = \frac{r^n - 1}{r - 1}.$$

Note the count: n terms, and the last exponent is $n - 1$.

Why — multiply by r and subtract. Multiplying shifts every term one place up; subtracting cancels everything in the middle:

$$\begin{aligned} S &= 1 + r + r^2 + \cdots + r^{n-1} \\ rS &= \quad r + r^2 + \cdots + r^{n-1} + r^n \\ rS - S &= r^n - 1 \quad \Longrightarrow \quad S = \frac{r^n - 1}{r - 1}. \end{aligned}$$

Example ($\sum_{i=1}^{20} 3^{i-1}$). Here $r = 3$, $n = 20$ terms ($i = 1, \ldots, 20$):

$$S = 1 + 3 + 9 + \cdots + 3^{19} = \frac{3^{20} - 1}{3 - 1} = \frac{3^{20} - 1}{2} = \frac{3\,486\,784\,401 - 1}{2} = 1\,743\,392\,200.$$

Small check: $1 + 3 + 9 = 13$ and $\frac{3^3 - 1}{2} = \frac{26}{2} = 13$ ✓
On the MC exam (no calculator) expect the answer in the form $\frac{3^{20} - 1}{2}$.

Link to recurrences: the terms $1, 3, 9, 27, \ldots$ satisfy $a_n = 3\,a_{n-1}$ — each term is the previous one times a fixed number, which is where the r^n in the recurrence recipe (3.5.2) comes from.

Chapter 4
Relations and Digraphs

4.1 Relations and digraphs

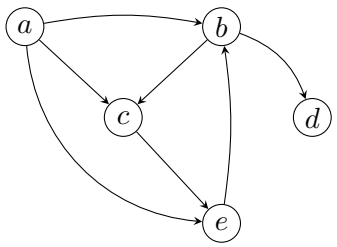
Source: Kolman 4.1–4.3; own notes

A **relation** R on a set A is a set of ordered pairs (x, y); $x\ R\ y$ means $(x, y) \in R$, "x is related to y". In the **digraph** (directed graph) each pair is an arrow $x \to y$. **First step for any digraph question: write out the list of pairs** — then every option is a lookup. (Think of R as a table of `(from, to)` rows.)

Notation	Meaning	Read from the digraph
$x\ R\ y$	x is related to y	arrow $x \to y$
$R(x)$	everything x is related to	where x's arrows point
$\mathrm{Dom}(R)$	domain	elements with an arrow **out**
$\mathrm{Ran}(R)$	range	elements with an arrow **in**
$x\ R^n\ y$	path of length n	walk $x \to \cdots \to y$ in exactly n arrows

$\mathrm{Dom}(R)$ is `SELECT DISTINCT from`, $\mathrm{Ran}(R)$ is `SELECT DISTINCT to`. For R^n one path of exactly n steps is enough; a shorter path does not count.

Example (Which statement is false for this digraph?).



$R = \{(a,b),(a,c),(a,e),(b,c),(b,d),(c,e),(e,b)\}$.

- $a\ R\ b$: arrow $a \to b$ — true.
- $R(a) = \{b, c, e\}$ — true.
- $\mathrm{Dom}(R) = \{a, b, c, e\}$ (d has no arrow out) — true.
- $\mathrm{Ran}(R) = \{a, b\}$ — **false**: the range is $\{b, c, d, e\}$ (nothing points to a).
- $b\ R^3\ b$: $b \to c \to e \to b$ — true.
- $a\ R^3\ b$: $a \to c \to e \to b$ — true.

4.1.1 Computing R^2 from a list of pairs (exam method)

$(x, z) \in R^2$ when there is a middle element y with $(x, y) \in R$ *and* $(y, z) \in R$ — two steps, $x \to y \to z$. (SQL: `SELECT r1.from, r2.to FROM R r1 JOIN R r2 ON r1.to = r2.from`.)

1. **One-step table:** for each element, list where it can go in one step. Check the total equals the number of pairs.
2. **Second step:** for each starting element, look up every landing place in the same table and collect the results.

For R^3, take one more step from the R^2 results.

Example ($R = \{(1,2),(2,2),(3,4),(4,3),(2,1),(4,1),(3,2)\}$ on $\{1,2,3,4\}$; find R^2). One-step table: $1 \to 2$; $2 \to 1, 2$; $3 \to 2, 4$; $4 \to 1, 3$ $(1+2+2+2 = 7$ pairs ✓).

- from 1: $\to 2 \to \{1, 2\}$
- from 2: $\to 1 \to \{2\}$ and $\to 2 \to \{1, 2\}$, together $\{1, 2\}$
- from 3: $\to 2 \to \{1, 2\}$ and $\to 4 \to \{1, 3\}$, together $\{1, 2, 3\}$
- from 4: $\to 1 \to \{2\}$ and $\to 3 \to \{2, 4\}$, together $\{2, 4\}$

$$R^2 = \{(1,1),(1,2),(2,1),(2,2),(3,1),(3,2),(3,3),(4,2),(4,4)\}.$$

Exam layout (pen and paper, about 2 minutes): one line per element, then replace each line by the merged lines of its targets — duplicates collapse. (Each line is a row of bits; the new row is the OR of the rows it points to — Kolman's Boolean matrix product, written as lists.)

R (one step)	R^2 (merge the targets' lines)
$1:2$	line $2 = \{1,2\}$
$2:1,2$	line 1 + line $2 = \{2\} \cup \{1,2\} = \{1,2\}$
$3:2,4$	line 2 + line $4 = \{1,2\} \cup \{1,3\} = \{1,2,3\}$
$4:1,3$	line 1 + line $3 = \{2\} \cup \{2,4\} = \{2,4\}$
7 pairs	$2+2+3+2 = 9$ pairs

Step 2 as "friends of friends": read each line as *who that element knows*. For line 3: (1) who does 3 know? 2 and 4; (2) ask each of them who *they* know — line 2 gives $\{1,2\}$, line 4 gives $\{1,3\}$; (3) put the answers together, each number once: $\{1,2,3\}$. Watch out:

- always look up in the **original** one-step table, never in the R^2 lines you are building;
- your own friends list is *not* part of the answer — only what comes out of step (2) ($3 \to 4$ directly does not give $(3,4) \in R^2$);
- a loop counts: $2 \in$ line 2, so asking 2 also returns 2.

Then match line by line: find the options whose 3-pairs are exactly $(3,1),(3,2),(3,3)$, and so on — usually one or two lines leave a single option.

Eliminating MC options fast: (1) count the pairs — here 9, which rules out options with another count; (2) test one or two easy pairs, e.g. $(1,1)$ via $1 \to 2 \to 1$ and $(2,1)$ via $2 \to 2 \to 1$. **Trap:** an option that is just R itself.

R^2 is built from scratch, not "R plus extra pairs". Think of R as one-way roads and R^2 as "reachable by driving *exactly* two roads". A pair can be in R but not in R^2: above, $4 \to 1$ is one road, but every two-road trip from 4 ends at 2 or 4, so $(4,1) \notin R^2$. A loop like $(2,2)$ counts as a road too ($1 \to 2 \to 2$ gives $(1,2) \in R^2$).

4.1.2 R^∞: the connectivity relation (exam method)

Picture R as a map of one-way roads between towns: each pair $(a,b) \in R$ is a road from a to b, and (a,a) is a loop road back to the same town. Then R^2 is "towns reachable by driving *exactly* two roads" (see above), and

$$R^\infty = R \cup R^2 \cup R^3 \cup \cdots$$

is **"towns reachable by driving one or more roads, any number"**: $(x,y) \in R^\infty$ means *you can get from x to y at all*. It is reachability in a directed graph (a graph search from each node), and the smallest transitive relation containing R (transitive closure).

Method (one town at a time):

1. Write the one-step table: for each town, where its roads lead.
2. Pick a starting town. Collect its neighbours; then the neighbours' neighbours; keep going and *keep a town only once*, until nothing new turns up.
3. Every town you collected gives a pair (start, town) in R^∞. Repeat for each starting town.

On n towns, paths longer than n never add anything, so the search always ends.

Example (MC practice: $R = \{(1,2),(2,1),(2,3),(3,3)\}$ on $\{1,2,3\}$; find R^∞). **Roads:** $1 \to 2$, $2 \to 1$, $2 \to 3$, $3 \to 3$ (a loop). One-step table: $1 \to 2$; $2 \to 1,3$; $3 \to 3$.

- **Start at 1:** the only road goes to 2. From 2 you can go to 1 and 3. So 2, 1 and 3 are all reachable (1 again via $1 \to 2 \to 1$). Reaches $\{1,2,3\}$: pairs $(1,1),(1,2),(1,3)$.
- **Start at 2:** roads to 1 and 3; from 1 you can drive back to 2. Reaches $\{1,2,3\}$: pairs $(2,1),(2,2),(2,3)$.
- **Start at 3:** the only road is the loop back to 3, so you can never leave. Reaches $\{3\}$: pair $(3,3)$.

$$R^\infty = \{(1,1),(1,2),(1,3),(2,1),(2,2),(2,3),(3,3)\} \quad (7 \text{ pairs}).$$

Two things that trip people up. (1) $(1,1) \in R^\infty$ although $(1,1) \notin R$: you can leave 1 and come back ($1 \to 2 \to 1$), and a path is allowed to return to its start. (2) $(3,1) \notin R^\infty$: 3 is a dead end that only loops on itself, so nothing outside $\{3\}$ is reachable from it. This eliminates every option containing $(3,1)$ or $(3,2)$.

4.1.3 Slide results for 4.1–4.3

- **Product set** $A \times B = \{(a, b) \mid a \in A, b \in B\}$; a relation from A to B is any subset $R \subseteq A \times B$ (on A when $B = A$). $|A_1 \times \cdots \times A_m| = |A_1| \cdots |A_m|$ (the multiplication principle again).
- **Partition** of A: nonempty blocks $A_1, \ldots, A_k$ with $A_1 \cup \cdots \cup A_k = A$ (nothing left out) and $A_i \cap A_j = \varnothing$ for $i \neq j$ (nothing in two blocks).
- **Matrix of** R from $A = \{a_1..a_m\}$ to $B = \{b_1..b_n\}$: $m \times n$, rows $= A$, columns $= B$, $m_{ij} = 1$ iff $(a_i, b_j) \in R$. Digraph (on one set): vertex per element, arrow $a_i \to a_j$ for $a_i R a_j$; **in-degree** = arrows entering, **out-degree** = arrows leaving.

4.2 Theorems 1 and 2

For R from A to B and $A_1, A_2 \subseteq A$, with $R(A_i)$ the set of partners of the elements of A_i: (1) $A_1 \subseteq A_2 \Rightarrow R(A_1) \subseteq R(A_2)$; (2) $R(A_1 \cup A_2) = R(A_1) \cup R(A_2)$; (3) $R(A_1 \cap A_2) \subseteq R(A_1) \cap R(A_2)$ — **only** $\subseteq$: with $R = \{(1, a), (2, a)\}$, $A_1 = \{1\}$, $A_2 = \{2\}$ the left side is $\varnothing$, the right side $\{a\}$.
If $R(a) = S(a)$ for every $a \in A$, then $R = S$.

4.3 Theorems 1 and 2. $M_{R^2} = M_R \odot M_R$ and, for $n \geq 2$, $M_{R^n} = M_R \odot \cdots \odot M_R$ (n factors): $(a, b) \in R^n$ iff there is a path of length n from a to b. Reachability: $R^\infty = R \cup R^2 \cup R^3 \cup \cdots$.

4.4 Properties of relations

Source: Kolman 4.4; own notes

Pen-and-paper tool: draw R as a 0/1 grid (the matrix M_R) — rows "from", columns "to", a 1 for each pair. Check the count of 1's against the number of pairs. Each property is then a visual check, and **one counterexample breaks any of them**.

Property	Definition	Check in the grid
reflexive	$x\ R\ x$ for all x	diagonal all 1
irreflexive	$x\ R\ x$ for no x	diagonal all 0
symmetric	$x\ R\ y \Rightarrow y\ R\ x$	mirror image across the diagonal
antisymmetric	$x\ R\ y$ and $y\ R\ x \Rightarrow x = y$	never 1 in both mirror positions off the diagonal
asymmetric	$x\ R\ y \Rightarrow$ not $y\ R\ x$	antisymmetric *and* diagonal all 0
transitive	$x\ R\ y$ and $y\ R\ z \Rightarrow x\ R\ z$	every chain $x \to y \to z$ has the shortcut $x \to z$

Traps: a relation can be *neither* reflexive nor irreflexive (diagonal mixed). Antisymmetric ignores the diagonal; asymmetric is the stricter version that also forbids it — any (x, x) makes a relation not asymmetric.

Special relations on A *Empty* $R = \varnothing$; *full* $R = A \times A$; *identity* (diagonal) $\Delta = I_A = \{(a, a) \mid a \in A\}$; *inverse* $R^{-1} = \{(b, a) \mid (a, b) \in R\}$. Asymmetric $= aRb \Rightarrow \neg bRa$, which also forbids every loop aRa; antisymmetric only restricts *distinct* pairs. In the matrix: antisymmetric means no $m_{ij} = m_{ji} = 1$ with $i \neq j$.

Example ($R = \{(1,1), (1,4), (2,1), (2,3), (3,1), (3,3), (3,4), (4,2)\}$ on $\{1, 2, 3, 4\}$: which is true?).

	1	2	3	4
1	1	0	0	1
2	1	0	1	0
3	1	0	1	1
4	0	1	0	0

- reflexive: diagonal $1, 0, 1, 0$ — no, $(2, 2)$ missing
- irreflexive: $(1, 1)$ present — no
- symmetric: $(1, 4)$ but no $(4, 1)$ — no
- **antisymmetric**: the six mirror pairs $(1, 2)/(2, 1)$, $(1, 3)/(3, 1)$, $\ldots$, $(3, 4)/(4, 3)$ never both 1 — **yes**
- asymmetric: $(1, 1)$ on the diagonal — no
- transitive: $1 \to 4 \to 2$ but no $(1, 2)$ — no

Example (MC practice: $x\ R\ y \iff x \neq y$ on $\mathbb{Z}^+$, how many properties hold?). Test each property with the smallest possible witnesses, 1 and 2:

- symmetric: $x \neq y \Rightarrow y \neq x$ — **yes**
- irreflexive: $x \neq x$ never holds, so no (x, x) — **yes**
- reflexive: $(1, 1) \notin R$ — no
- asymmetric: $(1, 2)$ and $(2, 1)$ are both in R — no (symmetric and nonempty always kills asymmetric)
- transitive: $1 \to 2 \to 1$ but $(1, 1) \notin R$ — no

Answer: **2** properties (symmetric and irreflexive).

4.5 Equivalence relations

Source: Kolman 4.5

The two exercises below (and in 4.7) are the ones worked at the end of the lecture. Row i, column j of M_R is 1 iff $(i, j) \in R$.

- **Reflexive**: all diagonal entries 1. **Symmetric**: $M_R = M_R^T$. **Transitive**: $(a, b), (b, c) \in R \Rightarrow (a, c) \in R$.
- **Equivalence relation** = reflexive + symmetric + transitive; it splits the set into equivalence classes.

4.5 Theorems 1, 2 and Lemma 1

Thm 1. If P is a partition of A and aRb means "a and b lie in the same block", then R is an equivalence relation. (Reflexive: a shares a block with itself. Symmetric: "same block" is symmetric. Transitive: a, b and b, c in the same block put a, b, c in one block.)
Lemma 1. For an equivalence relation, $aRb \iff R(a) = R(b)$, where the class of a is $[a] = R(a) = \{x \mid aRx\}$.
Thm 2. The distinct classes $\{[a] \mid a \in A\}$ form a **partition** of A. So equivalence relations and partitions are the same thing seen two ways.

- Mnemonic: **equivalence** = reflexive + symmetric + transitive; **partial order** = reflexive + antisymmetric + transitive.
- $a \leq b$ on $\mathbb{Z}$ is reflexive and transitive but **not symmetric** ($1 \leq 2$, $2 \not\leq 1$), so not an equivalence.
- **Congruence mod** n: $aRb \iff n \mid (a - b)$ (same remainder on division by n) is an equivalence relation on $\mathbb{Z}$. Mod 3 it has three classes $[0] = \{\ldots, -3, 0, 3, \ldots\}$, $[1] = \{\ldots, -2, 1, 4, \ldots\}$, $[2] = \{\ldots, -1, 2, 5, \ldots\}$, and $\{[0], [1], [2]\}$ is a partition of $\mathbb{Z}$.
- A set such as $A_3 \cup A_4 \cup A_5$ with blocks $\{a, c, e, g\}, \{b, d\}, \{f, h\}$ is a partition of $\{a, \ldots, h\}$ (union is everything, blocks disjoint); a family with overlapping blocks, like $\{a, b, c, d\}$ and $\{a, c, e, f, g\}$, is not.

Example (4.5.1). (1) $M_R = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}$ on $\{a, b, c\}$: reflexive, symmetric, transitive $\Rightarrow$ equivalence, classes $\{a\}$, $\{b, c\}$. (2) $M_R = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$: $(a, c) \in R$ but $(c, a) \notin R$, not symmetric $\Rightarrow$ not an equivalence (answer not on slides).

4.7 Operations on relations

Source: Kolman 4.7

Complement $\overline{R}$: flip every bit. $R \cap S$: AND. $R \cup S$: OR. Inverse R^{-1}: transpose, $M_{R^{-1}} = M_R^T$.

Example (4.7.9). $M_R = \begin{bmatrix} 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix}$, $M_S = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 \end{bmatrix}$:

$$M_{\overline{S}} = \begin{bmatrix}1&0&0&1\\0&1&1&0\\0&0&1&1\end{bmatrix},\ M_{R\cap S} = \begin{bmatrix}0&1&0&0\\0&0&0&1\\1&1&0&0\end{bmatrix},\ M_{R\cup S} = \begin{bmatrix}1&1&1&1\\1&0&0&1\\1&1&1&0\end{bmatrix},\ M_{R^{-1}} = \begin{bmatrix}1&0&1\\1&0&1\\0&0&1\\1&1&0\end{bmatrix}.$$

4.7 Theorems 1–4

Thm 1. $R \subseteq S \Rightarrow R^{-1} \subseteq S^{-1}$ and $\overline{S} \subseteq \overline{R}$; $(R \cap S)^{-1} = R^{-1} \cap S^{-1}$; $(R \cup S)^{-1} = R^{-1} \cup S^{-1}$; $\overline{R \cap S} = \overline{R} \cup \overline{S}$ (De Morgan).
Thm 2. R reflexive $\Rightarrow R^{-1}$ reflexive; R, S reflexive $\Rightarrow R \cap S$ and $R \cup S$ reflexive; R reflexive $\iff \overline{R}$ irreflexive.
Thm 3. R symmetric $\iff R = R^{-1}$; antisymmetric $\iff R \cap R^{-1} \subseteq \Delta$; asymmetric $\iff R \cap R^{-1} = \varnothing$.
Thm 4. R symmetric $\Rightarrow R^{-1}$ and $\overline{R}$ symmetric; R, S symmetric $\Rightarrow R \cap S$, $R \cup S$ symmetric.

Example (4.7 Ex. 4 and Ex. 6 (slides)). $M_R = \begin{bmatrix}1&1&0\\0&0&1\\0&0&1\end{bmatrix}$, $M_S = \begin{bmatrix}0&1&1\\1&0&0\\0&0&0\end{bmatrix}$: $M_{\overline{R}} = \begin{bmatrix}0&0&1\\1&1&0\\1&1&0\end{bmatrix}$, $M_{R^{-1}} = M_R^T = \begin{bmatrix}1&0&0\\1&0&0\\0&1&1\end{bmatrix}$, $M_{R\cap S} = \begin{bmatrix}0&1&0\\0&0&0\\0&0&0\end{bmatrix}$, $M_{R\cup S} = \begin{bmatrix}1&1&1\\1&0&1\\0&0&1\end{bmatrix}$.

On $\{1,2,3\}$ let $R = \{(1,1),(1,2),(2,1),(1,3),(3,1)\}$ and $S = \{(1,1),(1,2),(2,1),(2,2),(3,3)\}$ (both symmetric). Then $R^{-1} = R$, $\overline{R} = \{(2,2),(2,3),(3,2),(3,3)\}$, $R \cap S = \{(1,1),(1,2),(2,1)\}$, $R \cup S = \{(1,1),(1,2),(1,3),(2,1),(2,2),(3,1),(3,3)\}$ — all symmetric, as Theorem 4 predicts.

Closure of R for a property (reflexive, symmetric, transitive): the *smallest* relation containing R with that property — add only the pairs that are needed. **Composition**: for R from A to B and S from B to C, $a\,(S \circ R)\,c \iff \exists b:\ aRb \wedge bSc$; note the matrix order, $M_{S\circ R} = M_R \odot M_S$ (R first).
With the M_R, M_S above, $M_{S\circ R} = \begin{bmatrix}1&1&1\\0&0&0\\0&0&0\end{bmatrix}$.

Chapter 5
Functions

5.1 Functions

Source: Kolman 5.1

Definition. A **function** $f : A \to B$ is a relation where each $a \in A$ is related to *exactly one* $b \in B$; write $f(a) = b$. As a set of pairs: $f = \{(a, f(a)) \mid a \in A\}$. $f(x) = x^2$ on $\mathbb{R}$ is a function ($f(2) = f(-2) = 4$ is fine — two inputs may share an output, but one input may not have two outputs).

Identity: $1_A(a) = a$; e.g. $1_{\{1,2,3\}} = \{(1,1),(2,2),(3,3)\}$. **Composition**: $f : A \to B$, $g : B \to C$, $(g \circ f)(a) = g(f(a))$ — apply f first.

Special types, for $f : A \to B$

Injective (one-to-one): $f(a) = f(a') \Rightarrow a = a'$. **Surjective** (onto): every $b \in B$ is hit. **Bijective**: both.
Everywhere defined: every $a \in A$ has an image.

Example (5.1 Ex. 12). $f_1 = \{(a_1, b_2), (a_2, b_3), (a_3, b_1)\}$: everywhere defined, one-to-one, onto. $f_2 = \{(a_1, d_2), (a_2, d_1), (a_3, d_4)\}$ into $D = \{d_1, \ldots, d_4\}$: one-to-one, not onto. $f_3 = \{(b_1, c_2), (b_2, c_2), (b_3, c_1)\}$: onto, not one-to-one. $f_4 = \{(d_1, b_1), (d_2, b_2), (d_3, b_1)\}$: not everywhere defined (d_4), not one-to-one, not onto.

5.1.1 Checking a function from a table (exam method)

In Kolman a **function** f from A to B is a relation where each a has *at most one* partner $f(a)$ — a lookup table with no duplicate keys. Given as a list of pairs, write it as a two-row table first; every property below is then a glance at one of the rows. Kolman also says *one-to-one* for injective and *one-to-one correspondence* for bijection.

Property	Meaning	Check in the table
function at all	no a has two outputs	no repeat in the *top* row
everywhere defined	every $a \in A$ has an output	top row is all of A
injective	$a \neq a' \Rightarrow f(a) \neq f(a')$	no repeat in the *bottom* row
surjective (onto)	every $b \in B$ is hit	bottom row is all of B
bijection	all of the above	both rows full, no repeats

Inverse: f^{-1} swaps every pair. It is always a relation; it is a *function* exactly when f is injective (swapping the rows must not create a repeat in the new top row). To evaluate $f^{-1}(b)$, find b in the bottom row and read the a above it.

Preimage $f^{-1}(\{b\})$ (a *set* inside the braces): the set of inputs that f sends to b. Find b in the bottom row and collect *every* a above it. It always exists, even when f is not injective (then the set can have several elements) or not onto (then it can be $\emptyset$). So $f^{-1}(\{4\}) = \{3\}$ for $f(3) = 4$, not the output side $\{f(4)\}$: read *backwards*, bottom row to top row.

Composition: $(g \circ f)(a) = g(f(a))$ — **right to left**, innermost first, like nested calls `g(f(a))`. So $(f \circ f)(3)$ means look up 3, then look up the answer.

Finite shortcut: for $f : A \to A$ everywhere defined on a *finite* set, injective $\iff$ surjective (pigeonhole: n inputs with n distinct outputs fill all n slots). One check gives you bijection, and then f^{-1} is a function too.

Example ($f = \{(1,2),(2,1),(3,4),(4,5),(5,6),(6,3)\}$ on $\{1,\ldots,6\}$: which is false?).

x	1	2	3	4	5	6
$f(x)$	2	1	4	5	6	3

- bottom row $2, 1, 4, 5, 6, 3$: no repeats, all of 1..6 — bijection, so injective, onto, everywhere defined and "f^{-1} is a function" are all **true**
- $f^{-1}(3)$: 3 is under 6, so $f^{-1}(3) = 6$ — true
- $(f \circ f)(3) = f(f(3)) = f(4) = 5 \neq 3$ — **false**

Only the computed claims needed work; the structural ones fell together from one look at the bottom row. Cycle view: $f = (1\,2)(3\,4\,5\,6)$, and $f \circ f$ moves 3 two steps round its 4-cycle. Only elements in cycles of length 1 or 2 satisfy $(f \circ f)(x) = x$.

Example (MC practice: $f = \{(1,6),(2,3),(3,4),(4,1),(5,5),(6,2)\}$ on $\{1,\ldots,6\}$; which is false?). Table: $x = 1,\ldots,6$ gives $f(x) = 6, 3, 4, 1, 5, 2$. The bottom row has no repeat and is all of $\{1,\ldots,6\}$, so f is everywhere defined, injective and onto (a bijection). Then $f(5) = 5$ is true; $(f \circ f)(1) = f(f(1)) = f(6) = 2$ is true; but $f^{-1}(\{4\})$ is the input that gives 4, namely 3 ($f(3) = 4$), so $f^{-1}(\{4\}) = \{3\}$ and the claim "$= \{1\}$" is **false** (it confuses $f(4) = 1$ with $f^{-1}(4)$).

5.1.2 Injective or surjective from a formula (exam method)

For $f\colon \mathbb{Z} \to \mathbb{Z}$ given by a formula, test the two properties separately and hunt for a *counterexample* first.

Two quick tests

- **Not injective**: find $a \neq a'$ with $f(a) = f(a')$. Typical culprits: even powers and anything symmetric ($f(1) = f(-1)$), and a floor or ceiling, which squeezes several inputs into one output.
- **Not surjective**: find an integer that is never an output. Typical culprits: a bounded range ($x^2 + 1 \geq 1$), a coefficient that skips values ($2x$ misses every odd number), or a gap between outputs ($(x-1)^3$ gives only perfect cubes).

If the counterexample hunt fails, prove it: injective by solving $f(a) = f(a')$ and getting $a = a'$; surjective by solving $f(x) = y$ for x and checking that x is an integer for every integer y.

$f(x)$ on $\mathbb{Z}$	injective?	surjective?
$x + c$	yes	yes ($x = y - c$)
$2x$, $2x+1$, kx ($\lvert k\rvert \geq 2$)	yes	no (skips values)
$x^2 + 1$, x^2	no ($f(1) = f(-1)$)	no (nothing negative)
x^3, $(x-1)^3$	yes	no (only cubes)
$\lfloor x/2 \rfloor$	no ($f(2n) = f(2n+1)$)	yes (hits every n)

Floor check: $\lfloor x/2 + \frac{1}{2} \rfloor$ is n for both $x = 2n - 1$ and $x = 2n$ (not injective), and every n is reached (surjective). **Increasing functions** (strictly rising on $\mathbb{Z}$, such as $x \cdot 2^{x^2}$) are injective, but the outputs jump, so they are rarely onto.

5.1.3 Inverses

f^{-1} reverses the pairs: $f(a) = b \iff f^{-1}(b) = a$.

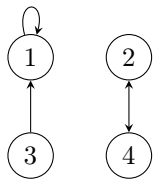
Thm 5.1.1: f^{-1} is a function $\iff$ f is bijective.
Thm 5.1.2: $1_B \circ f = f$, $f \circ 1_A = f$; for bijective f: $f^{-1} \circ f = 1_A$, $f \circ f^{-1} = 1_B$.
Thm 5.1.3: (a) if $g \circ f = 1_A$ and $f \circ g = 1_B$ then f, g are bijections and $g = f^{-1}$. (b) $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$ (reverse order, like $(AB)^T = B^T A^T$).

Example (5.1 Ex. 17). $f(x) = 2x^3 - 1$, $g(y) = \sqrt[3]{(y+1)/2}$ on $\mathbb{R}$: $g(f(x)) = \sqrt[3]{x^3} = x$ and $f(g(y)) = 2 \cdot \frac{y+1}{2} - 1 = y$, so both are bijections and $g = f^{-1}$.

Example (5.1.24–25, is f^{-1} a function?). $f = \{(1,a),(2,a),(3,c),(4,d)\}$: f^{-1} sends a to both 1 and 2 — **not** a function. $f = \{(1,a),(2,c),(3,b),(4,d)\}$: bijective, f^{-1} **is** a function.

Example (5.1 Ex. 9 and 5.1.10). $f(a) = a+1$, $g(b) = 2b$: $(g \circ f)(a) = 2a+2$.
$f(a) = a+1$, $g(b) = b^2+2$: $(g \circ f)(-2) = g(-1) = 3$; $(f \circ g)(-2) = f(6) = 7$; $(g \circ f)(x) = x^2+2x+3$; $(f \circ g)(x) = x^2+3$; $(f \circ f)(y) = y+2$; $(g \circ g)(y) = y^4+4y^2+6$.

Example (MC practice test 5: composition read off a digraph).



F on $A = \{1,2,3,4\}$ is given by the digraph: $F(1) = 1$, $F(3) = 1$, $F(2) = 4$, $F(4) = 2$ (the arrow between 2 and 4 points both ways). $G = \{(1,c),(2,d),(3,e),(4,a)\}$. Then $(G \circ F)(2) = G(F(2)) = G(4) = a$. Apply F *first*, then G.

5.2 Functions for computer science

Source: Kolman 5.2

Common examples: characteristic function χ_A, $m \bmod n$, $n!$, floor $\lfloor x \rfloor$, ceiling $\lceil x \rceil$, string length, power set, polynomials, 2^x, $\log_b x$, transpose, gcd, lcm. Boolean-valued functions have codomain $\{\text{true}, \text{false}\}$.

Hash function: $h(k) = k \bmod m$ maps key k to one of m table positions. A **collision** is $k_1 \neq k_2$ with $h(k_1) = h(k_2)$: with $m = 10$, $h(27) = h(37) = 7$.

Fuzzy set: membership $\mu_A(x) \in [0,1]$ (0 not in, 1 fully in, between = partly). E.g. "tall": 160 cm 0.1, 170 0.3, 180 0.7, 190 1.0.

Fuzzy operations (Theorem 5.2.1)

$(f \cap g)(x) = \min\{f(x), g(x)\}$ $\quad$ $(f \cup g)(x) = \max\{f(x), g(x)\}$ $\quad$ $\overline{f}(x) = 1 - f(x)$

Example (5.2 Ex. 13, answers not on slides). On $[0,10]$: $f(x) = \frac{x^2}{16}$ on $[0,4)$, 1 on $[4,6)$, $-\frac{x}{2}+4$ on $[6,8)$, 0 on $[8,10]$; $g(x) = \frac{x}{3}$ on $[0,3)$, $-\frac{x}{4}+\frac{7}{4}$ on $[3,5)$, $\frac{1}{2}$ on $[5,7)$, $-\frac{x}{6}+\frac{10}{6}$ on $[7,10]$.
$f(3.5) = \frac{49}{64}$, $g(3.5) = \frac{7}{8}$; $f(7) = g(7) = \frac{1}{2}$. So (a) 3.5 in $f \cap g$: $\frac{49}{64}$; (b) 7 in $f \cap g$: $\frac{1}{2}$; (c) 3.5 in $f \cup g$: $\frac{7}{8}$; (d) 7 in $f \cup g$: $\frac{1}{2}$.

Example (5.2.22–23, propositional functions). $P(x,y) = (x \vee y) \wedge \neg y$: $P(T,T) = F$, $P(F,T) = F$, $P(T,F) = T$.
$Q(x) : \exists y \in \mathbb{Z}^+,\ xy = 60$: $Q(3) = T$ $(y = 20)$, $Q(7) = F$, $Q(-6) = F$ $(y = -10 \notin \mathbb{Z}^+)$, $Q(15) = T$ $(y = 4)$.

5.3 Growth of functions

Source: Kolman 5.3

Growth order (slowest to fastest, for large n)

$$1 < \log n < n < n \log n < n^2 < n^3 < 2^n$$
constant → logarithmic → linear → linearithmic → polynomial → exponential

Definition (Big-O). $f = O(g)$ if there are constants $c > 0$, $k \in \mathbb{Z}^+$ with $|f(n)| \leq c\,|g(n)|$ for all $n \geq k$. An *upper bound*: f grows no faster than g. To prove it, exhibit a c and k.

Definition (Big-Θ). $f = \Theta(g)$ if $f = O(g)$ and $g = O(f)$; equivalently $c_1|g(n)| \leq |f(n)| \leq c_2|g(n)|$ for $n \geq k$. *Same order of growth.* Theorem 5.3.1: Θ is an equivalence relation (reflexive, symmetric, transitive), so functions fall into Θ-classes.

$100n^2 = O(n^3)$ but $100n^2 = \Theta(n^2)$, not $\Theta(n^3)$.

Finding the Θ-class: drop constant factors and lower-order terms; the highest power dominates a polynomial; floor/ceiling don't change the class; any exponential a^n $(a > 1)$ beats any polynomial.

Example (Big-O proofs). $100n^2 \le 100n^3$ for $n \ge 1$: $c = 100$, $k = 1$.
$2n^3 + n^2 + 7 \le 2n^3 + n^3 + n^3 = 4n^3$ for $n \ge 2$: $c = 4$, $k = 2$.
$n! = 1 \cdot 2 \cdots n \le n \cdot n \cdots n = n^n$: $c = 1$, $k = 1$.
$1 + \cdots + n = \frac{n(n+1)}{2} \le \frac{n \cdot 2n}{2} = n^2$: $c = 1$, $k = 1$.

Example (5.3.9: $5n^2 + 4n + 3$ and $n^2 + 100n$ have the same order). $f \le 5n^2 + 4n^2 + 3n^2 = 12n^2 \le 12g$, and $g \le n^2 + 100n^2 = 101n^2 \le \frac{101}{5} f$. So $f = \Theta(g) = \Theta(n^2)$.

Example (5.3 Ex. 7–8). $4n^4 - 6n^7 + 25n^3 \in \Theta(n^7)$; $\lfloor \lg n \rfloor - 3n \in \Theta(n)$; $1.1^n + n^{15} \in \Theta(1.1^n)$.
Ordering: $\Theta(1{,}000{,}000) = \Theta(1) < \Theta(n^{0.2}) < \Theta(n + 10^7) = \Theta(n) < \Theta(n \lg n) < \Theta(1000n^2 - n) = \Theta(n^2) < \Theta(1.3^n)$.

Example (MC practice test 5: highest order of growth). Options: $3^n + n^7$, $2^n + n^7$, n^7, $n^8 + \log_2 n$, $\log_2 n$, $n^2 \cdot 3^n + \log_2 n$. Keep each dominant term: 3^n, 2^n, n^7, n^8, $\log_2 n$, $n^2 3^n$. Since $\log n < n^k < a^n$ $(a > 1)$ and $2^n < 3^n < n^2\, 3^n$, the answer is $n^2 \cdot 3^n + \log_2 n$. A polynomial factor in front of an exponential still counts.

Example (MC practice: which option has the same order as n^2?). Reduce each option to its dominant term, then compare with n^2; only $\Theta(n^2)$ matches. $n^2 + 2^n \to 2^n$ and $n^2 + 1.1^n \to 1.1^n$ (any exponential wins, even base 1.1); $n^3 - n^2 \to n^3$; $8n^2 \log_2 n \to n^2 \log n$ (extra log factor, strictly faster); $4n \log_2 n \to n \log n$ (strictly slower); $4n^2 + n \log_2 n \to 4n^2$, since $\frac{n \log n}{n^2} = \frac{\log n}{n} \to 0$ — so it is $\Theta(n^2)$. Constant factors (the 4, the 8) never change the class; a log factor *does*.

Example (MC practice: which has the same order as $n^3 + 2^n \log_2 n$?). Dominant term: $2^n \log_2 n$ (n^3 is much smaller than any exponential). Two facts decide the options:

- **The log base never matters:** $\log_{10} n = \dfrac{\log_2 n}{\log_2 10}$ is a constant multiple of $\log_2 n$, so $2^n \log_{10} n = \Theta(2^n \log_2 n)$. (Constant factors, even $\frac{1}{4}$ or $\frac{1}{\log_2 10}$, never change the class.)
- **Exponential × log is its own class:** $2^n \log_2 n$ is strictly above 2^n (ratio $\log_2 n \to \infty$) and strictly below $2^n n^3$ and 4^n. So 2^n and $n^3 + 2^n$ (both $\Theta(2^n)$) do *not* match.

Answer: $2^n \log_{10} n$. The distractors: 2^n and $n^3 + 2^n$ (log factor dropped), $\frac{1}{4} 4^n \log_2 n$ (a bigger base), $2^n n^3$ (a polynomial factor, not a log), n^3 (too small).

Example (5.3.1: what doubling the input does).

1001	no change	$1.4 \lg n$	$f(2n) = f(n) + 1.4$
$3n$	$\times 2$	$n \lg n$	$f(2n) = 2f(n) + 2n$
$5n^2$	$\times 4$	$100n^4$	$\times 16$
$2.5n^3$	$\times 8$	2^n	$f(2n) = f(n)^2$ (not on slides)

5.3.1 Limits and L'Hôpital

$\lim\limits_{n\to\infty} \dfrac{f(n)}{g(n)} = L$ with $0 < L < \infty \Rightarrow f = \Theta(g)$; limit $= 0 \Rightarrow f$ grows strictly slower than g.

L'Hôpital: if $\frac{f}{g}$ has the form $\frac{0}{0}$ or $\frac{\infty}{\infty}$, then $\lim \dfrac{f(x)}{g(x)} = \lim \dfrac{f'(x)}{g'(x)}$ (if the right side exists).

$f(x)$	c	x	x^n	$\ln x$	e^x	$\sin x$	$\cos x$
$f'(x)$	0	1	nx^{n-1}	$1/x$	e^x	$\cos x$	$-\sin x$

e.g. $(5x^4)' = 20x^3$

Example (limits). $\lim \frac{100n^2}{n^3} = \lim \frac{100}{n} = 0$: $100n^2$ grows slower than n^3.
$\lim \frac{2n^3+n^2+7}{n^3} = \lim \left(2 + \frac{1}{n} + \frac{7}{n^3}\right) = 2$: $\Theta(n^3)$.
$\lim_{x\to 0} \frac{\sin x}{x}$ $\left(\frac{0}{0}\right) = \lim \frac{\cos x}{1} = 1$.
$\lim_{x\to\infty} \frac{\ln x}{x^k}$ $\left(\frac{\infty}{\infty}, k > 0\right) = \lim \frac{1/x}{kx^{k-1}} = \lim \frac{1}{kx^k} = 0$: $\ln x$ grows slower than every positive power of x.

Chapter 6
Order Relations and Structures

6.1 Partial orders and posets

Source: Kolman 6.1

Definition. A relation R on a set A is a **partial order** if it is reflexive, antisymmetric and transitive. The pair (A, R) (written $(A, \leq)$ when R is denoted $\leq$) is called a **poset** (partially ordered set).

Example (6.1 Ex. 1–3). $(\mathcal{P}(S), \subseteq)$ is a poset for any set S (Example 1); $(\mathbb{Z}^+, \leq)$ is a poset (Example 2); divisibility, $aRb \iff a \mid b$, is a poset on $\mathbb{Z}^+$ (Example 3) — each of reflexivity, antisymmetry and transitivity for $\leq$ and $\mid$ is the familiar number fact.

Strict order: why $<$ is not a partial order

$<$ fails reflexivity ($a < a$ is never true), so it is not a partial order on $\mathbb{Z}^+/\mathbb{R}$ — but it is a useful relation in its own right, a **strict partial order**. Two more relation properties: **Irreflexive**: $\forall x \in A,\ \neg(xRx)$ (no element related to itself). **Asymmetric**: $xRy \Rightarrow \neg(yRx)$ (never both ways). *Asymmetric and antisymmetric are not the same*: antisymmetric allows $xRy \wedge yRx$ when $x = y$ (as $\leq$ does); asymmetric forbids $xRy \wedge yRx$ outright, which is exactly why $<$ (irreflexive, transitive) is asymmetric rather than merely antisymmetric.

6.1.1 Comparability, chains, duals

Two elements $a, b \in A$ are **comparable** if $a \leq b$ or $b \leq a$; otherwise **incomparable**. Under divisibility, 2 and 6 are comparable ($2 \mid 6$) but 2 and 3 are incomparable ($2 \nmid 3$, $3 \nmid 2$). If *every* pair is comparable, $(A, \leq)$ is **linearly ordered** (a **chain**); e.g. $(\mathbb{Z}, \leq)$, but not divisibility in general.

Is it a linear order? Counterexample $\mathcal{P}(S)$

To show a poset is **not** linear, exhibit *one* incomparable pair. For $(\mathcal{P}(S), \subseteq)$ with $|S| \geq 2$, pick distinct $a, b \in S$: $\{a\} \not\subseteq \{b\}$ and $\{b\} \not\subseteq \{a\}$, so $\{a\}, \{b\}$ are incomparable and $\subseteq$ is **not** a linear order. (It is still a partial order — Example 1.) If $|S| \leq 1$ the power set *is* a chain ($\emptyset \subseteq \{a\}$), so state the case $|S| \geq 2$. To show it *is* linear you must prove *every* pair comparable, which needs a general argument, not an example.

The **dual poset** of $(A, \leq)$ is $(A, \geq)$ — same set, order reversed; e.g. the dual of $(\mathcal{P}(S), \subseteq)$ is $(\mathcal{P}(S), \supseteq)$.

6.1.2 Product and lexicographic order

Theorem 1 (product partial order)

For posets $(A, \leq)$, $(B, \leq)$, define an order on $A \times B$ by

$$(a, b) \leq (a', b') \iff a \leq a' \text{ and } b \leq b'.$$

$(2, 3) \leq (4, 5)$ since $2 \leq 4$ and $3 \leq 5$. But $(2, 5)$ and $(3, 4)$ are **incomparable** under the product order: $(2, 5) \leq (3, 4)$ needs $5 \leq 4$ (false), and $(3, 4) \leq (2, 5)$ needs $3 \leq 2$ (false).

Lexicographic (dictionary) order

$(a, b) \prec (a', b')$ iff $a < a'$, or $a = a'$ and $b < b'$ — compare the first component; only if it ties, compare the second. Unlike the product order, every pair of pairs is comparable this way.

$(2, 100)$ vs. $(3, 1)$: **product order** — $(2, 100) \leq (3, 1)$ needs $100 \leq 1$ (false), so incomparable. **Lexicographic order** — $2 < 3$ settles it immediately: $(2, 100) \prec (3, 1)$. Like sorting by surname, then first name: `ORDER BY a, b`.

Consequence: a larger first coordinate wins outright — the second coordinate is never even looked at. $(5, \emptyset)$ is above $(2, \{a, b, c\})$, however big that set is.

Trap when B is only a partial order (e.g. $\subseteq$ on a power set): two pairs with the same first coordinate can be *incomparable*. $\{a\}$ and $\{b\}$: neither $\subseteq$ the other, so $(3, \{a\})$ and $(3, \{b\})$ are incomparable.

Theorem 2 (no cycles)

A partial order's digraph has no cycle of length greater than 1. *Proof idea*: a cycle $2 \leq 5 \leq 8 \leq 2$ gives, by transitivity, $2 \leq 8$ and $8 \leq 2$; antisymmetry then forces $2 = 8$, contradiction (distinct elements).

6.1.3 Exam method: is a relation a partial order?

Given as a matrix, or as a formula on $\mathbb{Z}^+$:

Matrix test, cheapest check first — stop at the first failure:

1. **Reflexive:** the whole diagonal is 1. A single 0 on the diagonal and it is *not* an order. (Strict orders like $<$ have diagonal all 0 — not partial orders in Kolman's sense.)
2. **Antisymmetric:** for every off-diagonal 1 at (i, j), the mirror cell (j, i) is 0. Fold the matrix along the diagonal: no two 1's may land on top of each other. Shortcut: if everything below the diagonal is 0 (upper triangular), it is automatically antisymmetric.
3. **Transitive — row trick:** for every 1 at (i, j) with $i \neq j$, every 1 in row j must also be a 1 in row i ("if i reaches j, then i reaches everything j reaches"). Off-diagonal 1's only; the diagonal ones are trivially fine.

Do the transitivity check last: it is the slowest, and steps 1–2 usually eliminate most MC options.

Example (Is "divides" on $\{1, 2, 3, 6\}$ a partial order?).

	1	2	3	6
1	1	1	1	1
2	0	1	0	1
3	0	0	1	1
6	0	0	0	1

1. reflexive: diagonal $1, 1, 1, 1$ — yes
2. antisymmetric: everything below the diagonal is 0 — yes, no mirror pair can be $1/1$
3. transitive, row trick over the off-diagonal 1's:
   - $(1, 2)$, $(1, 3)$, $(1, 6)$: row 1 is all 1's, so it contains any row — fine
   - $(2, 6)$: row $6 = 0001 \subseteq$ row $2 = 0101$ — fine
   - $(3, 6)$: row $6 = 0001 \subseteq$ row $3 = 0011$ — fine

All three hold: **a partial order**. Had row 2 been 0100 instead, $(1, 2)$ would still pass but $(2, 6)$ would vanish — always recheck after a change.

Failing transitivity looks like this: rows 1: 1100, 2: 0110, 3: 0010, 4: 0001. Reflexive and antisymmetric pass, but $(1, 2)$ is a 1 and row 2 has a 1 in column 3 that row 1 lacks: $1 \to 2 \to 3$ with no shortcut $(1, 3)$.

Relations given by a formula ("which relation on $\mathbb{Z}^+$ is a partial order?"): first rewrite each condition into something you recognise, then run reflexive → antisymmetric → transitive.

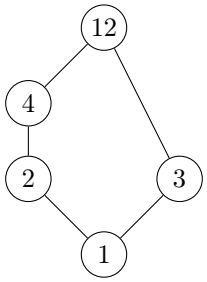
- $x - y \leq 0$ is just $x \leq y$ — the usual order, so a partial order. $x - y < 0$ is $x < y$: strict, so *not* reflexive.
- **Empty or everything:** on positive numbers $x + y \leq 0$ and $x + y < 0$ are never true (empty relation, not reflexive), and $xy \geq 0$ is always true (every pair related, so $1\,R\,2$ and $2\,R\,1$ breaks antisymmetry).
- $2 \mid (x + y)$ means "same parity": reflexive and symmetric (an equivalence relation), but $1\,R\,3$ and $3\,R\,1$ with $1 \neq 3$ kills antisymmetry.

A symmetric relation that relates any two *different* elements is never antisymmetric.

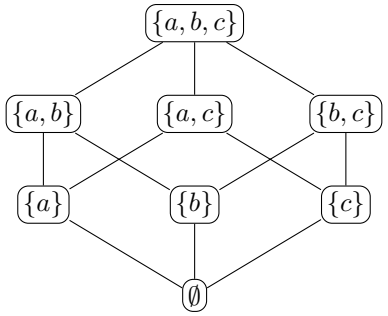
6.1.4 Hasse diagrams

Drawing every self-loop (reflexivity) and every transitive edge makes a poset's digraph unreadable. A **Hasse diagram** simplifies it: remove self-loops, remove edges implied by transitivity (keep only the **cover relations**), place larger elements higher, and draw no arrowheads — direction is "upward." Read it **bottom to top**: an upward path from a to b means $a \leq b$. The **dual** Hasse diagram is the same picture upside down (order reversed).

Example (6.1 Ex. 12: divisibility on $\{1, 2, 3, 4, 12\}$). Divisibility pairs: $1 \mid 2$, $1 \mid 3$, $1 \mid 4$, $1 \mid 12$, $2 \mid 4$, $2 \mid 12$, $3 \mid 12$, $4 \mid 12$. Dropping what transitivity already gives (e.g. $1 \mid 2 \mid 4 \Rightarrow 1 \mid 4$ is redundant, likewise $1 \mid 12$, $2 \mid 12$) leaves the cover relations $1 \prec 2,\ 1 \prec 3,\ 2 \prec 4,\ 3 \prec 12,\ 4 \prec 12$:



Example (power set of $\{a, b, c\}$ under $\subseteq$). The familiar cube: singletons cover $\emptyset$, pairs cover the singletons they contain, and $\{a, b, c\}$ covers every pair.



6.1.5 Exam method: from a matrix to a Hasse diagram

Given the matrix M_R of a partial order on $A = \{1, \ldots, n\}$ (row i, column j is 1 iff $i\,R\,j$, i.e. $i \leq j$):

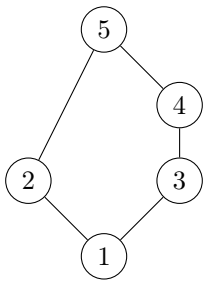
1. **Read the pairs.** Ignore the diagonal (loops). Every other 1 at (i, j) is a relation $i \leq j$.
2. **Cross out transitive edges.** Keep $i \to j$ only if there is *no* k ($k \neq i, j$) with $i \leq k$ and $k \leq j$. Matrix trick: $i \to j$ is redundant iff some $k \neq i, j$ has a 1 in row i (so $i \leq k$) *and* a 1 in column j (so $k \leq j$) — a k whose entry in row i and entry in column j are both 1.
3. **Find the bottom and top.** *Minimal* elements: the column has only its diagonal 1. *Maximal* elements: the row has only its diagonal 1.
4. **Assign heights.** Minimal elements at the bottom; every other element goes one level above the highest element it covers. Draw the covers as plain upward lines, no arrowheads.

Sanity check: row sum of i minus 1 = number of elements above i; column sum of j minus 1 = number below j. The picture must agree.

Example (a branch).

$$M = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

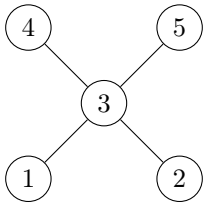
Pairs: $1 \leq 2, 3, 4, 5$; $2 \leq 5$; $3 \leq 4, 5$; $4 \leq 5$. Redundant: $1 \to 4$ (via 3), $1 \to 5$ (via 2), $3 \to 5$ (via 4). Covers: $1 \prec 2,\ 1 \prec 3,\ 3 \prec 4,\ 2 \prec 5,\ 4 \prec 5$. Minimal: 1 only (column 1 is $1, 0, 0, 0, 0$); maximal: 5 only. Heights: 1 bottom, 2 and 3 above it, 4 above 3, 5 on top.



Example (two minimal elements).

$$M = \begin{bmatrix} 1 & 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

Pairs: $1 \le 3,4,5$; $2 \le 3,4,5$; $3 \le 4,5$. Redundant: $1 \to 4,\ 1 \to 5,\ 2 \to 4,\ 2 \to 5$ (each goes through 3). Covers: $1 \prec 3,\ 2 \prec 3,\ 3 \prec 4,\ 3 \prec 5$. Minimal: 1 and 2 (columns $1,2$ have only a diagonal 1); maximal: 4 and 5 (rows $4,5$ likewise), so there is neither a least nor a greatest element. $4,5$ are incomparable (both zeros at $(4,5)$ and $(5,4)$), so they sit side by side.



6.1.6 Topological sorting

A poset only orders *some* pairs; a **topological sort** extends it to a full linear ordering that respects every $a \preceq b$ already fixed by the partial order. If $A \prec C$, $B \prec C$, $C \prec D$ with A, B incomparable, both A,B,C,D and B,A,C,D are valid topological sorts — **a topological sort need not be unique**.

6.1.7 Isomorphism of posets

Definition. A bijection $f : A \to A'$ between posets is an **isomorphism** if $a \le b \iff f(a) \le f(b)$ for all $a, b \in A$ — the element names differ, but the order structure is identical.

Example (6.1 Ex. 17). $A = \{1,2,3,6\}$ under $|$ and $A' = \{\emptyset, \{a\}, \{b\}, \{a,b\}\}$ under $\subseteq$ are isomorphic via $1 \mapsto \emptyset,\ 2 \mapsto \{a\},\ 3 \mapsto \{b\},\ 6 \mapsto \{a,b\}$.

Key points

Partial order = reflexive + antisymmetric + transitive; a set with one is a **poset**. Pairs are **comparable** or not; if all pairs are, the order is **linear** (a chain). A **Hasse diagram** drops loops and transitive edges. A **topological sort** extends a partial order to a linear one, not necessarily uniquely. **Isomorphic** posets share order structure regardless of element names.

Example (Ex. 21–24: divisibility posets, linearly ordered?).

$A_1 = \{1,2,3,5,6,10,15,30\}$ not linear ($2 \nmid 3,\ 3 \nmid 2$)
$A_2 = \{2,4,8,16,32\}$ **linear** — a chain, each divides the next
$A_3 = \{3,6,12,36,72\}$ **linear** — a chain, each divides the next
$A_4 = \{1,\ldots,6,10,12,15,30,60\}$ not linear ($2 \nmid 3,\ 3 \nmid 2$)

A set of divisors forms a chain exactly when it can be listed so each divides the next; branching (two elements dividing a common one but not each other) breaks linearity.

6.2 Extremal elements of posets

Source: Kolman 6.2

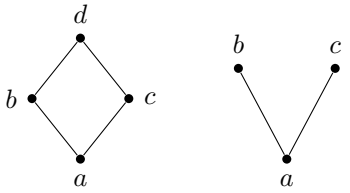
Definition. Let $(A, \leq)$ be a poset and $a \in A$. Going *up*:

- a is the **greatest** element if $x \leq a$ for all $x \in A$ — it is above everything;
- a is **maximal** if there is no $x \in A$ with $a < x$ — nothing is strictly above it.

Going *down*: a is the **least** element if $a \leq x$ for all $x \in A$, and **minimal** if there is no $x \in A$ with $x < a$.

Greatest vs. maximal

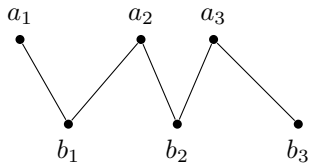
A greatest element is always maximal, but a maximal element need not be greatest. A greatest element, if it exists, is **unique** (Theorem 2); a poset may have **several** maximal elements, and then it has no greatest one. Left: d is greatest (and maximal). Right: b and c are both maximal, so there is no greatest element.



Reading a Hasse diagram: maximal = no edge leaves it upward; minimal = no edge leaves it downward.

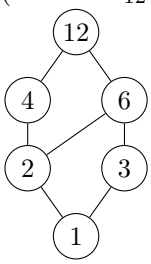
Not on the slides. In a *finite* poset, a greatest element exists exactly when there is only *one* maximal element (follow edges upward from anywhere and you must land on it). This fails for infinite posets.

Example (Ex. 1: a zigzag).



Maximal: a_1, a_2, a_3. Minimal: b_1, b_2, b_3. So there is no greatest and no least element *(answer not on slides)*.

Example (Ex. 2: $D_{12} = \{1, 2, 3, 4, 6, 12\}$ under divisibility).



Greatest element 12 (every element divides 12); least element 1 (1 divides everything).

6.2.1 Upper and lower bounds, LUB and GLB

Definition. Let $B \subseteq A$. An element $u \in A$ is an **upper bound** of B if $b \leq u$ for all $b \in B$; $\ell \in A$ is a **lower bound** of B if $\ell \leq b$ for all $b \in B$. The **least upper bound** $\mathrm{LUB}(B)$ (the **supremum**, $\sup B$) is the least of the upper bounds; the **greatest lower bound** $\mathrm{GLB}(B)$ (the **infimum**, $\inf B$) is the greatest of the lower bounds.

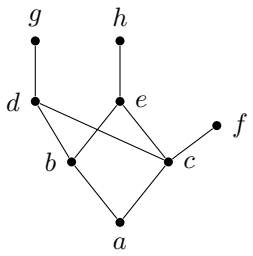
Bounds are elements of A, not necessarily of B. In a Hasse diagram: the upper bounds of B are the elements reachable by an upward path from *every* element of B.

Example (Ex. 3: $B = \{1, 2, 3\}$ in D_{12}). Upper bounds: 6 and 12, so $\mathrm{LUB}(B) = 6$. Lower bounds: just 1, so $\mathrm{GLB}(B) = 1$.

Extremal elements need not exist

A poset need not have a greatest or least element, and a subset need not have a LUB or GLB. $(\mathbb{Z}^+, \leq)$ has least element 1 but no greatest. A LUB can fail to exist in two ways: there are **no** upper bounds at all (Ex. 24 below), or there are upper bounds but **no least one** among them (Ex. 4 below: d and e are incomparable).

Example (Ex. 4).



Greatest: none. Least: a. Maximal: f, g, h. Minimal: a.
For $B = \{b, c\}$: upper bounds $\{d, e, g, h\}$; **no LUB**, since the two lowest upper bounds d and e are incomparable ($d \not\leq e$, $e \not\leq d$). Lower bounds $\{a\}$, so $\mathrm{GLB}(B) = a$.

Dual poset. Reversing the order, $(A, \leq) \to (A, \geq)$, swaps every "up" concept with its "down" twin: greatest ↔ least, maximal ↔ minimal, upper bound ↔ lower bound, LUB ↔ GLB. Prove one, get the other free.

6.2.2 Exam methods: greatest in a lexicographic order, bounds from a diagram, product order

Method for "which is greatest" in a lexicographic order:

1. Find the largest first coordinate among the candidates. Drop every candidate with a smaller one — they are all below.
2. Among the survivors (all with the same first coordinate), compare the second coordinates in B's order.
3. The greatest is the one whose second coordinate is above *all* the other survivors' second coordinates. If two survivors are incomparable and neither is beaten by a third, there is no greatest.

Example (Greatest element, lexicographic order). $\mathbb{Z}$ with $\leq$, $P(\{a, b\})$ with $\subseteq$, lexicographic order on the pairs. Which is greatest of $(1, \{a, b\})$, $(3, \{a\})$, $(3, \{b\})$, $(3, \{a, b\})$?

1. Largest first coordinate is 3: drop $(1, \{a, b\})$, despite its big set.
2. Survivors' second coordinates: $\{a\}$, $\{b\}$, $\{a, b\}$.
3. $\{a\} \subseteq \{a, b\}$ and $\{b\} \subseteq \{a, b\}$: $\{a, b\}$ is above both.

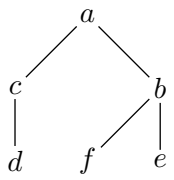
Greatest: $(3, \{a, b\})$. *Without* that candidate, $(3, \{a\})$ and $(3, \{b\})$ would both be maximal but incomparable — no greatest.

Method for "lower bounds of S" from a Hasse diagram:

1. Start at each element of S and walk *downwards* along the lines, collecting everything you reach.
2. **Include the element itself.** $x \leq x$, so every element of S is a bound of S whenever it is below all the others. For a one-element set $S = \{b\}$, the lower bounds are b and everything below b.
3. For a bigger S, keep only what is below *every* element of S. Upper bounds: the same, walking upwards.

Trap: answering "everything strictly below". That drops b itself — the usual wrong MC option.

Example (Lower bounds of $\{b\}$).



Below b: the lines go down to e and f. Add b itself: lower bounds of $\{b\}$ are b, e, f. Not d (d hangs under c, not b), and not a or c (above or beside b).

- **GLB** of $\{b\} = b$ (the greatest lower bound).
- Upper bounds of $\{b\}$: a, b; **LUB** $= b$.

e, f alone is the wrong answer — it forgets $b \leq b$.

Product order on pairs: $(x, y) \leq (x', y') \iff x \leq x'$ *and* $y \leq y'$. Picture points on a grid: "below" means down and to the left. $(1, 4)$ and $(3, 2)$ are incomparable — one coordinate is larger,

the other smaller. LUB of a set = (largest x, largest y), GLB = (smallest x, smallest y), each taken separately.

Example (Product order on $\mathbb{Z}^+ \times \mathbb{Z}^+$). $S = \{(1,2),(2,1),(2,3),(4,1),(1,5)\}$.

- **Minimal:** $(1,2)$ and $(2,1)$ — nothing in S is below either. Incomparable, so **no least element**.
- **Maximal:** $(2,3)$, $(4,1)$, $(1,5)$ — nothing in S is above any of them. $(1,2) \leq (2,3)$ and $(2,1) \leq (2,3)$, so those two are not maximal. Three incomparable maximal elements, so **no greatest element**.
- **LUB** $= (\max x, \max y) = (4,5)$, not in S — fine.
- **GLB** $= (\min x, \min y) = (1,1)$.

Trap: $(0,0)$ is below everything too, but $0 \notin \mathbb{Z}^+$ — it isn't in the poset, so it can't be a bound. Always check the ambient set. And if one element of S is $\leq$ all the others, it *is* the least element and the GLB at once; a least element is always unique.

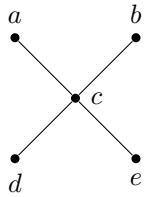
6.2.3 Topological sorting: algorithm SORT

In a finite poset you can always pick a minimal element (Theorem 1), so this loop terminates with every element listed:

1. Select a minimal element and append it to the ordering.
2. Remove it from the poset.
3. Repeat on what is left until nothing remains.

It is the textbook "Kahn's algorithm" for build-order/dependency resolution: repeatedly emit a node with no unmet prerequisites.

Example.



Minimal: d, e — take d. Then e is minimal; take it. Then c. Then a and b are both minimal; take b, then a. One ordering: d, e, c, b, a. Each tie is a free choice, so the answer is not unique.

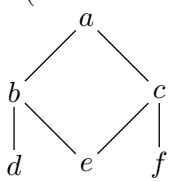
6.2.4 Exam method: which list is *not* a topological sorting?

Method for "which is *not* a topological sorting":

1. From the Hasse diagram, write down the edges as "lower < upper" (each line, bottom end first).
2. For each element with more than one line going *up* from it, note everything above it — these are the constraints most easily broken.
3. In each option, check that every element appears after everything directly below it. The first violation rules the option out.

Check which way the options run: if they all end with the top element, the list goes bottom-up.

Example (Which is *not* a topological sorting?).



Constraints: b after d, e; c after e, f; a last. e sits under *both* b and c, so look for e first.

- e, f, c, d, b, a — fine
- d, e, b, f, c, a — fine (b may come before f: they are incomparable)
- d, f, b, c, e, a — **not**: e comes after b and c

6.2.5 Theorems

6.2 Theorems 1–4

Th. 1. A finite nonempty poset has at least one maximal and at least one minimal element.
Th. 2. A poset has at most one greatest and at most one least element. *Proof*: if a, b are both greatest, then $a \le b$ and $b \le a$, so $a = b$ by antisymmetry.
Th. 3. A subset B has at most one LUB and at most one GLB. It says *if* they exist they are unique — not that they exist.
Th. 4. An isomorphism $f : (A, \le) \to (A', \le')$ ($x \le y \iff f(x) \le' f(y)$) carries maximal/minimal, greatest/least, upper/lower bounds and LUB/GLB to the same thing on the other side.

6.2.6 Exercises covered

Figure numbers are Kolman's; the slides redraw the same diagrams.
Ex. 1–4 (maximal/minimal) and **Ex. 9–12 (greatest/least)** *(answers not on slides)*:

Ex.	Maximal	Minimal	Ex.	Greatest	Least
1 (Fig. 27)	$3, 5$	$1, 6$	9 (Fig. 31)	f	a
2 (Fig. 28)	f, g	a, b, c	10 (Fig. 32)	e	none (a, b minimal)
3 (Fig. 29)	e, f	a	11 (Fig. 33)	none	none
4 (Fig. 30)	$4, 7$	$1, 8, 9$	12 (Fig. 34)	5	none (1, 3 minimal)

Ex. 23–26 (bounds of B):

Ex.	B	Upper bounds	Lower bounds	LUB	GLB
23 (Fig. 35)	$\{c, d, e\}$	$\{f, g, h\}$	$\{a, b, c\}$	f	c
24 (Fig. 36)	$\{1, 2, 3, 4, 5\}$	none	none	none	none
25 (Fig. 37)	$\{b, c, d\}$	$\{d, e, f\}$	$\{a, b\}$	d	b
26 (Fig. 38)	$\{3, 4, 6\}$	$\{5\}$	$\{1, 2, 3\}$	5	3

Ex. 29. $A = \mathbb{R}$ with $\le$, $B = (1, 2)$: upper bounds $[2, \infty)$, lower bounds $(-\infty, 1]$, $\mathrm{LUB}(B) = 2$, $\mathrm{GLB}(B) = 1$ — even though $1 \notin B$ and $2 \notin B$.
Ex. 33. SORT on Figure 35 (the poset of Ex. 23) gives, for example, a, b, c, d, e, f, g, h. Another valid ordering is b, a, c, e, d, f, h, g.

6.3 Lattices

Source: Kolman 6.3

Definition. A **lattice** is a poset $(L, \le)$ in which every pair a, b has both a LUB and a GLB, written

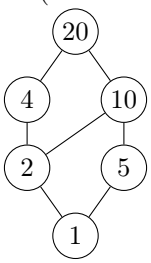
$$a \vee b = \mathrm{LUB}(a, b) \text{ (the } \textbf{join}\text{)}, \qquad a \wedge b = \mathrm{GLB}(a, b) \text{ (the } \textbf{meet}\text{)}.$$

In a Hasse diagram: $a \vee b$ = go *up* to the first element both reach; $a \wedge b$ = go *down* to the first element both reach.
Not on the slides. The symbols are chosen on purpose: in $(\mathcal{P}(S), \subseteq)$ join is $\cup$ and meet is $\cap$, and in logic they are OR and AND. That correspondence is where 6.4 (Boolean algebras) is heading.

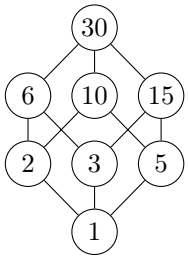
Example (Ex. 1: $(\mathcal{P}(S), \subseteq)$). For $A, B \subseteq S$: $A \cup B$ contains both and is the smallest set that does, so $A \vee B = A \cup B$; dually $A \wedge B = A \cap B$. Every pair has both, so $(\mathcal{P}(S), \subseteq)$ is a lattice. (The slide stops mid-sentence; this is the conclusion.)

Example (Ex. 2: $(\mathbb{Z}^+, |)$). An upper bound u of $\{a, b\}$ needs $a \mid u$ and $b \mid u$ — a common multiple — so $a \vee b = \mathrm{lcm}(a, b)$. A lower bound ℓ needs $\ell \mid a$, $\ell \mid b$, so $a \wedge b = \gcd(a, b)$. Every pair has both, so it is a lattice. For $\{4, 6\}$: upper bounds $12, 24, 36, \ldots$, LUB 12; lower bounds $1, 2$, GLB 2 *(answer not on slides)*.

Example (Ex. 3–4: divisor lattices D_{20} and D_{30}).



$2 \vee 5 = 10$
$4 \wedge 10 = 2$
$4 \vee 5 = 20$
$4 \wedge 5 = 1$



$2 \vee 3 = 6$
$6 \wedge 10 = 2$
$6 \vee 15 = 30$
$10 \wedge 15 = 5$
Lattice? **Yes** *(answer not on slides)*

Join is lcm, meet is gcd — read the diagram or just compute. Every D_n is a lattice, since gcd and lcm of two divisors of n are again divisors of n.

Common lattices at a glance (join, meet, bottom/top):

Lattice	Order	$a \vee b$	$a \wedge b$	0 / I
$P(S)$	$\subseteq$	$A \cup B$	$A \cap B$	$\emptyset$ / S
D_n (divisors of n)	$\mid$ (divides)	$\text{lcm}(a,b)$	$\gcd(a,b)$	1 / n
$\mathbb{Z}$ (any linear order)	$\leq$	$\max(a,b)$	$\min(a,b)$	none
$L_1 \times L_2$ product order	$\leq$ in both	per coord.	per coord.	$(0,0)$ / (I,I)
B_n (bit strings)	bit by bit	bitwise OR	bitwise AND	$0\ldots0$ / $1\ldots1$

Example (Product order on $\mathbb{Z} \times \mathbb{Z}$: $(x,y) \leq (x',y') \iff x \leq x'$ and $y \leq y'$. Find $(3,-2) \vee (-1,4)$ and $(3,-2) \wedge (-1,4)$.). Join takes the larger value in *each* coordinate, meet the smaller:

$$(3,-2) \vee (-1,4) = (\max(3,-1), \max(-2,4)) = (3,4), \qquad (3,-2) \wedge (-1,4) = (-1,-2).$$

Trap: the answer is usually *neither* of the two given pairs. Don't confuse this with the lexicographic order (first coordinate decides), and don't add the coordinates.

Example (In D_{12}: $4 \vee 6$ and $4 \wedge 6$). $4 \vee 6 = \text{lcm}(4,6) = 12$ and $4 \wedge 6 = \gcd(4,6) = 2$. See the gcd/lcm methods in 1.4.5.

6.3.1 Properties of lattices

Not on the slides. Kolman 6.3 pp. 267–268. These are the tools for proving anything about a lattice *in general* (no Hasse diagram to look at).
The definition of join and meet, written out as four facts:

(1) $a \leq a \vee b$ and $b \leq a \vee b$	($a \vee b$ is an upper bound)
(2) if $a \leq c$ and $b \leq c$, then $a \vee b \leq c$	(it is the *least* one)
(1′) $a \wedge b \leq a$ and $a \wedge b \leq b$	($a \wedge b$ is a lower bound)
(2′) if $c \leq a$ and $c \leq b$, then $c \leq a \wedge b$	(it is the *greatest* one)

6.3 Theorem 2 (order vs. join and meet)

For all a, b in a lattice L: (a) $a \vee b = b \iff a \leq b$; (b) $a \wedge b = a \iff a \leq b$; (c) $a \wedge b = a \iff a \vee b = b$.
So "$a \leq b$" can always be swapped for an equation, and back.

6.3 Theorem 3 (algebraic laws)

Idempotent	$a \vee a = a$	$a \wedge a = a$
Commutative	$a \vee b = b \vee a$	$a \wedge b = b \wedge a$
Associative	$a \vee (b \vee c) = (a \vee b) \vee c$	$a \wedge (b \wedge c) = (a \wedge b) \wedge c$
Absorption	$a \vee (a \wedge b) = a$	$a \wedge (a \vee b) = a$

These hold in *every* lattice. The distributive laws do *not* — see below.

6.3 Theorem 4 (monotonicity)

For all a, b, c, d in a lattice L:

1. If $a \leq b$, then (a) $a \vee c \leq b \vee c$ and (b) $a \wedge c \leq b \wedge c$.
2. $a \leq c$ and $b \leq c \iff a \vee b \leq c$.
3. $c \leq a$ and $c \leq b \iff c \leq a \wedge b$.
4. If $a \leq b$ and $c \leq d$, then (a) $a \vee c \leq b \vee d$ and (b) $a \wedge c \leq b \wedge d$.

Method: proving an equation $u = v$ in a general lattice. Either rewrite with the laws of Theorem 3 (and Theorem 2 to turn a given $\leq$ into an equation), or prove $u \leq v$ and $v \leq u$ separately and conclude $u = v$ by **antisymmetry** — each inequality from (1)–(2′) and Theorem 4. This is how Kolman proves Theorem 3 itself:

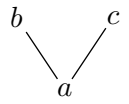
Example (Absorption $a \vee (a \wedge b) = a$ (Kolman's proof of Theorem 3, 4(a))). $\leq$: $a \wedge b \leq a$ by (1′) and $a \leq a$, so a is an upper bound of a and $a \wedge b$; by (2), $a \vee (a \wedge b) \leq a$.
$\geq$: $a \leq a \vee (a \wedge b)$ by (1).
Both directions, so $a \vee (a \wedge b) = a$ by antisymmetry.

Example (A two-line proof with Theorem 2: if $a \leq b$, then $a \vee (a \wedge b) = a \wedge b$). $a \leq b$ gives $a \wedge b = a$ (Theorem 2(b)). Then $a \vee (a \wedge b) = a \vee a = a = a \wedge b$ (idempotent).

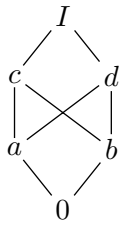
Watch out: a law that holds in D_n or $\mathcal{P}(S)$ need not hold in every lattice. To prove a statement about *all* lattices, use only the definition and Theorems 2–4; to disprove one, a single lattice (one Hasse diagram, one choice of elements) is enough.

6.3.2 Exam method: is this Hasse diagram a lattice?

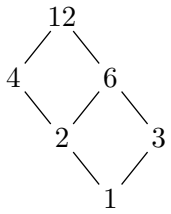
1. **Exactly one top and one bottom?** Every finite lattice has a greatest element I and a least element 0 (Kolman 6.3, Theorem 5). Two maximal elements at the top: **not** a lattice — those two have no upper bound at all.
2. **Every incomparable pair has a *unique* lowest common point above, and a unique highest common point below.** Comparable pairs (connected by an upward path) never fail. The failure to look for is the **bow-tie**: two elements that both sit below the same two incomparable elements.



two tops: no $b \vee c$



bow-tie: no $a \vee b$



D_{12}: a lattice

In the bow-tie, a and b have upper bounds c, d and I; c and d are both minimal among them and incomparable, so there is no *least* one. Note that it has a top and a bottom — step 1 is necessary, not sufficient.

6.3.3 Sublattice

A nonempty $S \subseteq L$ is a **sublattice** if it is closed under the lattice's own $\vee$ and $\wedge$: $a, b \in S \Rightarrow a \vee b \in S$ and $a \wedge b \in S$. Example: $(D_n, |)$ is a sublattice of $(\mathbb{Z}^+, |)$.

6.3 Theorem 1 (product of lattices)

If $(L_1, \leq)$ and $(L_2, \leq)$ are lattices, then $L_1 \times L_2$ with the product order, $(a_1, a_2) \leq (b_1, b_2) \iff a_1 \leq b_1$ and $a_2 \leq b_2$, is a lattice. (Join and meet are taken componentwise.)

Exam method: is S a sublattice?

- **Method:** for every incomparable pair in S, compute $a \vee b$ and $a \wedge b$ in L (for D_n: lcm and gcd) and check that both are in S. Comparable pairs pass automatically.
- **Trap:** S can be a lattice on its own and still not be a sublattice, because inside S the "join" may be a higher element than the real join in L (Kolman 6.3, Example 9).

Example (In D_{36}: are $\{1, 4, 9, 36\}$ and $\{2, 4, 6, 36\}$ sublattices?).

- $\{1, 4, 9, 36\}$: the only incomparable pair is $4, 9$: lcm $= 36 \in S$, gcd $= 1 \in S$. **Sublattice.**
- $\{2, 4, 6, 36\}$: pair $4, 6$: gcd $= 2 \in S$, but lcm $= 12 \notin S$. **Not a sublattice** — even though inside S alone, 4 and 6 have the least upper bound 36, so S by itself is a lattice.

6.3.4 Bounded lattice

L is **bounded** if it has a least element 0 and a greatest element I, so $0 \leq a \leq I$ for all a. Then

$$a \vee 0 = a, \qquad a \wedge 0 = 0, \qquad a \vee I = I, \qquad a \wedge I = a.$$

Not on the slides. Every finite lattice is bounded (Kolman 6.3 Theorem 5): $I = a_1 \vee \cdots \vee a_n$ and $0 = a_1 \wedge \cdots \wedge a_n$.

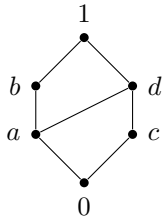
6.3.5 Distributive lattice

L is **distributive** if for all a, b, c:

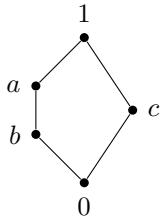
$$a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c) \quad \text{and} \quad a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c).$$

To show a lattice is *not* distributive, **one counterexample is enough**: one triple a, b, c where either identity fails. $\mathcal{P}(S)$, every D_n and every linear order are distributive.

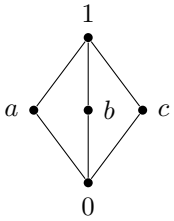
Example (6.3 Ex. 17 and 18: distributive or not?).



Ex. 17: distributive



Ex. 18a: pentagon N_5



Ex. 18b: diamond M_3

Ex. 17 satisfies both distributive laws for every triple. (Not on the slides: it is the 3×2 grid, the product of a 3-chain and a 2-chain, and a product of chains is always distributive.)
Ex. 18a: $a \wedge (b \vee c) = a \wedge 1 = a$, but $(a \wedge b) \vee (a \wedge c) = b \vee 0 = b \neq a$. Not distributive.
Ex. 18b: $a \wedge (b \vee c) = a \wedge 1 = a$, but $(a \wedge b) \vee (a \wedge c) = 0 \vee 0 = 0 \neq a$. Not distributive.

Not on the slides: where to find the counterexample (Kolman 6.3 Theorem 6)

A lattice is nondistributive **if and only if** it contains a sublattice shaped like one of the two in Ex. 18: the **pentagon** N_5 or the **diamond** M_3. So scan the Hasse diagram for an element with three upper covers meeting again above (diamond), or a square where one side has an extra element (pentagon). Then use its labels as the triple: the element on the long side of the pentagon, or any middle element of the diamond, as a in $a \wedge (b \vee c)$.

6.3.6 Modular lattice

Not on the slides. Not in the lectures or the text proper: Kolman 6.3 Exercises 26 and 39 (p. 272).

Definition. A lattice $(L, \leq)$ is **modular** if for all $a, b, x \in L$:

$$a \leq b \implies a \vee (x \wedge b) = (a \vee x) \wedge b.$$

Reading it. Only pairs with $a \le b$ are tested, and for them the two ways of "combining a, x and b" must agree: on the left, cut x down below b first and then join with a; on the right, join x with a first and then cut down below b. Set it side by side with the second distributive law $a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c)$: the shapes are close, and the modular law is the weaker demand of the two. (Kolman's Exercise 26 writes the same law with $a \le c$ and b in the middle.)

What Kolman asks (p. 272), no proofs given: Ex. 26(a) every **distributive** lattice is modular; Ex. 26(b) the diamond in Figure 52 (M_3) is **nondistributive but modular**; Ex. 39 prove or disprove: a sublattice of a modular lattice is modular.

Method: checking the modular law.

- **To disprove:** one triple $a \le b$, x where the two sides differ is enough. Compute both sides from the Hasse diagram (go up for $\vee$, down for $\wedge$).
- **To prove for a given finite lattice:** the law must hold for *every* triple with $a \le b$. List the pairs $a \le b$, and for each one the possible x; use Theorems 2–4 to dispose of whole families of cases at once rather than computing every triple by hand. State clearly why each family is covered.
- **To prove for a whole class** (e.g. all distributive lattices): work with letters only — the given laws plus Theorems 2–4, never a picture.

Useful inequality (proved here, so it can be cited by this proof)

In any lattice, if $a \le b$ then

$$a \vee (x \wedge b) \;\le\; (a \vee x) \wedge b.$$

Proof. $a \le a \vee x$ by (1) and $a \le b$, so $a \le (a \vee x) \wedge b$ by (2′). Also $x \wedge b \le x \le a \vee x$ and $x \wedge b \le b$, so $x \wedge b \le (a \vee x) \wedge b$ by (2′). Both a and $x \wedge b$ are below $(a \vee x) \wedge b$, so their join is too, by (2). □
Consequence: to show the modular law for a triple you only need the *other* direction, "$\ge$".

Example (The mechanics in D_{12} (a distributive lattice)). Take $a = 2 \le b = 12$ and $x = 3$. Left: $2 \vee (3 \wedge 12) = \mathrm{lcm}(2, \gcd(3, 12)) = \mathrm{lcm}(2, 3) = 6$. Right: $(2 \vee 3) \wedge 12 = \gcd(\mathrm{lcm}(2, 3), 12) = \gcd(6, 12) = 6$. Equal. With $a = 2 \le b = 4$, $x = 6$: left $2 \vee \gcd(6, 4) = 2 \vee 2 = 2$; right $\gcd(\mathrm{lcm}(2, 6), 4) = \gcd(6, 4) = 2$. Equal again.

6.3.7 Complement

In a bounded lattice, a' is a **complement** of a if

$$a \vee a' = I \quad \text{(together they reach the top)} \qquad \text{and} \qquad a \wedge a' = 0 \quad \text{(they share only the bottom)}.$$

Not on the slides. L is **complemented** if it is bounded and *every* element has at least one complement (Kolman 6.3) — needed for Ex. 28–29 below. In a nondistributive lattice complements can be non-unique: in M_3, a has both b and c as complements.

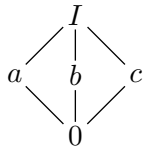
Example (Ex. 19: $L = \mathcal{P}(\{x, y\})$). Here $0 = \emptyset$, $I = \{x, y\}$, $\vee = \cup$, $\wedge = \cap$. Since $\{x\} \cup \{y\} = I$ and $\{x\} \cap \{y\} = \emptyset = 0$, the complement of $\{x\}$ is $\{y\}$. All of them: $\emptyset' = \{x, y\}$, $\{x\}' = \{y\}$, $\{y\}' = \{x\}$, $\{x, y\}' = \emptyset$ — exactly the ordinary set complement.

6.3 Theorem 7

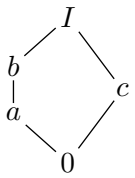
In a bounded distributive lattice, a complement, if it exists, is unique — so we can write it a' unambiguously.

Slide erratum. The slide says only "a distributive lattice". Complements are only defined in a *bounded* lattice, and Kolman's Theorem 7 says "bounded distributive".

Complements, in practice. Always $0' = I$ and $I' = 0$; in a lattice with two or more elements nothing is its own complement. An element can have **no**, **one** or **several** complements; in a bounded *distributive* lattice at most one.



diamond M_3



pentagon N_5

Example (Complements in the pentagon N_5 above). c': try every element. $a \vee c = I$ and $a \wedge c = 0$; the same for b. So c has **two** complements, a and b; and $a' = c$, $b' = c$. Two complements for one element prove that N_5 is not distributive (Theorem 7). In M_3, each of a, b, c has the other two as complements.

Complements in D_n: the only candidate for d' is n/d (since gcd = 1 and lcm = n force $d \cdot d' = n$). It works exactly when $\gcd(d, n/d) = 1$.

Example (Complements of 5 and 2 in D_{30} and in D_{20}). D_{30}: $5' = 30/5 = 6$, and $\gcd(5, 6) = 1$ — yes. D_{20}: $5' = 4$, $\gcd(5, 4) = 1$ — yes; but for 2 the candidate is 10 and $\gcd(2, 10) = 2 \neq 1$ — 2 has **no complement**, so D_{20} is not complemented.

6.3.8 Exercises covered

Ex. 1 (lattice?) *(answers not on slides).* Diagram 1 (Fig. 45): **yes** — it is a 3×3 grid, the product of two 3-element chains, so a lattice by Theorem 1. Diagram 2: **no** — e, f are both maximal, so $e \vee f$ does not exist (and a, b minimal, so no $a \wedge b$). Diagram 3: **no** — a, b are both minimal, so $a \wedge b$ does not exist.

Ex. 7. $A = \{2, 3, 6, 12, 24, 36, 72\}$ under $|$: **not a lattice**. The pair $2, 3$ has no lower bound in A (1 is missing), so $2 \wedge 3$ does not exist. (Every other pair is fine: $24 \vee 36 = 72$, $24 \wedge 36 = 12$.)
Slide erratum. The solution slide's bullets say "every pair has a unique LUB" and "every pair has a unique GLB", then conclude "not a lattice" — contradictory. The first bullet is true; the second is false, because of $2 \wedge 3$, and that is the reason it fails.

Ex. 28–29 (distributive? complemented?) Both are **neither**. The slides work Ex. 28, with the same triple c, f, g and the same complement argument for c; Ex. 29 is *(answer not on slides).*

- **28** (Fig. 53; $a < b < c < e$, $a < d$, $d < c, f, g < e$): d has three upper covers c, f, g meeting at e — a diamond $\{d, c, f, g, e\}$. $c \wedge (f \vee g) = c \wedge e = c$ but $(c \wedge f) \vee (c \wedge g) = d \vee d = d$: not distributive. c has no complement (every element is either below c or above d, so $c \vee x = c$ or $c \wedge x \geq d$): not complemented.
- **29** (Fig. 54; $a < b < c < d < f$, $a < e < d$): $\{a, b, c, d, e\}$ is a pentagon. $c \wedge (b \vee e) = c \wedge d = c$ but $(c \wedge b) \vee (c \wedge e) = b \vee a = b$: not distributive. Only f joins with anything to give $I = f$, and $b \wedge f = b \neq a$, so b has no complement: not complemented.

Key points (6.2–6.3)

Greatest/least (above/below everything, unique if it exists) vs. **maximal/minimal** (nothing strictly above/below, can be several). **Upper/lower bounds** of B live in A; **LUB/GLB** are unique if they exist, but may not. **SORT**: repeatedly remove a minimal element. A **lattice** has $a \vee b$ (join, LUB) and $a \wedge b$ (meet, GLB) for every pair; $\mathcal{P}(S)$: $\cup, \cap$; divisibility: lcm, gcd. **Sublattice**: closed under $\vee, \wedge$. **Bounded**: has 0 and I. **Distributive**: one counterexample disproves it. **Complement**: $a \vee a' = I$, $a \wedge a' = 0$; unique in a bounded distributive lattice.

6.4 Finite Boolean algebras

Source: Kolman 6.4

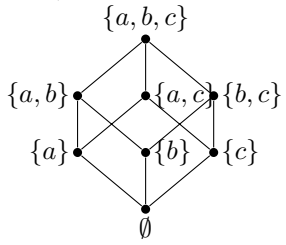
Three ideas from 6.3 — **lattice**, **complement**, **distributive** — combine into one structure, the **Boolean algebra**.

6.4 Theorem 1

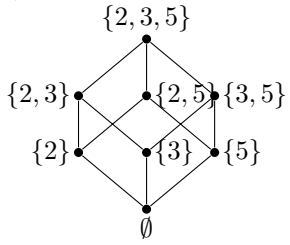
If $S_1 = \{x_1, \ldots, x_n\}$ and $S_2 = \{y_1, \ldots, y_n\}$ have the same number of elements, then $(\mathcal{P}(S_1), \subseteq)$ and $(\mathcal{P}(S_2), \subseteq)$ are isomorphic lattices.

Why: pair the elements up, $x_i \leftrightarrow y_i$. That induces a map on subsets, e.g. $f(\{x_1, x_3\}) = \{y_1, y_3\}$, and it preserves inclusion: $A \subseteq B \iff f(A) \subseteq f(B)$. Only the *number* of elements matters, not what they are — so call the lattice of subsets of any n-element set B_n.

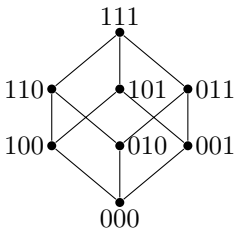
Example (6.4 Ex. 1–2: three faces of B_3).



(a) $\mathcal{P}(\{a, b, c\})$



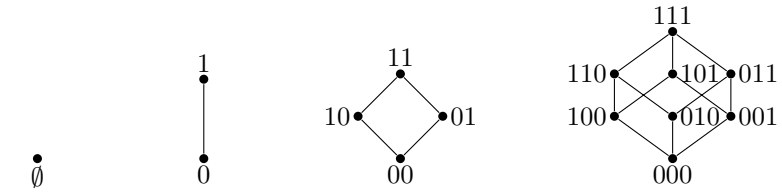
(b) $\mathcal{P}(\{2, 3, 5\})$



(c) bit strings of length 3

Same Hasse diagram, different labels. In (c) bit k says whether element k is in the subset, and $\leq$ is "bit by bit".

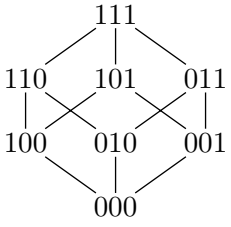
Definition. A finite lattice is a **Boolean algebra** if it is isomorphic to B_n for some $n \geq 0$. Since $|B_n| = 2^n$, **a finite Boolean algebra always has 2^n elements** — a quick test to rule one out.



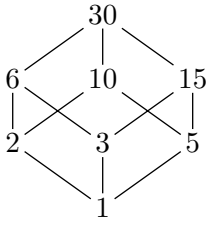
Properties: a Boolean algebra is **bounded**, **complemented** and **distributive**. So every element has a complement (unique, by 6.3 Theorem 7) and both distributive laws hold.

6.4.1 Exam method: is it a Boolean algebra?

B_n is the set of bit strings of length n, ordered bit by bit ($x \leq y$ when every bit of x is $\leq$ the same bit of y). Then $x \vee y$ is **bitwise OR**, $x \wedge y$ is **bitwise AND**, and the complement x' is **bitwise NOT**. This is exactly $P(S)$ with $|S| = n$, each subset stored as a bit mask (bit k is 1 when element k is in the set) — a `[Flags]` enum: $\cup$ is `|`, $\cap$ is `&`, complement is `~`.



B_3



D_{30}, same shape

The isomorphism $D_{30} \to B_3$: one bit per prime $(2, 3, 5)$ — $1 \mapsto 000$, $2 \mapsto 100$, $6 \mapsto 110$, $30 \mapsto 111$, and so on.

Is L a Boolean algebra? Ways to say no, quickest first:

1. $|L|$ is not a power of 2 $(1, 2, 4, 8, 16, \ldots)$. No Boolean algebra has 3, 5, 6 or 7 elements.
2. Some element has no complement, or two complements.
3. It contains a diamond M_3 or pentagon N_5 (not distributive).

To say **yes**: match the diagram to B_n (for 8 elements, the cube above).

6.4.2 Which D_n are Boolean algebras?

Example (6.4 Ex. 3: D_{20} and D_{30}). $D_{20} = \{1, 2, 4, 5, 10, 20\}$ has 6 elements, and $6 \neq 2^n$ for any n: **not** a Boolean algebra $(20 = 2^2 \cdot 5)$.
$D_{30} = \{1, 2, 3, 5, 6, 10, 15, 30\}$ has $8 = 2^3$ elements — the right count, so it *may* be one. It is: $D_{30} \cong B_3$, and is a Boolean algebra.
Not on the slides. The count alone doesn't prove it — Ex. 6 below has 8 elements and fails. What settles D_{30} is Theorem 2: $30 = 2 \cdot 3 \cdot 5$.

6.4 Theorem 2

If $n = p_1 p_2 \cdots p_k$ with $p_1, \ldots, p_k$ **distinct** primes, then D_n is a Boolean algebra, isomorphic to B_k.

Not on the slides. The converse holds too (Kolman 6.4 Example 8): if $p^2 \mid n$ for some prime p, then p has no complement in D_n, so D_n is *not* a Boolean algebra. Together: D_n is a Boolean algebra $\iff$ no prime divides n twice. E.g. D_{40} $(40 = 2^3 \cdot 5)$ and D_{75} $(75 = 3 \cdot 5^2)$ are not.

Rule: D_n is a Boolean algebra $\iff$ n is **square-free** — a product of *distinct* primes, no prime appearing twice in the factorization. (Kolman 6.4, Theorem 2 and Example 8.)

Method: prime-factorize n (trial division, see 1.4.2) and look for any exponent ≥ 2. Watch for 4, 9, 25, 49 hiding inside.

Why: each divisor is a choice of which primes to include, so divisors $\leftrightarrow$ subsets of the primes, lcm $= \cup$ and gcd $= \cap$. With $p^2 \mid n$, the divisor p has no complement: the only candidate n/p still contains a factor p, so $\gcd(p, n/p) \neq 1$.

Example (Which of D_{78}, D_{98}, D_{91}, D_{60}, D_{17} are Boolean algebras?).

- $78 = 2 \cdot 3 \cdot 13$ — distinct primes, **yes** $(\cong B_3$, 8 divisors)
- $98 = 2 \cdot 7^2$ — **no**
- $91 = 7 \cdot 13$ — **yes** $(\cong B_2)$. 91 looks prime but isn't.
- $60 = 2^2 \cdot 3 \cdot 5$ — **no** (12 divisors, not a power of 2)
- 17 is prime: $D_{17} = \{1, 17\} \cong B_1$ — **yes**. A prime is square-free; don't skip it.

Example (6.4 Ex. 5: D_{210}, D_{66}, D_{646}). $210 = 2 \cdot 3 \cdot 5 \cdot 7$, $66 = 2 \cdot 3 \cdot 11$, $646 = 2 \cdot 17 \cdot 19$ — all distinct primes, so $D_{210} \cong B_4$, $D_{66} \cong B_3$, $D_{646} \cong B_3$: all three are Boolean algebras.

6.4.3 Shape of the Hasse diagram of D_n (exam method)

Factorize $n = p_1^{e_1} \cdots p_k^{e_k}$. Each prime is one "direction": its exponent runs $0, 1, \ldots, e_i$, a chain of $e_i + 1$ elements, and D_n is the *product* of these chains (join = lcm, meet = gcd, as for any product order). So

- **dots** = number of divisors $= (e_1 + 1) \cdots (e_k + 1)$;
- **levels** $= e_1 + \cdots + e_k + 1$ (the height from 1 up to n);
- one prime $\Rightarrow$ a single vertical **chain**.

n looks like	shape of D_n	example
p	chain of 2 dots	$D_7 = \{1, 7\}$
p^2	chain of 3 dots	$D_{25} = \{1, 5, 25\}$, D_9, D_{49}
p^3	chain of 4 dots	D_8, D_{27}
pq	diamond (square, 4 dots)	D_6, D_{15}, D_{91}
p^2q	3×2 grid, 6 dots	D_{12}, D_{20}, D_{18}
p^2q^2	3×3 grid, 9 dots	D_{36}
pqr	cube B_3, 8 dots	D_{30}, D_{42}, D_{66}

A prime power gives a chain, so D_{25} is *not* a diamond: $\{1, 5, 25\}$ has $1 \mid 5 \mid 25$ and nothing incomparable. **To match an MC picture:** count the dots first (that is $d(n)$), then check chain, grid or cube. Only the cube-shaped 8-dot diagram is Boolean (n square-free), while an 8-dot chain (e.g. D_{128}) or the 4×2 grid (D_{24}) is not.

Example (MC practice: $(\mathcal{P}(\{1, 2, 3\}), \subseteq)$ vs. D_{30}, D_9; which statement is false?). $\mathcal{P}(S)$ with $|S| = k$ is B_k, and $D_n \cong B_k$ exactly when n is a product of k *distinct* primes. So with $|S| = 3$: D_{30} ($2 \cdot 3 \cdot 5$), D_{42}, D_{66} all match, with $2^3 = 8$ elements. $D_9 = \{1, 3, 9\}$ ($9 = 3^2$) is a 3-element *chain*: not isomorphic, and the false statement.
In any lattice the words change but the operations do not: **LUB = join** $= \cup$, **GLB = meet = "gcd"** $= \cap$ in a power set (gcd is only the name of the meet in D_n). So $\text{LUB}(\{1\}, \{2, 3\}) = \{1, 2, 3\}$ and $\text{GLB}(\{1\}, \{2, 3\}) = \emptyset$ are both true, and $|\mathcal{P}(S)| = 2^3 = 8$.

6.4.4 Fast square-free check (no factoring needed)

To decide Boolean or not you do *not* need the factorization, only whether some prime square divides n. Test the squares of primes with $p^2 \leq n$, each by a quick rule:

square	divides n when ...	only relevant when
4	last two digits are a multiple of 4	always
9	digit sum is a multiple of 9	$n \geq 9$
25	ends in 00, 25, 50 or 75	$n \geq 25$
49	n is a multiple of 49 $(49, 98, 147, 196, \dots)$	$n \geq 49$
121	multiples $121, 242, 363, \dots$	$n \geq 121$
169	multiples $169, 338, \dots$	$n \geq 169$

If none of them divides n, then n is square-free and D_n *is* a Boolean algebra. For $n < 49$ only 4, 9 and 25 can matter.

Not square-free, $n \leq 100$ (so D_n is not Boolean)

Every multiple of 4: $4, 8, 12, \dots, 96, 100$.
Plus the odd multiples of 9: $9, 27, 45, 63, 81, 99$, and the even ones not already above: $18, 54, 90$.
Plus $25, 50, 75$ and $49, 98$.
Every other $n \leq 100$ is square-free, so D_n *is* Boolean.

Example ($D_{37}, D_{38}, D_{39}, D_{40}, D_{41}, D_{42}$). Only 4, 9, 25 can divide these. 37, 41: odd, digit sums 10 and 5, no. 38: last two digits 38, not a multiple of 4; digit sum 11. 39: digit sum 12, not 9. 42: 42 is not a multiple of 4, digit sum 6. But $40 = 4 \cdot 10$, so D_{40} is the one that is **not** Boolean ($40 = 2^3 \cdot 5$).

6.4.5 Lookup: every D_n seen in questions and examples

Boolean $\iff$ n square-free; dots = number of divisors; shape from the exponents (see the two subsections before).

D_n	$n =$	dots	Boolean?	shape
D_9	3^2	3	**no**	chain of 3
D_{12}	$2^2 \cdot 3$	6	**no**	3×2 grid
D_{17}	17	2	**yes** (B_1)	chain of 2
D_{20}	$2^2 \cdot 5$	6	**no**	3×2 grid
D_{25}	5^2	3	**no**	chain of 3
D_{30}	$2 \cdot 3 \cdot 5$	8	**yes** (B_3)	cube B_3
D_{36}	$2^2 \cdot 3^2$	9	**no**	3×3 grid
D_{37}	37	2	**yes** (B_1)	chain of 2
D_{38}	$2 \cdot 19$	4	**yes** (B_2)	diamond
D_{39}	$3 \cdot 13$	4	**yes** (B_2)	diamond
D_{40}	$2^3 \cdot 5$	8	**no**	4×2 grid
D_{41}	41	2	**yes** (B_1)	chain of 2
D_{42}	$2 \cdot 3 \cdot 7$	8	**yes** (B_3)	cube B_3
D_{60}	$2^2 \cdot 3 \cdot 5$	12	**no**	$3 \times 2 \times 2$ product of chains
D_{63}	$3^2 \cdot 7$	6	**no**	3×2 grid
D_{66}	$2 \cdot 3 \cdot 11$	8	**yes** (B_3)	cube B_3
D_{75}	$3 \cdot 5^2$	6	**no**	3×2 grid
D_{78}	$2 \cdot 3 \cdot 13$	8	**yes** (B_3)	cube B_3
D_{84}	$2^2 \cdot 3 \cdot 7$	12	**no**	$3 \times 2 \times 2$ product of chains
D_{90}	$2 \cdot 3^2 \cdot 5$	12	**no**	$3 \times 2 \times 2$ product of chains
D_{91}	$7 \cdot 13$	4	**yes** (B_2)	diamond
D_{92}	$2^2 \cdot 23$	6	**no**	3×2 grid
D_{98}	$2 \cdot 7^2$	6	**no**	3×2 grid
D_{105}	$3 \cdot 5 \cdot 7$	8	**yes** (B_3)	cube B_3
D_{210}	$2 \cdot 3 \cdot 5 \cdot 7$	16	**yes** (B_4)	B_4
D_{646}	$2 \cdot 17 \cdot 19$	8	**yes** (B_3)	cube B_3

6.4.6 Calculating in a Boolean algebra

6.4 Theorem 4: Boolean algebra $\leftrightarrow$ set algebra

Boolean algebra	$a \vee b$	$a \wedge b$	a'	0	I
Set algebra	$A \cup B$	$A \cap B$	$\overline{A}$	$\emptyset$	S

In a Boolean algebra $\vee$, $\wedge$, $'$ behave exactly like $\cup$, $\cap$, complement. So any set identity carries over — useful when simplifying Boolean expressions.

Example (6.4 Ex. 6). For $x, y \in L$, a Boolean algebra: **involution** $(x')' = x$; **De Morgan** $(x \wedge y)' = x' \vee y'$ and $(x \vee y)' = x' \wedge y'$.
Proof: the matching set laws $\overline{\overline{A}} = A$, $\overline{A \cap B} = \overline{A} \cup \overline{B}$, $\overline{A \cup B} = \overline{A} \cap \overline{B}$ hold, and Theorem 4 translates them.

Boolean properties (from the slides; $x, y, z \in L$, and the primed column is the set version in $\mathcal{P}(S)$):

1. $x \le y \iff x \vee y = y$	$1'$. $A \subseteq B \iff A \cup B = B$
2. $x \le y \iff x \wedge y = x$	$2'$. $A \subseteq B \iff A \cap B = A$
3. $x \vee x = x,\ x \wedge x = x$	$3'$. $A \cup A = A,\ A \cap A = A$
4. $x \vee y = y \vee x,\ x \wedge y = y \wedge x$	$4'$. $A \cup B = B \cup A,\ A \cap B = B \cap A$
5. $x \vee (y \vee z) = (x \vee y) \vee z$	$5'$. $A \cup (B \cup C) = (A \cup B) \cup C$
$x \wedge (y \wedge z) = (x \wedge y) \wedge z$	$A \cap (B \cap C) = (A \cap B) \cap C$
6. $x \vee (x \wedge y) = x,\ x \wedge (x \vee y) = x$	$6'$. $A \cup (A \cap B) = A,\ A \cap (A \cup B) = A$
7. $0 \le x \le I$ for all x	$7'$. $\emptyset \subseteq A \subseteq S$
8. $x \vee 0 = x,\ x \wedge 0 = 0$	$8'$. $A \cup \emptyset = A,\ A \cap \emptyset = \emptyset$
9. $x \vee I = I,\ x \wedge I = x$	$9'$. $A \cup S = S,\ A \cap S = A$
10. $x \wedge (y \vee z) = (x \wedge y) \vee (x \wedge z)$	$10'$. $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
$x \vee (y \wedge z) = (x \vee y) \wedge (x \vee z)$	$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
11. unique x': $x \vee x' = I,\ x \wedge x' = 0$	$11'$. unique $\overline{A}$: $A \cup \overline{A} = S,\ A \cap \overline{A} = \emptyset$

3 idempotent, 4 commutative, 5 associative, 6 absorption, 10 distributive, 11 complement — plus involution and De Morgan from Ex. 6.

Substitution rule (Kolman 6.4, Theorem 3): any identity for sets using $\cup$, $\cap$ and complement holds in every Boolean algebra with $\vee$ for $\cup$ and $\wedge$ for $\cap$. They are also the laws of propositional logic and of `||`, `&&`, `!` on `bool` ($= B_1$).
Every law comes in a pair: swap $\vee \leftrightarrow \wedge$ and $0 \leftrightarrow I$ to get the other one (*duality*). Note that *both* distributive laws hold — unlike ordinary arithmetic, where $a + bc \neq (a + b)(a + c)$.

Absorption laws (exam method).

$$x \vee (x \wedge y) = x \qquad\qquad x \wedge (x \vee y) = x$$

Recognise it: the same variable stands *alone* outside the brackets and again *inside*, with the *other* operator in the brackets. The whole bracket vanishes — y can be anything, even a long expression.

Why it works — three ways to see it:
- *Sets:* $A \cap B \subseteq A$, so adding it to A adds nothing: $A \cup (A \cap B) = A$. Dually, $A \subseteq A \cup B$, so $A \cap (A \cup B) = A$.
- *Order:* $x \wedge y \le x$, and the order law "$a \le b \iff a \vee b = b$" gives $x \vee (x \wedge y) = x$.
- *Code:* `x || (x && y)`: if `x` is true the result is true; if `x` is false, `x && y` is false too, so the result is false. Either way it equals `x`.

Check by truth table (the proof that always works):

x	y	$x \wedge y$	$x \vee (x \wedge y)$	$x \vee y$	$x \wedge (x \vee y)$
0	0	0	0	0	0
0	1	0	0	1	0
1	0	0	1	1	1
1	1	1	1	1	1

Both result columns equal the x column.

Example (Absorption with bigger pieces).
- $(x \wedge y) \vee (x \wedge y \wedge z) = x \wedge y$ — take $x \wedge y$ as "the variable": it appears alone, and again inside $(x \wedge y) \wedge z$.
- $z' \wedge (z' \vee x \vee y) = z'$ — the alone part is z'; the bracket is $z' \vee$ (anything).
- $(x \vee y) \wedge \big(x \vee y \vee (w \wedge z)\big) = x \vee y$.

Trap — the look-alike with a complement: $x \vee (x' \wedge y)$ has x' inside, not x, so absorption does *not* apply. It simplifies to $x \vee y$, not x — see the next example. Similarly $x \wedge (x' \vee y) = x \wedge y$.

Example (Simplify $x \vee (x' \wedge y)$).

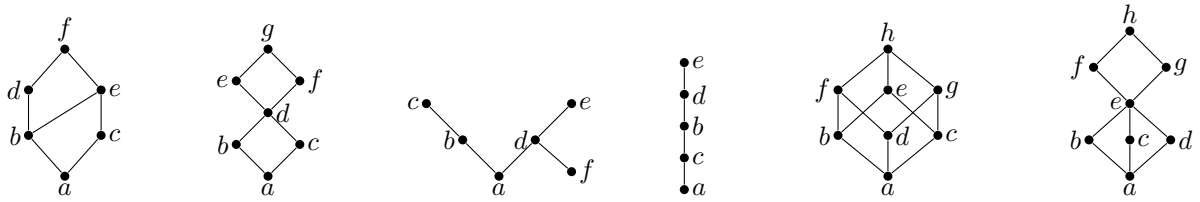
$$x \vee (x' \wedge y) \overset{\text{distr.}}{=} (x \vee x') \wedge (x \vee y) \overset{\text{compl.}}{=} I \wedge (x \vee y) \overset{\text{bounds}}{=} x \vee y.$$

Name the law at each step, as in a proof. The usual trick is the distributive law that turns a complement pair $x \vee x'$ into I.

Atoms: the elements directly above 0 (in B_n the strings with exactly one 1; in D_n the primes). B_n has n atoms, and every element is the join of the atoms below it — like every subset being the union of its one-element subsets.

6.4.7 Exercises covered

6.4 Ex. 1–6: is the poset a Boolean algebra?



1. **No** — 6 elements, $6 \neq 2^n$.
2. **No** — 7 elements, $7 \neq 2^n$.
3. **No** — not even a lattice: c and e have no common upper bound, so $c \vee e$ does not exist.
4. **No** — a lattice (every chain is), but 5 elements, $5 \neq 2^n$.
5. **Yes** — $8 = 2^3$ elements, and the diagram has the same structure as B_3, so $L \cong B_3$.
6. **No** — also 8 elements, which shows that having 2^n elements is not enough on its own.

Slide erratum. For Ex. 6 the slide argues that b has two complements, c and d, "since $b \wedge c = a$, $b \vee c = e$". But the top of this lattice is h, not e, so $b \vee c = e$ makes c *no* complement of b. The right argument: b has **no complement at all** — every element is either $\leq e$ (then $b \vee x \leq e \neq h$) or $\geq e$ (then $b \wedge x = b \neq a$). So the lattice isn't complemented and can't be a Boolean algebra. The slide's calculation *is* valid inside the lower diamond $\{a, b, c, d, e\}$, where e is the top: there b has two complements, c and d. That diamond is M_3, so the lattice isn't distributive either — a second valid reason.

6.5 Functions on Boolean algebras

Source: Kolman 6.5

Definition. With $B = \{0, 1\}$, a **Boolean function** is a function $f : B^n \to B$: the input is a bit string of length n, the output a single bit. E.g. $f : B^3 \to B$ takes (x_1, x_2, x_3) and returns 0 or 1.

A Boolean function is completely described by its **truth table**, one row per input, so 2^n rows. Many different polynomials give the same function; simplifying with the laws means fewer gates in the circuit.

Not on the slides. The slide breaks off at "There are". Each of the 2^n rows can be 0 or 1, so there are 2^{2^n} Boolean functions of n variables — 16 for $n = 2$, 256 for $n = 3$.

Definition. Boolean polynomials (Boolean expressions) in $x_1, \ldots, x_n$ are built by these rules: (1) each variable x_i is one; (2) the constants 0 and 1 are; (3) if p and q are, so are $p \vee q$ and $p \wedge q$; (4) if p is, so is p'.

Example: from x, y, z build $x \wedge y$, then $(x \wedge y)'$, then $(x \wedge y)' \vee z$ — each step uses one rule on polynomials already built.

Simplification rules	
$x \vee 0 = x, \quad x \wedge 1 = x$	$x \vee 1 = 1, \quad x \wedge 0 = 0$
$x \vee x = x, \quad x \wedge x = x$	$x \vee x' = 1, \quad x \wedge x' = 0$
Absorption:	$x \vee (x \wedge y) = x, \quad x \wedge (x \vee y) = x$
De Morgan:	$(x \vee y)' = x' \wedge y', \quad (x \wedge y)' = x' \vee y'$

Truth table method: 2^n rows, inputs listed by counting in binary $(000, 001, \ldots, 111)$; one helper column per subexpression, innermost first; the last column is f.

Example (Truth table of $p(x, y, z) = (x' \vee y) \wedge z$).

x	y	z	x'	$x' \vee y$	f
0	0	0	1	1	0
0	0	1	1	1	1
0	1	0	1	1	0
0	1	1	1	1	1
1	0	0	0	0	0
1	0	1	0	0	0
1	1	0	0	1	0
1	1	1	0	1	1

Shortcut for checking: $\wedge z$ means every row with $z = 0$ gives 0 — fill those first, then work out only the $z = 1$ rows.

Example (6.5 Ex. 1).

1a. $p_1 = (x \vee y) \wedge z$:

x	y	z	p_1
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1

1b. $p_2 = (x \vee y') \vee (y \wedge 1)$. Since $y \wedge 1 = y$, $p_2 = x \vee y' \vee y = x \vee 1 = \mathbf{1}$ — always true.

1c. $p_3 = x \vee (y' \wedge z) \vee (x \wedge y) = \big(x \vee (x \wedge y)\big) \vee (y' \wedge z) = \mathbf{x} \vee (\mathbf{y}' \wedge \mathbf{z})$ by absorption.

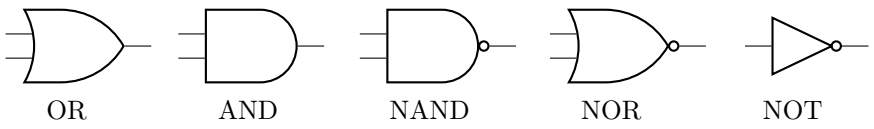
1d. $p = (x_1 \wedge x_2) \vee \big(x_1 \vee (x_2' \wedge x_3)\big)$: build it up one column at a time (table below).

x_1	x_2	x_3	$x_1 \wedge x_2$	$x_2' \wedge x_3$	$x_1 \vee (x_2' \wedge x_3)$	p
0	0	0	0	0	0	0
0	0	1	0	1	1	1
0	1	0	0	0	0	0
0	1	1	0	0	0	0
1	0	0	0	0	1	1
1	0	1	0	1	1	1
1	1	0	1	0	1	1
1	1	1	1	0	1	1

Not on the slides. The last two columns are equal: $x_1 \wedge x_2$ is absorbed by the x_1 inside the second term, so $p = x_1 \vee (x_2' \wedge x_3)$.

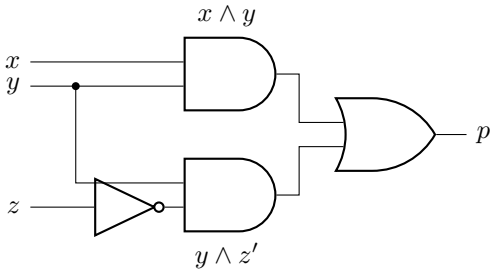
6.5.1 Logic diagrams

Each operation is a **gate**: $x \vee y$ is an **OR** gate (1 if at least one input is 1), $x \wedge y$ an **AND** gate (1 only if both are), x' a **NOT** gate (flips $0 \leftrightarrow 1$). Two more are common: $\mathbf{NAND}(A, B) = (A \wedge B)'$ and $\mathbf{NOR}(A, B) = (A \vee B)'$ — an AND or OR with a NOT bubble on the output.



Example (Truth table and circuit of $p(x, y, z) = (x \wedge y) \vee (y \wedge z')$).

x	y	z	p
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1



Reading the diagram back: the two AND gates give $x \wedge y$ and $y \wedge z'$, and the OR gate joins them, which gives back $(x \wedge y) \vee (y \wedge z')$.

6.5.2 Exercises covered

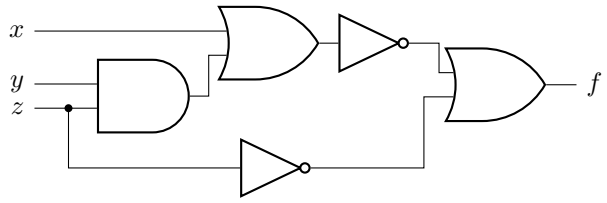
6.5 Ex. 3. $p(x,y,z) = (x \wedge y') \vee \big(y \wedge (x' \vee y)\big)$. Simplify first:

$$p = (x\wedge y')\vee\big((y\wedge x')\vee(y\wedge y)\big) = (x\wedge y')\vee\big((x'\wedge y)\vee y\big) = (x\wedge y')\vee y = (x\vee y)\wedge(y'\vee y) = (x\vee y)\wedge 1 = x\vee y.$$

So $f(x,y,z) = x \vee y$: the truth table is 0 exactly on the rows with $x = y = 0$ (000 and 001) and 1 on the other six. z doesn't matter.
Not on the slides. A shortcut: $y \wedge (x' \vee y) = y$ straight away by absorption.

6.5 Ex. 18 (Kolman Fig. 75). Read the circuit left to right:

1. AND: $y \wedge z$
2. OR: $x \vee (y \wedge z)$
3. NOT: $\big(x \vee (y \wedge z)\big)'$
4. lower branch NOT: z'
5. final OR:
   $f(x,y,z) = \big(x \vee (y \wedge z)\big)' \vee z'$.



Not on the slides. De Morgan simplifies it further: $\big(x \vee (y \wedge z)\big)' = x' \wedge (y' \vee z')$, so $f = (x' \wedge y') \vee (x' \wedge z') \vee z' = (x' \wedge y') \vee z'$, since z' absorbs $x' \wedge z'$.

Key points (6.4–6.5)

Boolean algebra = finite lattice $\cong B_n$ (subsets of an n-set, or bit strings of length n): bounded, complemented, distributive, 2^n elements. 2^n **elements is necessary, not sufficient.** D_n is a Boolean algebra $\iff$ n is a product of distinct primes. $\vee, \wedge, '$ behave like $\cup, \cap$, complement, so every set law carries over — especially absorption and De Morgan. A **Boolean function** $B^n \to B$ is a truth table with 2^n rows; a **Boolean polynomial** is built from variables, 0, 1, $\vee$, $\wedge$, $'$. **Logic diagrams**: OR $= \vee$, AND $= \wedge$, NOT $= '$; NAND $= (\wedge)'$, NOR $= (\vee)'$. Read circuits left to right, one gate at a time.

6.6 Circuit design

Source: Kolman 6.6, pp. 284–292

6.5 goes from an expression to a truth table. This section goes the other way: given only the truth table of $f : B^n \to B$, find an expression — and so a circuit — that produces it. For $f : B^n \to B$ let

$$S(f) = \{\, b \in B^n \mid f(b) = 1 \,\} \qquad \text{(the rows where } f \text{ is 1).}$$

6.6 Theorem 1

Let $f, f_1, f_2 : B^n \to B$. (a) If $S(f) = S(f_1) \cup S(f_2)$, then $f = f_1 \vee f_2$. (b) If $S(f) = S(f_1) \cap S(f_2)$, then $f = f_1 \wedge f_2$.

Definition. For an input $b = (c_1, \ldots, c_n) \in B^n$, the **minterm** E_b is $\bar{x}_1 \wedge \bar{x}_2 \wedge \cdots \wedge \bar{x}_n$, where $\bar{x}_k = x_k$ if $c_k = 1$ and $\bar{x}_k = x_k'$ if $c_k = 0$. It is 1 on the single input b and 0 everywhere else.

For two variables: $E_{00} = x' \wedge y'$, $E_{01} = x' \wedge y$, $E_{10} = x \wedge y'$, $E_{11} = x \wedge y$. For three: $E_{011} = x' \wedge y \wedge z$, and so on — a 0 in the input becomes a primed variable.

6.6 Theorem 2: every Boolean function has an expression

Any $f : B^n \to B$ is produced by a Boolean expression: if $S(f) = \{b_1, \ldots, b_k\}$, then

$$f = E_{b_1} \vee E_{b_2} \vee \cdots \vee E_{b_k}$$

— one minterm per row where $f = 1$, joined by $\vee$. (Proof: each E_{b_i} has $S = \{b_i\}$; apply Theorem 1(a).) If f is never 1, $f = 0$.

Method: truth table → circuit.

1. Mark the rows with $f = 1$; write the minterm of each.
2. Join the minterms with $\vee$ (Theorem 2). This is already a correct answer.
3. Simplify with the laws (distributive, $x \vee x' = 1$, $x \wedge 1 = x$, absorption, De Morgan) or a Karnaugh map (below). Fewer terms means fewer gates.
4. Draw it: one NOT per primed variable, one AND per term, one OR joining the terms.
5. **Verify**: a truth table of the circuit, one helper column per gate output, and check the last column against the given table row by row.

Allowed gates. If a task says "only NOT, OR and AND", a NAND or NOR gate is not allowed — draw an AND or OR followed by a NOT instead. An AND (or OR) of three terms can be drawn as two two-input gates in a row (associativity).

Example (6.6 Ex. 3: simplifying a sum of minterms).

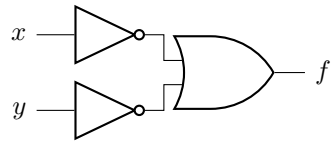
x	y	z	f
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

$S(f) = \{011, 111\}$, so by Theorem 2

$$\begin{aligned} f &= (x' \wedge y \wedge z) \vee (x \wedge y \wedge z) \\ &= (x' \vee x) \wedge (y \wedge z) && \text{distributive} \\ &= 1 \wedge (y \wedge z) = y \wedge z. && x \vee x' = 1,\ 1 \wedge p = p \end{aligned}$$

The circuit is a single AND gate.

Example (6.6 Fig. 83: a full worked circuit, with the check).
$f(x, y)$ is 1 on 00, 01, 10 and 0 on 11.
Minterms: $(x' \wedge y') \vee (x' \wedge y) \vee (x \wedge y')$.
First two: $x' \wedge (y' \vee y) = x'$. Then $x' \vee (x \wedge y') = (x' \vee x) \wedge (x' \vee y') = x' \vee y'$.
So $f = \mathbf{x' \vee y'}$.



x	y	x'	y'	$x' \vee y'$
0	0	1	1	1
0	1	1	0	1
1	0	0	1	1
1	1	0	0	0

The last column matches f on every row, so the circuit is correct. (It is a NAND, but drawn with NOT and OR only.)

6.6.1 Karnaugh maps

A **Karnaugh map** writes the values of f in a grid where *neighbouring squares differ in exactly one variable*; then a block of 1's collapses to a short term, because $x \vee x' = 1$ cancels the variable that changes inside the block.

two variables

	y'	y
x'	00	01
x	10	11

three variables (Kolman's layout)

	$y'z'$	$y'z$	yz	yz'
x'	000	001	011	010
x	100	101	111	110

The column order $00, 01, 11, 10$ (Gray code, not counting order) is what makes neighbours differ in one bit. The left and right edges are **neighbours too** (glue them into a cylinder).

Reading a Karnaugh map. Cover every 1 with rectangles of 1's of size $1, 2, 4$ or 8 (powers of two; rectangles may overlap and may wrap around the edge). Each rectangle gives one term: the variables that are *constant* on it (primed if constant 0). Join the terms with $\vee$. **Bigger rectangles give shorter terms**: in the 3-variable map, a single square is a 3-variable minterm, two squares give 2 variables, a full row or two adjacent columns give 1 variable.

Example (6.6 Ex. 6).

$S(f) = \{000, 011, 100, 101, 111\}$

	$y'z'$	$y'z$	yz	yz'
x'	**1**	0	**1**	0
x	**1**	**1**	**1**	0

Rectangles: the column $y'z'$ (both rows) gives $y' \wedge z'$; the column yz gives $y \wedge z$; the pair $100, 101$ in row x gives $x \wedge y'$. So

$$f = (y' \wedge z') \vee (x \wedge y') \vee (y \wedge z).$$

(The pair $101, 111$, giving $x \wedge z$, would do instead of $x \wedge y'$ — the cover is not unique.)

6.6.2 Grouping rules and slide examples

Grouping rules. (1) Group adjacent cells containing 1. (2) A group has $1, 2, 4, 8, \ldots$ cells (a power of 2). (3) Groups are rectangles (squares included). (4) Make groups as large as possible. (5) Groups may overlap. (6) Opposite edges are adjacent — the map wraps around. (7) Every 1 must be covered by at least one group. Reading a group: variables that stay constant stay (value 1: plain, value 0: primed); the variable that changes is dropped.

The slides draw the three-variable map with AB across the top ($00, 01, 11, 10$) and C down the side ($0, 1$) — Kolman's layout above is the same grid turned a quarter-turn; use whichever you like, but keep the Gray order on the two-variable axis.

- **Two variables.** $f(0,0) = f(0,1) = 1$, $f(1,0) = f(1,1) = 0$: one group of two in row x'; y changes (dropped), $x = 0$ stays, so $f = x'$. (Check by laws: $x'y' \vee x'y = x'(y' \vee y) = x'$.)
- **Three variables.** $F(A, B, C) = 1$ exactly on $010, 100, 101, 110$. Horizontal group $\{010, 110\}$: A changes, $B = 1$, $C = 0$ gives BC'. Vertical group $\{100, 101\}$: $A = 1$, $B = 0$, C changes gives AB'. So $F = BC' + AB'$.
- **Exercise 1.** Rows $x = 0, 1$, columns $yz = 00, 01, 11, 10$, both rows 1 1 0 0: one group of four (columns $00, 01$), only $y = 0$ is constant, so $f = y'$.
- **Exercise 2.** Row x': 1 0 0 1; row x: 1 0 1 1. Columns 00 and 10 are neighbours across the edge, so the block $\{x', x\} \times \{00, 10\}$ has $z = 0$ constant: z'. The leftover 1 at $xyz = 111$ pairs with 110 (row x, columns $11, 10$): $x \wedge y$. So $f = z' \vee (x \wedge y)$.

Not on the slides. Four variables w, x, y, z (Kolman pp. 291–292; on the 2024 MC2 paper): a 4×4 grid, rows wx and columns yz both in Gray order $00, 01, 11, 10$; left–right *and* top–bottom edges wrap. Rectangles of $1, 2, 4, 8$ squares give terms of $4, 3, 2, 1$ variables. To check an MC option quickly, evaluate it on a few squares (a 1 and a 0) rather than deriving the expression from scratch.

Key points (6.6)

$S(f)$ = the inputs where $f = 1$. **Minterm** E_b: AND of all variables, primed where b has a 0; it is 1 only at b. **Theorem 2**: f = OR of the minterms of its 1-rows — every truth table has an expression, hence a circuit (NOT/AND/OR suffice). Simplify with the laws or a **Karnaugh map** (Gray-order columns, edges wrap, biggest power-of-two rectangles). Always **verify** the circuit with a truth table, one column per gate.

Chapter 7
Trees

7.1 Rooted trees

Source: Kolman 7.1

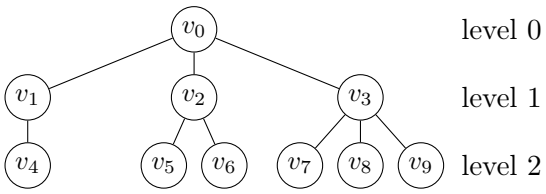
Definition. Let T be a relation on a set A. T is a **tree** if there is a vertex $v_0 \in A$ such that there is a *unique path* from v_0 to every other vertex, and there is *no path* from v_0 to itself. v_0 is the **root**, and (T, v_0) is a **rooted tree**. Elements of A are **vertices**.

Slide erratum. The slide says "a unique path from v_0 to *every* vertex $v \in A$", which would include v_0 itself and contradict the second bullet. Kolman says every *other* vertex, $v \neq v_0$.

7.1 Theorem 1

In a rooted tree (T, v_0): (a) T has no cycles; (b) v_0 is the unique root; (c) $\text{indeg}(v_0) = 0$ and every other vertex has $\text{indeg}(v) = 1$.

Terminology (draw the root at the top, edges point down):
- **parent**: the vertex directly above; **children**: the vertices directly below; **siblings**: same parent;
- **leaf**: a vertex with no children;
- **level**: the root is level 0, its children level 1, and so on; the **height** of the tree is its highest level.



Here the leaves are $v_4, \ldots, v_9$ and the height is 2.

Example (7.1 Ex. 2 (answer not on slides)). $A = \{v_1, \ldots, v_{10}\}$ and

$$T = \{(v_2, v_3), (v_2, v_1), (v_4, v_5), (v_4, v_6), (v_5, v_8), (v_6, v_7), (v_4, v_2), (v_7, v_9), (v_7, v_{10})\}.$$

Show T is a rooted tree and find the root.
Only v_4 is never the second coordinate of a pair (in-degree 0), so it is the only candidate root. Unique paths from v_4: $v_4 \to v_5 \to v_8$; $v_4 \to v_6 \to v_7 \to v_9$ and v_{10}; $v_4 \to v_2 \to v_3$ and v_1. All ten vertices are reached exactly once, so T is a rooted tree with **root** v_4. (Check: 10 vertices, 9 edges, every other vertex has in-degree 1.)

Ordered tree: a rooted tree whose children of each vertex have a specified order ($v_1 < v_2 < v_3$). Swapping the left-to-right order of the children gives a *different* ordered tree.

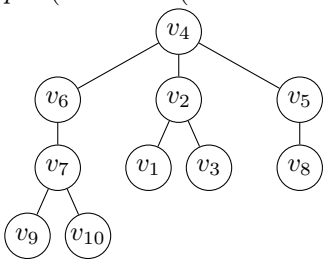
7.1 Theorem 2: a tree is like a parent relation

Read $(a, b) \in T$ as "a is the parent of b". Then T is **irreflexive** (no one is their own parent), **asymmetric** (if a is b's parent, b is not a's) and, for $(a, b), (b, c) \in T$, $(a, c) \notin T$: the grandparent is not the parent — so T is **not transitive**.

Definition. A rooted tree is an **n-tree** if every vertex has *at most* n children, and a **complete n-tree** if every non-leaf has *exactly* n children (so 0 or n children each). For $n = 2$: a **binary tree**, whose children are told apart as *left* and *right*.

A vertex v_2 is a **descendant** of v_1 if there is a path $v_1 \rightsquigarrow v_2$. **7.1 Theorem 3:** v together with all its descendants is again a tree, the **subtree** $T(v)$ rooted at v.

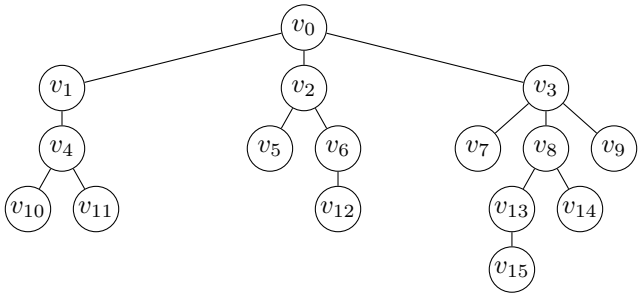
Example (Subtrees (7.1 Ex. 3 on the slides)).



- $T(v_6) = \{v_6, v_7, v_9, v_{10}\}$
- $T(v_7) = \{v_7, v_9, v_{10}\}$
- $T(v_2) = \{v_2, v_1, v_3\}$
- $T(v_8) = \{v_8\}$ (a leaf is a one-vertex subtree)

Example (7.1 Ex. 3–4). Is R a tree? If so, find the root.
3. $A = \{a, b, c, d, e, f\}$, $R = \{(a, b), (c, e), (f, a), (f, c), (f, d)\}$. Only f has in-degree 0; paths $f \to a \to b$, $f \to c \to e$, $f \to d$ reach every vertex once. Tree, **root** f.
4. $A = \{1, \ldots, 6\}$, $R = \{(2, 1), (3, 4), (5, 2), (6, 5), (6, 3)\}$. Only 6 has in-degree 0; paths $6 \to 5 \to 2 \to 1$ and $6 \to 3 \to 4$. Tree, **root** 6.

Example (7.1 Ex. 9–13).



- **9.** Level-3 vertices: $v_{10}, v_{11}, v_{12}, v_{13}, v_{14}$. Leaves: $v_5, v_7, v_9, v_{10}, v_{11}, v_{12}, v_{14}, v_{15}$.
- **10.** Siblings of v_8 (same parent v_3): v_7, v_9. Descendants of v_8: v_{13}, v_{14}, v_{15}.
- **11.** $T(v_2) = \{v_2, v_5, v_6, v_{12}\}$; $T(v_3) = \{v_3, v_7, v_8, v_9, v_{13}, v_{14}, v_{15}\}$.
- **12.** Height of (T, v_0): longest path $v_0 - v_3 - v_8 - v_{13} - v_{15}$ has 4 edges, so 4. Height of $T(v_3)$: $v_3 - v_8 - v_{13} - v_{15}$, so 3.
- **13.** The most children any vertex has is 3, so it is a **3-tree**; but v_1 has only one child, so it is **not complete**.

7.1.1 Exam method: which relation (or matrix) is a (directed) tree?

A "directed tree" or "rooted directed tree" in an MC question is just Kolman's *tree* (7.1): a relation T on A, read as a digraph with an arrow $a \to b$ for each $(a, b) \in T$, with one root and a unique path from it to every other vertex. It must be a tree *on all of* A.

Facts to use: no cycles (so no loops (x, x) and no symmetric pair $(x, y), (y, x)$); the root has in-degree 0 and every other vertex in-degree exactly 1; so n vertices means exactly $n - 1$ edges.

Method for "which relation is a tree?":

1. Cross out any relation with a loop (x, x) or a pair $(x, y), (y, x)$.
2. Count edges: must be (number of vertices) -1.
3. Tally the *second* coordinates (in-degrees). Exactly one vertex of A may be missing — the root; none may appear twice. A vertex of A that appears in no pair at all is unreachable.
4. From the root, follow the arrows and confirm every vertex is reached (rules out a cycle sitting apart from the root).

Example (On $A = \{1, 2, 3, 4, 5\}$: tree or not?).

- $\{(1, 2), (2, 3), (3, 1), (4, 5)\}$: cycle $1 \to 2 \to 3 \to 1$, and 4 and 5 cannot be reached from it. Not a tree.
- $\{(4, 1), (4, 2), (2, 3), (1, 3)\}$: 3 appears twice as a second coordinate (two parents), 5 never appears. Not a tree.
- $\{(3, 1), (3, 5), (5, 2), (5, 4)\}$: $4 = 5 - 1$ edges; second coordinates $1, 5, 2, 4$ — each once, only 3 missing, so the root is 3. $3 \to 1$, $3 \to 5 \to 2$, $5 \to 4$ reaches everything. **A tree**, root 3.

Same question with a matrix ("which matrix defines a tree?"): row i, column j is a 1 for the pair (i, j), i.e. an arrow $i \to j$. The method above becomes:

1. **Diagonal all** 0 — any 1 on the diagonal is a loop.
2. **Count the** 1**'s:** must be (number of vertices) -1.
3. **Column sums are the in-degrees:** exactly one column of zeros (the root), every other column sums to 1.
4. **Follow the arrows from the root** (read each row for where an arrow leads) and check every vertex is reached.

Steps 1–3 can all pass and the relation still fail: a cycle can sit apart from the root and satisfy the in-degree test by itself.

Example (Which 5×5 matrix is a tree?).

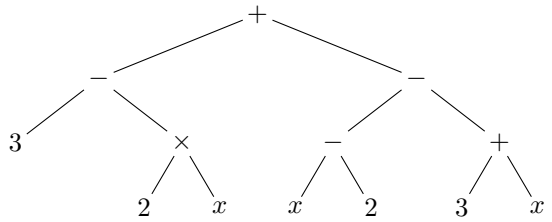
- **Pairs** $(1,2),(2,1),(3,4),(5,3)$: 4 ones, column sums $1,1,1,1,0$, so the root would be 5. But from 5 you only reach 3 and 4; $1 \leftrightarrow 2$ is a cycle cut off from the root. **Not a tree.**
- **Pairs** $(1,2),(3,1),(3,4),(4,5)$: 4 ones, column sums $1,1,0,1,1$, root 3. $3 \to 1 \to 2$ and $3 \to 4 \to 5$ reach everything. **A tree.**

7.2 Labeled trees

Source: Kolman 7.2

Vertices can carry labels (numbers, words, operators). The *structure* says how the objects are related, the *labels* say what they are: categories, decisions, expressions, programs, data structures.

Expression trees. Each operator is the *central operator* of its subexpression: it sits at the root of that subtree, with the left and right operands as its children. For $(3-(2\times x))+((x-2)-(3+x))$:



Positional trees. A *positional* tree is an ordered tree where each child has a fixed position: $1,2,\ldots,n$ in an n-tree (left, middle, right for $n=3$). The position stays fixed *even if some children are missing*: a vertex may have only its position-3 child. A **positional binary tree** has positions L and R; a vertex may have both, only L, only R, or neither.

Storing a binary tree. One cell per vertex: `LEFT | DATA | RIGHT`, where LEFT and RIGHT are the *indices* of the child cells, and 0 means "no child" (null). The first cell (*Start*) holds the index of the root in its LEFT field.

Example (7.2 Ex. 1: the expression tree above, as an array).

Index	L	Data	R
1	2	Start	0
2	3	$+$	8
3	4	$-$	5
4	0	3	0
5	6	$\times$	7
6	0	2	0
7	0	x	0
8	9	$-$	12
9	10	$-$	11
10	0	x	0
11	0	2	0
12	13	$+$	14
13	0	3	0
14	0	x	0

Read it: Start points to index 2 (the root $+$); its LEFT is 3 (the left $-$), its RIGHT is 8, and so on.

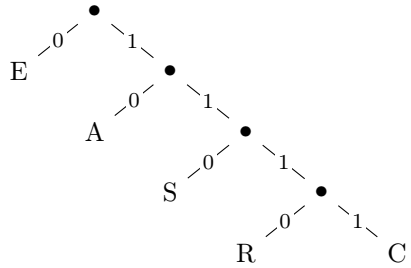
Example (7.2 Ex. 17 (the table on the slide)).

Index	LEFT	DATA	RIGHT
1	8	Start	0
2	5	D	7
3	9	E	0
4	2	C	3
5	0	F	0
6	0	B	4
7	0	G	0
8	6	A	0
9	0	H	0

Draw the digraph by the same reading rule; the slides give no solution.

Huffman codes. In a fixed-length code every character uses the same number of bits; in a **Huffman code** frequent characters get *shorter* codes. A positional binary tree stores the code: 0 = go left, 1 = go right, and the characters sit at the leaves.

Example (7.2 Ex. 4: decoding).



Codes: E = 0, A = 10, S = 110, R = 1110, C = 1111. Decode by walking from the root; at a leaf, write the letter and return to the root.

$$0101100 = \underbrace{0}_{E}\ \underbrace{10}_{A}\ \underbrace{110}_{S}\ \underbrace{0}_{E} \Rightarrow \text{EASE}.$$

Not on the slides. The question lists a second string, 0101110, but the slides only solve the first: $0\,|\,10\,|\,1110$ gives **EAR**. One flipped bit turns EASE into EAR — the weakness of a variable-length code (Kolman Example 4).

7.3 Tree searching

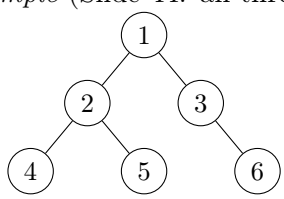
Source: Kolman 7.3

To do something with *every* vertex (compute, print, count, search) we need a systematic order: **tree traversal**. For a vertex v with left child v_L and right child v_R there are three actions: V (visit v), L (traverse the left subtree), R (traverse the right subtree).

	first	second	third	the root is visited
Preorder	V	L	R	*before* the subtrees
Inorder	L	V	R	*between* the subtrees
Postorder	L	R	V	*after* the subtrees

Each step is recursive: "traverse the left subtree" means apply the same rule to the subtree rooted at v_L, if it exists.

Example (Slide 44: all three traversals).



- Preorder: $1, 2, 4, 5, 3, 6$
- Inorder: $4, 2, 5, 1, 3, 6$
- Postorder: $4, 5, 2, 6, 3, 1$

Polish (prefix) form writes arithmetic without parentheses by putting the operator *before* its operands: infix $6 - 4$ becomes prefix $-\,6\,4$.

Not on the slides. Traversing an expression tree in *preorder* gives its Polish (prefix) form, *inorder* gives the ordinary infix form (without parentheses it can be ambiguous), and *postorder* gives the postfix (reverse Polish) form.

Evaluating a prefix expression: find an operator that is immediately followed by two numbers, replace the three symbols by the result, and repeat.

Example (Evaluate $\times\ -\ 6\,4\ +\ 5\ \div\ 2\,2$). $-\,6\,4 = 2$ and $\div\,2\,2 = 1$, giving $\times\,2\ +\ 5\,1$; then $+\,5\,1 = 6$, giving $\times\,2\,6 = \mathbf{12}$.

Example (7.3 Ex. 19: $\times\ -\ +\,3\,4\ -\ 7\,2\ \div\ 12\ \times\ 3\ -\ 6\,4$).

$\times\ -\ +\,3\,4\ -\ 7\,2\ \div\ 12\ \times\ 3\ -\ 6\,4$	$(+\,3\,4 = 7)$
$\times\ -\ 7\ -\ 7\,2\ \div\ 12\ \times\ 3\ -\ 6\,4$	$(-\,7\,2 = 5)$
$\times\ -\ 7\,5\ \div\ 12\ \times\ 3\ -\ 6\,4$	$(-\,7\,5 = 2)$
$\times\,2\ \div\ 12\ \times\ 3\ -\ 6\,4$	$(-\,6\,4 = 2)$
$\times\,2\ \div\ 12\ \times\ 3\,2$	$(\times\,3\,2 = 6)$
$\times\,2\ \div\ 12\,6$	$(\div\,12\,6 = 2)$
$\times\,2\,2 = \mathbf{4}$	

7.3.1 Exam method: traversal and Polish notation

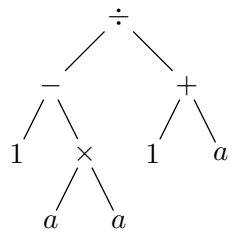
An algebraic expression is a **positional binary tree**: operators at the inner vertices, numbers and variables at the leaves, left child = left operand. The three traversals differ only in *when the root is written* — same recursion, the `visit(v)` line moved:

Algorithm	Order	Gives	Operator sits
PREORDER	root, left, right	prefix / **Polish** form	first
INORDER	left, root, right	infix (ordinary, needs parentheses)	middle
POSTORDER	left, right, root	postfix / **reverse Polish** form	last

Postfix is what a stack machine (an HP calculator, the JVM) executes: push operands, and each operator pops two and pushes the result.

Method: write the expression with full parentheses, one pair per operator, then recurse: for postfix, "left part, right part, operator". **Sanity checks on MC options:** postfix must *start with a leaf* and *end with the root operator*; prefix the reverse. Every leaf appears exactly once, and the leaves keep their left-to-right order in all three forms — only the operators move.

Example (Postfix form of $\big(1 - a \times a\big) \div \big(1 + a\big)$).



- left subtree $1 - (a \times a)$: $1\ \ a\,a\ \times\ \ -$
- right subtree $1 + a$: $1\,a\,+$
- root last: $\mathbf{1\,a\,a} \times - \mathbf{1\,a} +\ \div$

Prefix (PREORDER) would be $\div - 1 \times a\,a + 1\,a$. **Traps:** the prefix form is offered as a distractor; the infix $1 - a \times a \div 1 + a$ without parentheses means something else; and the evaluated value $\frac{1-a^2}{1+a}$ (or $1 - a$ simplified) is not a "form" at all.

Key points (7.1–7.3)

A **rooted tree**: a unique path from the root to every other vertex, in-degree 0 at the root and 1 everywhere else, no cycles. Terms: parent, child, sibling, leaf, level, **height** (the highest level), descendants, subtree $T(v)$. n**-tree**: at most n children; **complete**: exactly 0 or n. A tree relation is irreflexive, asymmetric and *not* transitive. **Labeled** trees: expression trees, positional trees, Huffman codes (0 = left, 1 = right, letters at leaves), and the `LEFT|DATA|RIGHT` array (index 0 = null). **Traversals**: preorder VLR, inorder LVR, postorder LRV — ask "when is the root visited?" Prefix expressions: evaluate an operator followed by two numbers, repeat.

7.4 Undirected trees

Source: Kolman 7.4

Take a rooted tree and forget the hierarchy: no parent/child, only the underlying shape. Formally, an **undirected tree** is the **symmetric closure** of a tree: for every $(a, b) \in T$ add (b, a) as well. The pair $a \to b$, $b \to a$ is drawn as one line $a - b$, the **undirected edge** $\{a, b\} = \{b, a\}$. So each line stands for two ordered pairs, and $(x, y) \in T \iff (y, x) \in T$.

Definition. If $\{a, b\}$ is an undirected edge, a and b are **adjacent vertices** (**neighbors**). A path $v_0, v_1, \ldots, v_n$ is **simple** if no undirected edge is used more than once (going $a \to b$ and later $b \to a$ uses the edge $\{a, b\}$ twice). A simple path with $v_0 = v_n$ is a **simple cycle**. A symmetric relation with no simple cycles is **acyclic**.

Slide erratum. Slide 10 (7.4 Ex. 2) switches to "simple = no *vertex* repeated", contradicting the definition two slides earlier. Use the definition above (no *edge* repeated). The example's verdict survives either way: f, e, d, c, d, a walks $d \to c \to d$, which uses the edge $\{c, d\}$ twice.

7.4 Theorems 1–3: what makes an undirected tree

Let R be a symmetric relation on a set A with n elements.

(1) **Theorem 1.** R is an undirected tree $\iff$ R is **connected and acyclic**. TREE = CONNECTED + ACYCLIC.

(2) **Theorem 2.** R is an undirected tree $\iff$ R is acyclic, but adding *any* new edge creates a cycle (maximally acyclic) $\iff$ R is connected, but removing *any* edge disconnects it (minimally connected).

(3) **Theorem 3.** A tree with n vertices has exactly $\mathbf{n - 1}$ **edges** (10 vertices: 9 edges; 100 vertices: 99).

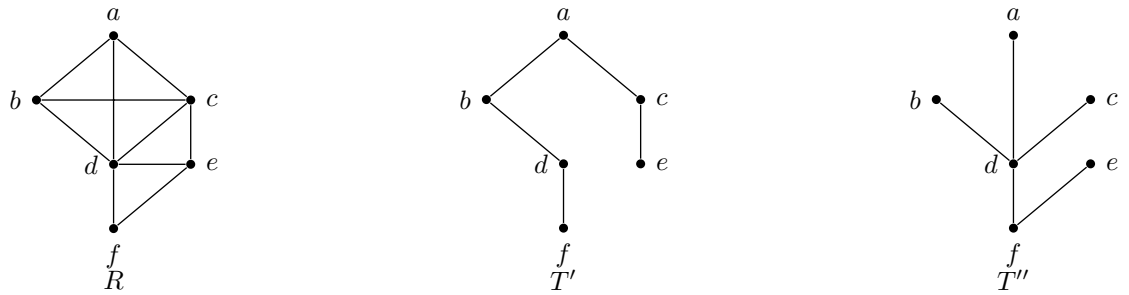
Slide erratum. Slide 13 states Theorem 2 as "tree iff (1) and (2)". Each condition *on its own* already characterises a tree, so read it as "tree iff (1), and tree iff (2)".
The four cases (slide 12): connected + no cycle = tree; connected + cycle, disconnected + no cycle (a *forest*) and disconnected + cycle are not trees. Example: $A = \{1, 2, 3, 4\}$, $R = \{\{1, 2\}, \{2, 3\}, \{3, 4\}\}$ is the path $1 - 2 - 3 - 4$: connected, acyclic, $3 = 4 - 1$ edges, so a tree.

7.4.1 Spanning trees

Definition. Let R be symmetric and connected. A **spanning tree** T of R is a tree with the same vertices that uses only edges of R: $V(T) = V(R)$ and $E(T) \subseteq E(R)$.

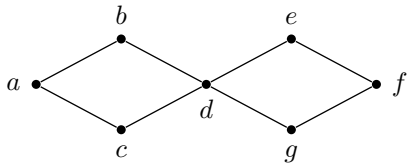
A graph usually has *many* spanning trees, but every one of them contains all vertices, is connected, has no cycles and has exactly $n - 1$ edges. **By deletion:** while there is a cycle, remove one of its edges (removing an edge *on a cycle* never disconnects the graph). Count: you delete $|E(R)| - (n - 1)$ edges in total.

Example (7.4 Ex. 3).



R has 6 vertices and 10 edges. Every spanning tree has $6 - 1 = \mathbf{5}$ edges, so 5 edges must go. T' and T'' are both spanning trees: **not unique**.

Example (7.4 Exercise, slide 30 (answer not on slides)).



7 vertices, 8 edges, so delete $8-6=2$ edges: one from each of the two 4-cycles $a\,b\,d\,c$ and $d\,e\,f\,g$. E.g. delete $\{a,b\},\{e,f\}$, or delete $\{c,d\},\{d,g\}$. Since any one of 4 edges can go from each cycle, there are $4{\cdot}4=16$ spanning trees.

7.4.2 Spanning trees by vertex merging

Instead of deleting edges, *build* the tree. **Merging** a and b replaces them by one new vertex $a'=\{a,b\}$ that is adjacent to everything a or b was adjacent to (two edges to the same vertex become one). **Record** the original edge $\{a,b\}$ at each merge — the recorded edges *are* the spanning tree.

Spanning tree by merging (R symmetric, connected):

1. Pick a start vertex v_1 and an adjacent vertex v_2, $\{v_1,v_2\}\in R$.
2. Record the original edge $\{v_1,v_2\}$; merge v_1,v_2 into one vertex.
3. Repeat (merged vertex + any neighbour) until one vertex is left. The $n-1$ recorded edges form a spanning tree.

In the matrix: merging rows/columns i and j = OR the two rows into one, OR the two columns into one, set the diagonal entry to 0, delete the leftover row and column. The matrix shrinks by one each step: $n\times n\to\cdots\to 1\times 1$.

Why it cannot make a cycle: each merge joins a vertex *outside* the merged blob to the blob, and an edge between two different groups never closes a cycle. (It is Prim's algorithm, below, without weights.)

Example (7.4 Ex. 7: merging in the matrix). The 4-cycle $a{-}c{-}b{-}d{-}a$ (edges $\{a,c\},\{a,d\},\{b,c\},\{b,d\}$):

	a	b	c	d
a	0	0	1	1
b	0	0	1	1
c	1	1	0	0
d	1	1	0	0

$\xrightarrow[\text{record } \{a,c\}]{a'=\{a,c\}}$

	a'	b	d
a'	0	1	1
b	1	0	1
d	1	1	0

$\xrightarrow[\text{record } \{c,b\}]{a''=\{a,b,c\}}$

	a''	d
a''	0	1
d	1	0

$\xrightarrow[\text{record } \{a,d\}]{a'''}$

	a'''
a'''	0

Row a' = row a OR row $c=(0,1,1)$ for (a',b,d). Spanning tree $\{a,c\},\{c,b\},\{a,d\}$: $3=4-1$ edges. When merging a' with b, record an *original* edge between the two groups: $\{c,b\}$ (the only one, since a–b is not an edge). At the last step $\{a,d\}$ or $\{b,d\}$ both work. The slides draw the merged vertex without saying which original edge to record.

The slides' second example (slides 24–25) merges v_0,v_1, then v_2, in a 7-vertex graph where v_0 and v_2 are both adjacent to v_5: after the second merge the two edges to v_5 become *one* undirected edge, which is exactly how the cycle $v_0-v_2-v_5$ is broken.

7.5 Minimal spanning trees

Source: Kolman 7.5

A **weighted graph** has a number, the **weight**, on each edge (distance, cost, travel time). For a vertex u, a **nearest neighbor** is a vertex joined to u by a minimum-weight edge (there may be ties). A **minimal spanning tree** (MST) is a spanning tree whose total weight

$$w(T)=\sum_{e\in E(T)} w(e)$$

is as small as possible. Trying every spanning tree is hopeless on a big graph; the algorithms below build an MST directly.

Prim's algorithm (7.5 Theorem 1: it always gives an MST)

R symmetric, connected, weighted, n vertices.

1. Start: $V = \{v_1\}$, $E = \emptyset$.
2. Choose a minimum-weight edge $\{v_i, v_j\}$ with $v_i \in V$ and $v_j \notin V$ (one end *in* the tree, one *out*). Ties: pick any.
3. Add v_j to V and $\{v_i, v_j\}$ to E.
4. Repeat until $|E| = n - 1$.

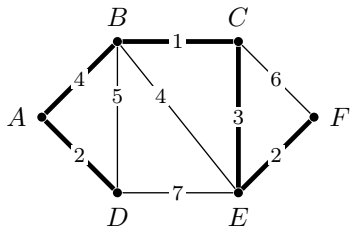
The rule is "the cheapest edge from *my current tree* to a new vertex" — not the globally cheapest remaining edge. It is **greedy**: the cheapest valid choice now, never looking ahead, and Theorem 1 says that is still optimal.

Kruskal's algorithm

Go through *all* edges from lightest to heaviest. Keep an edge if it does not close a cycle with the edges kept so far; otherwise skip it. Stop at $n - 1$ edges. Unlike Prim, the kept edges may form *several* separate trees (a forest) along the way; they join up at the end.

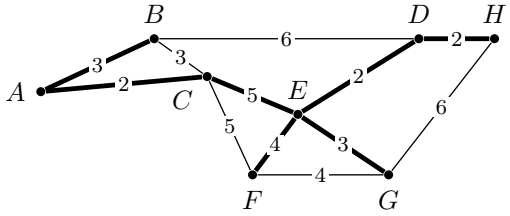
Both give the same total weight. The MST itself need not be unique when weights tie, but its weight is.

Example (Prim vs. Kruskal (slide 47)).



Prim from A: $A-D$ (2); from $\{A, D\}$ the cheapest way out is $A-B$ (4) (not $B-C$ (1): neither end is in the tree yet); then $B-C$ (1), $C-E$ (3), $E-F$ (2).
Kruskal: $B-C$ (1), $A-D$ (2), $E-F$ (2), $C-E$ (3), $A-B$ (4) — five edges, done; $B-E$ (4) would have closed a cycle.
Same tree (thick), total **12**.

Example (7.5 Ex. 1 and Ex. 5: walking trails (answer not on slides)).



8 vertices, so the MST has 7 edges. **Prim from A:** $A-C$ 2; $A-B$ 3 (tie with $B-C$); $C-E$ 5 (tie with $C-F$); $E-D$ 2; $D-H$ 2; $E-G$ 3; $E-F$ 4 (tie with $F-G$).
Prim from E: $E-D$ 2; $D-H$ 2; $E-G$ 3; $E-F$ 4; $C-E$ 5; $A-C$ 2; $A-B$ 3.
Kruskal: the 2's ($A-C$, $D-E$, $D-H$), the 3's ($A-B$, $E-G$; skip $B-C$), $E-F$ 4 (skip $F-G$), $C-E$ 5.
Minimum total distance $2+3+5+2+2+3+4 =$ **21** km. The ties mean several different MSTs, all of weight 21.

7.5.1 Prim in the matrix

A weighted graph's matrix holds the weight w at (i, j) and 0 for "no edge"; $M = M^T$. Prim becomes the merging of 7.4 with weights:

Pick the smallest non-zero entry in the merged vertex's row; record that edge. To merge rows i and j: entry by entry keep the **minimum non-zero** weight (a 0 means no edge, so it does not count as smaller), do the same for the columns, set the diagonal to 0 and delete the extra row and column.

Example (Slide 43 (answer not on slides)). Vertices 1–4:

$$M = \begin{bmatrix} 0 & 4 & 3 & 0 \\ 4 & 0 & 5 & 3 \\ 3 & 5 & 0 & 2 \\ 0 & 3 & 2 & 0 \end{bmatrix}.$$

Start at 1: row 1's smallest is 3 at column 3, record $\{1, 3\}$. Merge rows $1, 3$ into $1'$: $(\cdot, \min(4, 5), \cdot, \min(-, 2)) = (\cdot, 4, \cdot, 2)$; smallest 2 (it came from row 3), record $\{3, 4\}$. Merge with 4: weight to 2 is $\min(4, 3) = 3$ (from row 4), record $\{2, 4\}$. MST $\{1, 3\}, \{3, 4\}, \{2, 4\}$, weight $3 + 2 + 3 = \mathbf{8}$.

7.5.2 Exam method: spanning-tree and MST questions

- **How many edges?** Always $n - 1$. An option with the wrong count is out at once.
- **Is this a spanning tree of G?** $n - 1$ edges, all taken from G, every vertex touched, no cycle (with $n - 1$ edges, "connected" and "no cycle" imply each other — check the easier one).
- **MST weight:** run Kruskal on paper — list the edges by weight and tick them off, skipping any that join two vertices already connected. Fastest by hand, and the total is the same as Prim's.
- **"Which edge does Prim pick next?"** Only edges with exactly one end in the current tree count. The trap option is the cheapest edge *anywhere* (that is Kruskal's choice).
- **Is the MST unique?** If all weights are different, yes. With ties it can differ, the weight cannot.

Key points (7.4–7.5)

Undirected tree = symmetric closure of a tree = **connected** + **acyclic**; equivalently maximally acyclic, or minimally connected; n vertices $\Rightarrow$ $n - 1$ edges. **Simple** path: no edge used twice. **Spanning tree**: all vertices, edges from R, connected, acyclic, $n - 1$ edges, usually not unique; build it by deleting cycle edges or by **merging** vertices (OR the rows, record the edge). **MST**: spanning tree of least total weight. **Prim** (greedy): grow one tree, always the cheapest edge from the tree to a new vertex. **Kruskal**: cheapest edge overall that closes no cycle, may be a forest on the way. Same total weight.

Exam review and past-paper practice

The last lecture reviews the course and works questions from the 2023 and 2024 MC2 papers. The answers below are the ones highlighted on the slides; the reasoning is added here.

Relations	Equivalence = R+S+T; partial order = R+A+T.
Linear order	$\forall a, b: \ a \le b \vee b \le a$. $(\mathbb{Z}, \le)$ is linear; $(\mathbb{Z}^+, \mid)$ and $(\mathcal{P}(A), \subseteq)$ are partial only.
Hasse diagram	lower = smaller; edges are covering pairs; no loops, no shortcuts. Minimal/maximal: nothing strictly below/above. Least/greatest: $\le$ / $\ge$ *every* element.
Lattice	every pair has a join (lub, $\vee$) and a meet (glb, $\wedge$). **One pair without a join or meet and it is not a lattice.** Product order on $\mathbb{Z} \times \mathbb{Z}$: join = coordinatewise max, meet = coordinatewise min.
Boolean algebra	bounded (least 0, greatest 1), distributive, complemented ($a \vee a' = 1$, $a \wedge a' = 0$).
Functions	injective $f(a) = f(b) \Rightarrow a = b$; surjective every y hit; bijective both; invertible iff bijective. Boolean matrix: **total** (everywhere defined) = exactly one 1 per *row*; **injective** = at most one 1 per column; **surjective** = at least one 1 per column.
Trees	undirected tree: connected, no cycles, n vertices $\Rightarrow n-1$ edges. Rooted directed tree: exactly one root, every other vertex exactly one parent, all reachable from the root. Preorder root–left–right; inorder left–root–right; postorder left–right–root.
Growth	$1 \ll \log n \ll n \ll n \log n \ll n^2 \ll n^3 \ll 2^n$; O is an upper bound, Θ a tight one.

Paper	Question	Answer and why
2023	Q3 Hasse	(a) g is a minimal element. f, g, h are the bottom elements; there is no single least element, a is not above b, and $g \not\le h$.
2023	Q4 D_n	(f) D_{105}. Boolean iff n is square-free: $105 = 3 \cdot 5 \cdot 7$; the others have a square factor ($63 = 3^2 \cdot 7$, $75 = 3 \cdot 5^2$, $84 = 2^2 \cdot \cdots$, $90 = 2 \cdot 3^2 \cdot 5$, $92 = 2^2 \cdot 23$).
2023	Q5 linear	(e) $(\{1, 2, 4\}, \mid)$ is the chain $1 \mid 2 \mid 4$; the others contain incomparable elements.
2023	Q7 growth	(b) $n^3 \lg(n^2) + n = 2n^3 \lg n + n$ has the same order as $n^3 \lg n$.
2023	Q8 join	(d) $(-1, 1) \vee (2, -3) = (\max(-1, 2), \max(1, -3)) = (2, 1)$.
2023	Q9 functions	(d) $f(2) = f(5) = 1$. Not everywhere defined (6 has no image), not onto, not one-to-one, f^{-1} not a function.
2023	Q12 MST	(f) 14 (Kruskal/Prim on the weighted graph; the slide's marked answer).
2024	Q2 tree	(a) R_1: root 1, every other vertex one parent, all reachable. R_2: 3 has two parents; R_3: $5, 6, 7$ unreachable; R_4: 5 has two parents; R_5: loop $(3, 3)$; R_6: cycle $1 \to 2 \to 3 \to 4 \to 1$.
2024	Q9 growth	(c) $4n^2 + n \lg n$ is $\Theta(n^2)$, like $n^2 + n + \lg n$.
2024	Q10 meet	(b) $(-1, 1) \wedge (2, -3) = (\min, \min) = (-1, -3)$.
2024	Q11 function	(d) $\{(1, 3), (2, 1), (3, 2)\}$: a function with $f(2) = 1 < f(3) = 2$; the others have an element with two images or $f(2) \not< f(3)$.

Multiplication table 1–30 (grey diagonal = squares)

×	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
1	**1**	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
2	2	**4**	6	8	10	12	14	16	18	20	22	24	26	28	30	32	34	36	38	40	42	44	46	48	50	52	54	56	58	60
3	3	6	**9**	12	15	18	21	24	27	30	33	36	39	42	45	48	51	54	57	60	63	66	69	72	75	78	81	84	87	90
4	4	8	12	**16**	20	24	28	32	36	40	44	48	52	56	60	64	68	72	76	80	84	88	92	96	100	104	108	112	116	120
5	5	10	15	20	**25**	30	35	40	45	50	55	60	65	70	75	80	85	90	95	100	105	110	115	120	125	130	135	140	145	150
6	6	12	18	24	30	**36**	42	48	54	60	66	72	78	84	90	96	102	108	114	120	126	132	138	144	150	156	162	168	174	180
7	7	14	21	28	35	42	**49**	56	63	70	77	84	91	98	105	112	119	126	133	140	147	154	161	168	175	182	189	196	203	210
8	8	16	24	32	40	48	56	**64**	72	80	88	96	104	112	120	128	136	144	152	160	168	176	184	192	200	208	216	224	232	240
9	9	18	27	36	45	54	63	72	**81**	90	99	108	117	126	135	144	153	162	171	180	189	198	207	216	225	234	243	252	261	270
10	10	20	30	40	50	60	70	80	90	**100**	110	120	130	140	150	160	170	180	190	200	210	220	230	240	250	260	270	280	290	300
11	11	22	33	44	55	66	77	88	99	110	**121**	132	143	154	165	176	187	198	209	220	231	242	253	264	275	286	297	308	319	330
12	12	24	36	48	60	72	84	96	108	120	132	**144**	156	168	180	192	204	216	228	240	252	264	276	288	300	312	324	336	348	360
13	13	26	39	52	65	78	91	104	117	130	143	156	**169**	182	195	208	221	234	247	260	273	286	299	312	325	338	351	364	377	390
14	14	28	42	56	70	84	98	112	126	140	154	168	182	**196**	210	224	238	252	266	280	294	308	322	336	350	364	378	392	406	420
15	15	30	45	60	75	90	105	120	135	150	165	180	195	210	**225**	240	255	270	285	300	315	330	345	360	375	390	405	420	435	450
16	16	32	48	64	80	96	112	128	144	160	176	192	208	224	240	**256**	272	288	304	320	336	352	368	384	400	416	432	448	464	480
17	17	34	51	68	85	102	119	136	153	170	187	204	221	238	255	272	**289**	306	323	340	357	374	391	408	425	442	459	476	493	510
18	18	36	54	72	90	108	126	144	162	180	198	216	234	252	270	288	306	**324**	342	360	378	396	414	432	450	468	486	504	522	540
19	19	38	57	76	95	114	133	152	171	190	209	228	247	266	285	304	323	342	**361**	380	399	418	437	456	475	494	513	532	551	570
20	20	40	60	80	100	120	140	160	180	200	220	240	260	280	300	320	340	360	380	**400**	420	440	460	480	500	520	540	560	580	600
21	21	42	63	84	105	126	147	168	189	210	231	252	273	294	315	336	357	378	399	420	**441**	462	483	504	525	546	567	588	609	630
22	22	44	66	88	110	132	154	176	198	220	242	264	286	308	330	352	374	396	418	440	462	**484**	506	528	550	572	594	616	638	660
23	23	46	69	92	115	138	161	184	207	230	253	276	299	322	345	368	391	414	437	460	483	506	**529**	552	575	598	621	644	667	690
24	24	48	72	96	120	144	168	192	216	240	264	288	312	336	360	384	408	432	456	480	504	528	552	**576**	600	624	648	672	696	720
25	25	50	75	100	125	150	175	200	225	250	275	300	325	350	375	400	425	450	475	500	525	550	575	600	**625**	650	675	700	725	750
26	26	52	78	104	130	156	182	208	234	260	286	312	338	364	390	416	442	468	494	520	546	572	598	624	650	**676**	702	728	754	780
27	27	54	81	108	135	162	189	216	243	270	297	324	351	378	405	432	459	486	513	540	567	594	621	648	675	702	**729**	756	783	810
28	28	56	84	112	140	168	196	224	252	280	308	336	364	392	420	448	476	504	532	560	588	616	644	672	700	728	756	**784**	812	840
29	29	58	87	116	145	174	203	232	261	290	319	348	377	406	435	464	493	522	551	580	609	638	667	696	725	754	783	812	**841**	870
30	30	60	90	120	150	180	210	240	270	300	330	360	390	420	450	480	510	540	570	600	630	660	690	720	750	780	810	840	870	**900**

Primes up to 2500 (367 of them): 2 3 5 7 11 13 17 19 23 29 31 37 41 43 47 53 59 61 67 71 73 79 83
89 97 101 103 107 109 113 127 131 137 139 149 151 157 163 167 173 179 181 191 193 197 199 211
223 227 229 233 239 241 251 257 263 269 271 277 281 283 293 307 311 313 317 331 337 347 349 353
359 367 373 379 383 389 397 401 409 419 421 431 433 439 443 449 457 461 463 467 479 487 491 499
503 509 521 523 541 547 557 563 569 571 577 587 593 599 601 607 613 617 619 631 641 643 647 653
659 661 673 677 683 691 701 709 719 727 733 739 743 751 757 761 769 773 787 797 809 811 821 823
827 829 839 853 857 859 863 877 881 883 887 907 911 919 929 937 941 947 953 967 971 977 983 991
997 1009 1013 1019 1021 1031 1033 1039 1049 1051 1061 1063 1069 1087 1091 1093 1097 1103 1109 1117
1123 1129 1151 1153 1163 1171 1181 1187 1193 1201 1213 1217 1223 1229 1231 1237 1249 1259 1277 1279
1283 1289 1291 1297 1301 1303 1307 1319 1321 1327 1361 1367 1373 1381 1399 1409 1423 1427 1429 1433
1439 1447 1451 1453 1459 1471 1481 1483 1487 1489 1493 1499 1511 1523 1531 1543 1549 1553 1559 1567
1571 1579 1583 1597 1601 1607 1609 1613 1619 1621 1627 1637 1657 1663 1667 1669 1693 1697 1699 1709
1721 1723 1733 1741 1747 1753 1759 1777 1783 1787 1789 1801 1811 1823 1831 1847 1861 1867 1871 1873
1877 1879 1889 1901 1907 1913 1931 1933 1949 1951 1973 1979 1987 1993 1997 1999 2003 2011 2017 2027
2029 2039 2053 2063 2069 2081 2083 2087 2089 2099 2111 2113 2129 2131 2137 2141 2143 2153 2161 2179
2203 2207 2213 2221 2237 2239 2243 2251 2267 2269 2273 2281 2287 2293 2297 2309 2311 2333 2339 2341
2347 2351 2357 2371 2377 2381 2383 2389 2393 2399 2411 2417 2423 2437 2441 2447 2459 2467 2473 2477

Index

absorption law, 51, 52
  lattice, 43
acyclic, 63
addition principle, 17
adjacent vertices, 63
AND gate, 54
antisymmetric, 28, 35
  proving equality, 44
array representation of a tree, 60
associative law, 6
asymmetric, 28, 35
at least one (complement trick), 19
atom, 53
atom of a Boolean algebra, 48

B_n, 48
backtracking, 22
base conversion, 10
  hex to octal, 11
biconditional, 13
Big-O, 33
Big-Θ, 33
bijection, 31
bijective, 31
binary, 10
binary addition, 11
binary tree, 58
binomial coefficient, 18
block of a partition, 28
Boolean algebra, 48
  D_n as a Boolean algebra, 49
  D_n test, 49–51
  laws, 51
  not a Boolean algebra, 48
Boolean function, 53
  expression from a truth table, 55
Boolean matrix, 11
Boolean polynomial, 53
  simplification rules, 53
Boolean product of matrices, 11
bounded lattice, 45

cardinality, 5
Cartesian product, 28
ceiling function, 21
chain (linear order), 35
characteristic equation, 22, 23
  working backwards, 24
characteristic function, 6
child, 58
circuit design, 55
combination $C(n, k)$, 18
  table, 18
  with repetition, 19
common denominator, 22
commutative law, 6
comparable elements, 35, 44
complement
  in a lattice, 46
  of a set, 5
  of an event, 21
complemented lattice, 46
complete n-tree, 58
composition of functions, 31
composition of relations, 26
conditional
  equivalent forms, 14
conditional statement, 13
conjecture, 16
conjunction, 13
connected, 63
connectivity relation R^∞, 27
constant coefficients, 23
contingency, 13
contradiction, 13
contrapositive, 13
  of a statement about sets, 13
converse, 13
counterexample, 28
counting
  size of a union, 6
  the four cases, 20
cover relation, 37
  from a matrix, 37

D_n (divisors of n), 43, 49, 50
  lookup table, 51
De Morgan's laws, 52
  Boolean algebra, 51
  logic, 14
  sets, 6
degree (recurrence), 23
derivative
  table of derivatives, 34
descendant, 58
determinant formula (2×2 inverse), 11
diamond lattice (M_3), 45
digraph, 26
directed tree, 59
disjunction, 13
disjunctive syllogism, 15
distinct primes, 49

distinguishable permutations, 19
MISSISSIPPI, 20
distributive lattice, 45
distributive law
logic, 14
sets, 6
dividend and divisor, 7
divides (|), 7
divisibility, 7
divisibility rules, 7
division with remainder, 7
domain (relation), 26
dominant term, 34
domination law, 6
double negation, 14
doubling the input, 33
dual poset, 35
duality, 52

element of (∈), 5
empty relation, 36
empty set, 5
equally likely outcomes, 21
equivalence class, 29
equivalence relation, 29, 36
Euclidean algorithm, 9
event, 21
everywhere defined, 31
everywhere defined function, 31
exactly / at least / at most, 19
exam review, 67
expected value, 21
explicit formula, 6
exponent rule, 23
exponential times logarithm, 34
exponents
dividing powers, 23
multiplying powers, 20
expression tree, 60

$f^{-1}(\{b\})$, 31
factorial, 17
matching MC options, 17
unrolling, 17
floor and ceiling, 33
floor function
injective or surjective, 32
for all (∀), 14
fractions, adding, 22
function, 31
formula on $\mathbb{Z}$, 32
fundamental theorem of arithmetic, 7
fuzzy set, 33
min and max, 33

GCD and LUB in a power set, 50
$\gcd \cdot \operatorname{lcm} = ab$, 10
general solution, 23
generalized pigeonhole principle, 21
geometric series, 24
greatest common divisor (gcd), 8
Euclidean algorithm, 9
written as $sa + tb$, 9
greatest element, 39
versus maximal, 39, 40
greatest lower bound (GLB), 39
greedy algorithm, 64
growth of functions, 33

hash function, 33
Hasse diagram, 37
from a matrix, 37
of D_n, 49
reading bounds, 40
topological sorting, 41
height of a tree, 58
hexadecimal, 10
homogeneous (recurrence), 23
Horner's rule, 10
Huffman code, 60
hypothetical syllogism, 15

idempotent law, 6
lattice, 43
identity function, 31
identity law, 6
identity matrix, 11
image of a set under a relation, 28
in-degree, 58
column sums, 59
inclusion–exclusion, 6
incomparable elements, 35, 44
counterexample, 35
induction
base case, 16
hypothesis, 16
strong, 16
inference rules, 15
identifying from an English argument, 15
infix form, 62
initial conditions, 22
injective, 31, 32
injective (one-to-one), 31
inorder, 61
integers, 5
properties, 7
intersection, 5
inverse
of a conditional, 13

of a matrix, 11
inverse function, 31, 32
of a composition, 32
inverse image, 31
involution, 51
irreflexive, 28, 35
isomorphism, 48
of posets, 38
$\mathcal{P}(S)$ and D_n, 50

join, 42
of matrices, 11

Karnaugh map, 56
grouping rules, 57

labeled tree, 60
lattice, 42
consistency ($a \leq b \iff a \wedge b = a$), 43
Hasse diagram test, 44
is it a lattice?, 44
modular, 45
monotonicity of $\vee$, $\wedge$, 43
properties, 43
proving an identity, 44
test from a Hasse diagram, 42
leaf, 58
least common multiple (lcm), 8
least element, 39
least upper bound (LUB), 39
level, 58
lexicographic order, 35, 36, 40
L'Hôpital's rule, 34
limit, 34
linear (recurrence), 23
linear homogeneous recurrence, 22
recognising, 23
linear order
$\mathcal{P}(S)$ is not, 35
logarithm
change of base, 34
logic diagram, 54
logical equivalence, 13
logical equivalences, 14
lower bound, 39
method, 40
of a one-element set, 39, 40

mathematical induction, 16
matrix, 11
matrix multiplication, 11
matrix of a relation
to Hasse diagram, 37
maximal element, 39
MC options as products of primes, 20
meet, 42
of matrices, 11
merging vertices, 64
minimal element, 39
minimal spanning tree, 64
exam method, 66
minimum spanning tree, *see* minimal spanning tree
minterm, 55
mod function, 10
modular lattice, 45
checking, 46
modulo, 7, 10
modus ponens, 15
modus tollens, 15
MST, *see* minimal spanning tree
multiplication principle, 17
license plates, 17
password, 20
multiplication table, 68
mutually exclusive events, 21

n choose k, *see* combination
n-tree, 58
NAND gate, 54
natural numbers, 5
${}_nC_k$, ${}_nP_k$ notation, 19
nearest neighbor, 64
negating a quantified statement, 14
negation, 13
negation of a conditional, 14
neighbor, 63
non-distributive lattice, 45
NOR gate, 54
NOT gate, 54
number bases, 10
number of divisors, 8
number sets, 5

one-to-one, *see* injective
onto, *see* surjective
OR gate, 54
order matters / does not matter, 18
order of growth, 33
ranking of common functions, 33
order relation
lexicographic, 36
ordered tree, 58

parent, 58
partial order, 35
given by a formula, 36
matrix test, 36
matrix to Hasse diagram, 37
which relation on $\mathbb{Z}^+$?, 36

partition, 28
Pascal's triangle, 18
past MC questions
  2023, 67
  2024, 67
path
  R^∞ reachability, 27
  computing R^2, 26
    exam layout, 27
  of length n (R^n), 26
pentagon lattice (N_5), 45
permutation $P(n,k)$, 18
  with repeated letters, 19
pigeonhole principle, 21
Polish notation, 61, 62
polynomial versus exponential, 33
poset, 35
positional tree, 60
postfix form, 62
postorder, 61
power set, 5
  isomorphic to D_n, 50
  not linearly ordered, 35
predicate, 5
prefix form, 61, 62
  evaluating, 61
preimage, 31
preorder, 61
Prim's algorithm, 64
  matrix version, 65
prime, 7
  testing (square-root rule), 8
prime factorization, 7
  by trial division, 8
prime numbers
  list of primes up to 2500, 68
primes
  list, 68
probability, 21
probability rules, 21
product of chains, 49
product order, 35, 40, 41, 43
product set, 28
proof, 15
  by contradiction, 15
  by contrapositive, 15
  direct, 15
proposition, 13

quantifier, 14
quotient, 7

$R(x)$, relative set, 26
range (relation), 26
recurrence relation, 22
  degree 2 solution, 22
  four-step recipe, 23
  repeated root example, 23
  shifted indices, 24
recurrence.py (checker script), 24
reflexive, 28, 29, 35
relation, 26
  complement, 29
  given by a formula, 28, 36
  inverse, 29
  matrix of a relation, 28, 29
  properties, 28
  union and intersection, 29
relatively prime, 9
remainder, 7
repeated root, 22, 23
reverse Polish notation, *see* postfix
root, 58
rooted tree, 58

$S(f)$, 55
same order of growth, 33
  log base, 34
  picking from options, 34
same parity, 36
sample space, 21
sequence, 6
set, 5
set algebra, 51
set difference, 5
set laws, 6
set-builder notation, 5
sibling, 58
sigma notation (Σ), 24
simple cycle, 63
simple path, 63
SORT algorithm, 41
spanning tree, 63
  exam method, 66
  vertex merging, 64
square-free, 49
  fast check, 50
stars and bars, 19
strict order, 35
sublattice, 44
subset, 5
subsets
  number of (2^n), 20
subtree, 58
sum, *see* sigma notation
surjective, 31, 32
surjective (onto), 31
symmetric, 28, 29

symmetric closure
    undirected tree, 63
symmetric difference, 5

tautology, 13
there exists (∃), 14
Θ-class, 33
topological sorting, 38, 41
transitive, 28, 29, 35
    row trick, 36
transitive closure, 27
transpose of a matrix, 11
tree, 58
    from a matrix, 59
    is this relation a tree?, 58
    spanning, 63
    traversal, 62
    undirected, 63
    which matrix is a tree?, 59
    which relation is a tree?, 59
tree traversal, 61
trial division, 8
truth table, 13
    of a Boolean function, 53
    to Boolean expression, 55

undirected edge, 63
undirected tree, 63
union, 5
unique factorisation, 7
universal set, 5
upper bound, 39
    method, 40

valid argument form, 15
Venn diagram, 5
vertex merging, 64

weight of an edge, 64
weighted graph, 64
weighted matrix, 65